Reconstruction of periodic nanostructures using grazing incidence X-ray fluorescence

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Kurzfassung

In dieser Arbeit wird eine neue Methode zur elementsensitiven Charakterisierung von periodischen Nanostrukturen entwickelt und metrologisch detailliert analysiert. Sie basiert auf der Erweiterung der bekannten Technik der Röntgenfluoreszenzspektrometrie unter streifendem Einfall auf 2D-Strukturen. Dabei wird die Funktionalität der Nanostrukturen von der Materialverteilung bestimmt. Material in der Größenordnung mehrere Nanometer unterscheidet sich in seinen Eigenschaften von Bulkmaterial, einzelnen Atomen oder Molekülen. Diese Nanostrukturen werden beispielsweise bereits für Displays und elektronischen Mikrochips kommerziell produziert. Die Leistung und gleichbleibende Qualität von Nanostrukturen werden durch eine gut kontrollierte Herstellung gewährleistet. Zur Charakterisierung werden Messtechniken im Nanobereich benötigt. Die Nanometrologie ist ein Zweig der Metrologie, der sich mit der Messung und Charakterisierung von Nanostrukturen befasst.

Die in dieser Arbeit entwickelte Methode verwendet Röntgenfluoreszenzmessungen bei einer Energie von 680 eV (~1,83 nm) unter streifenden Einfall. Dabei entsteht ein stehendes Röntgenwellenfeld aus der Interferenz zwischen der einfallenden und der reflektierten Röntgenstrahlung. Diese Messung ist sensitiv auf das Linienprofil der Strukturen, weil das sich ausbildende Wellenfeld im Nahfeld von dem Profil der Nanostruktur und dem Einfallswinkel abhängt. Die resultierenden Röntgenfluoreszenz hängt von der Position des entsprechenden Elementes im Wellenfeld ab. Die entwickelte Methode verwendet einen auf finiten Elementen basierenden Maxwell-Solver zur Berechnung des stehendes Röntgenwellenfeldes. In dieser Arbeit wurde die bekannte Sherman-Gleichung zur Berechnung der Fluoreszenzintensität von einer Probe mit bekannter Materialzusammensetzung zum ersten Mal auf den Rekonstruktionsprozess von Linienprofilen angewendet. Dazu wurde die Sherman-Gleichung verwendet, um die erwartete Fluoreszenz von periodischen 2D-Nanostrukturen mit einem bestimmten Profil für verschiedene Einfallswinkel zu bestimmen. Im Bereich der weichen Röntgenstrahlung sind die Fundamentalparameter wie z.B. die optischen Konstanten aktuell nicht gut bekannt. Zusätzlich können Nanostrukturen eine reduzierte Dichte und damit optischen Konstanten aufweisen, die sich vom Bulkmaterial unterscheiden. Wegen des großen Einflusses auf den Rekonstruktionsprozess ist es notwendig, die optischen Konstanten der Materialien so genau wie möglich zu bestimmen. In dieser Arbeit werden Messungen der weichen Röntgenreflexion analysiert, um die optischen Konstanten der für den Maxwell-Solver verwendeten Materialien zu bestimmen. Es werden verschiedene Parametrisierungen der Form der Nanostrukturen verwendet, um die Struktur so genau wie möglich zu beschreiben. Um das Ergebnis und die Effizienz der Rekonstruktion zu verbessern, werden verschiedene Optimierungsmethoden benutzt und miteinander verglichen. Die Verwendung einer Finite-Elemente-Methode im weichen Röntgenbereich stellt eine Herausforderung für die Simulation der Fluoreszenz komplexer Nanostrukturen dar auf Grund der kleinen Wellenlänge der einfallenden Strahlung. Surrogates, ein Werkzeug des maschinellen Lernens, können die zeitaufwändige Finite-Elemente Rechnung approximieren und ersetzen. In dieser Arbeit wurde ein Surrogat erfolgreich trainiert, um die Unsicherheiten des rekonstruierten Linienprofils abzuschätzen. Die berechneten Ergebnisse werden mit mikroskopischen Bildern von Monitorproben und Rasterkraftmikroskopie-Messungen validiert. Dabei wurde eine gute Übereinstimmung festgestellt. Schließlich wird der Einfluss von Modellfehlern wie

einer kontaminierten Oberfläche im Detail analysiert, um ihren Einfluss auf die Rekonstruktion besser zu verstehen.

Die im Rahmen dieser Arbeit entwickelte Methode liefert elementsensitive Informationen mit einer Auflösung im sub-nm-Bereich und ist eine vielfältige Ergänzung zur Charakterisierung von Nanostrukturen, die an komplexere reale Proben angepasst werden kann.

Abstract

In this thesis, a new method for element sensitive characterization of periodic nanostructures is developed and metrologically analyzed in detail. It is based on the extension of the well-known technique grazing incidence X-ray fluorescence spectrometry to 2D structures. To function properly, nanostructures require the control of matter at the nanoscale, where the properties of matter differ from those of bulk materials, single atoms, or molecules. These kinds of structures are present in everyday objects such as displays and electronic microchips. The performance of nanostructures is ensured with a well-controlled fabrication. Therefore, metrology techniques in the nano-regime are needed. Nanometrology is a branch of metrology concerned with the measurement and characterization of nanostructures.

The in this work developed method uses grazing incidence X-ray fluorescence measurements with an incident energy of 680 eV (~1.83 nm). The sensitivity of the measurements to the line profile of the structure is due to the X-ray standing wave field resulting from the interference between the incident and reflected X-rays. The angle of incidence and shape of the nanostructure affect the field within the structure. The X-ray fluorescence intensity of an element depends on its spatial position within the electromagnetic field distribution. The method uses a finite element-based Maxwell solver to calculate the X-ray standing wave field. In this work, the Sherman equation for the calculation of the fluorescence intensity of a sample with known material composition was applied to the reconstruction process of the shape of nanostructures for the first time. The Sherman equation was adapted to determine the expected fluorescence of periodic 2D nanostructures with a certain profile for different angles of incidence. In the soft X-ray regime, the fundamental parameters like the optical constants are not well studied, and nanostructures may have a reduced density with optical constants that differ from the bulk material. Because of the high impact on the reconstruction process, it is necessary to determine the optical constant of the materials as accurately as possible. In this work, soft X-ray reflectivity measurements are analyzed to determine the optical constants of the materials used for the Maxwell solver. Different parameterizations of the shape of the nanostructures are used to describe them as accurately as possible. To improve the performance and the efficiency of the reconstruction, different optimization methods are applied and compared to each other. Using a finite elements method in the soft X-ray range is a challenge for the simulation of the fluorescence of complex nanostructures because of the small wavelength of the incident beam. Surrogates, a machine learning tool, can approximate and replace the time-consuming finite-element calculations as an alternative. In this work, a surrogate was successfully trained to estimate the uncertainties of the reconstructed shape. The compute results are validated with microscopic images of sister samples and atomic force microscopy measurements. The observed results met the expectations of these validations. Finally, the influence of model imperfections such as a small contamination layer is analyzed in detail to better understand its influence on the reconstruction.

The method provides element sensitive information with sub-nm resolution and is a versatile supplement to the characterization of nanostructures that can be adapted to more complex real samples.

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List of Abbreviation

A

AFM – *Atomic Force Microscopy* ...1, 4, 17, 21, 22, 27, 66, 67, 78, 79, 81, 83

APT - *Atom Probe Tomography* 1

B

BI – *Bayesian Inference* 92

BO - *Bayesian Optimization* ..3, 42, 50, 51, 52, 53, 54, 55, 56, 57, 60, 62, 63, 65, 67, 71, 72, 73, 74, 75, 78, 79, 81, 92, 93

C

CDSAXS - *Critical Dimension Small Angle X-ray Scattering* 2

CDSAXS - *Critical Dimension Small Angle X-Ray Scattering* 2

CG - *Conjugated Gradient* 91

CXRO - *Center for X-ray Optics* ...30, 36, 37, 72, 75, 78

CXRO - *Center for X-Ray Optics* ..30, 36, 37, 72, 75, 78

D

DE – *Differential Evolution* ..28, 29, 33, 37, 46, 48, 74, 91

DoF - Degrees of Freedom 15, 56

DUV - *Deep Ultraviolet* 2

E

EBL – *Electron-Beam Lithography* 16, 21

EDX - *Energy-Dispersive X-ray Spectroscopy* 1

EDX - *Energy-Dispersive X-Ray Spectroscopy* 1

EI – *Expected Improvement* 50, 93

EM - *Error Model* 13

EUV – *Extreme Ultraviolet* 2, 9, 82, 85, 86, 90

F

FEM – *Finite Element Method* ...3, 8, 9, 11, 26, 27, 43, 60, 63, 72, 73, 76, 77, 78, 79, 84

FIB - *Focused Ion Beam* 17, 18, 86

FLM - *First Local Maximum* 40, 41, 57

G

GEXRF - *Grazing Exit X-ray Fluorescence* 82

GEXRF - *Grazing Exit X-Ray Fluorescence* 82

GISAXS - *Grazing Incidence Small Angle X-ray Scattering* 2

GISAXS - *Grazing Incidence Small Angle X-Ray Scattering* 2, 83

GIXRF - *Grazing Incidence X-ray Fluorescence* 2, 3, 23, 26, 27, 38, 42, 52, 53, 54, 66, 67, 68, 78, 81

GIXRF - *Grazing Incidence X-Ray Fluorescence*2, 3, 23, 26, 27, 38, 42, 52, 53, 54, 66, 67, 68, 78, 81, 84

GP - *Gaussian Process* 50, 92, 93

H

HSQ - *Hydrogen silsesquioxane* .16, 18, 20, 21, 22, 27, 28, 31, 32, 34, 35, 37, 38, 41, 42, 44, 46, 47, 48, 49, 50, 55, 56, 58, 66, 67, 68, 73, 75, 76, 78, 79, 89, 90

HZB – *Helmholtz-Zentrum Berlin* 16, 19, 21

L

L-BFGS-B - *Limited-memory-Broyden–Fletcher–Goldfarb–Shanno-Bounds* 29, 33, 35, 91

M

MBDDT – *Many-beam dynamical diffraction theory* 9

MBDDT – *Many-Beam Dynamical Diffraction Theory* 9

MCMC - *Markov Chain Monte Carlo* 28, 29, 34, 35, 36, 37, 59, 62, 63, 64, 65, 70, 71, 79, 81, 82, 89, 92

ML - *Machine Learning* 92

N

NIST - *National Institute of Standards and Technology* 29

O

OCD - *Optical Critical Dimension* 2

P

PCE – *Polynomial Chaos Expansion* 11, 59, 60, 64, 65, 70, 74, 79

PGM – *Plane-Grating Monochromator* . 23, 33, 34, 36

PI - *Probability of Improvement* 56

PI - *Probability of mprovement* 56

PSI – *Paul Scherrer Institut*.....16, 20, 90
PSO - *Particle Swarm Optimization*.....50, 51, 78, 91
PTB – *Physikalisch-Technische Bundesanstalt* ..23, 24, 25, 29, 83, 86, 87

R

RCWA – *Rigorous coupled wave analysis* 9
RCWA – *Rigorous Coupled Wave Wnalysis*..... 9
RMSE – *Root Mean Square Error* 11, 75, 76

S

SDD – *Silicon Drift Detector* 23, 24, 25
SEM - *Scanning Electron Microscopy* ..1, 4, 16, 17, 18, 19, 20, 21, 27, 43, 66, 78, 79, 83, 86
SIMs - *Secondary Ion Mass Spectroscopy* 1, 83
STEM - *Scanning Transmission Electron Microscopy* 1, 83
STM - S*canning Tunneling Microscope* 21

T

TEM - *Transmission Electron Microscopy* .1, 4, 16, 17, 18, 27, 42, 44, 45, 46, 47, 48, 49, 55, 57, 58, 66, 67, 78, 79, 86
TMM – *Transfer Matrix Method* 8, 9, 26, 35
TNC - *Truncated Newton Constrained* 33, 35, 91

U

UHV - *Ultrahigh vacuum* 23

X

XRF - *X-ray Fluorescence* 2, 23, 24, 78
XRF - *X-Ray Fluorescence* 2, 23, 24, 78
XRR – *X-ray Reflectivity*2, 4, 23, 24, 26, 27, 31, 34, 39, 43, 68, 78, 81
XRR – *X-Ray Reflectivity* 2, 4, 23, 24, 26, 27, 31, 32, 34, 39, 43, 68, 78, 81
XSW – *X-ray Standing Wave* 2, 7, 9, 10, 39, 40, 41, 54, 64, 70, 78
XSW – *X-Ray Standing Wave* 2, 7, 9, 10, 39, 40, 41, 54, 64, 70, 78

1 Introduction and Motivation

Nowadays, nanostructures can not only be found in nature but also in everyday objects like any display or electronic microchip. The word “Nano” is a unit prefix and means one billionth (10^{-9}) and originated from the Greek word for dwarf. A nanostructure is a “composition of inter-related constituent parts in which one or more of those parts is a nanoscale region”, where nanoscale is a “length range approximately from 1 nm to 100 nm” (ISO/TS 80004-1:2015(en)). The properties of matter at the nanoscale differ from bulk materials, single atoms, or molecules [1]. This difference can occur in the magnetic or electric properties, heat conductivity, chemical reactivity, or reflectivity. The color of nanostructures can depend on the size of the structure without changing the material. Materials in the nanoscale regime have been around quite a long time. For example, gold and silver nanoparticles have been used in the stained-glass windows in old medieval churches, even without the ancient craftsmen knowing about their existence.

Semiconductor devices with their complex 2D and 3D nanostructures with feature sizes in the single-digit nanometer regime [2], [3] are today’s most prevalent application of nanotechnology. These structures are a strong driver for research that is continually pushing Moore’s Law [4] to new limits. The performance of these nanostructures crucially depends on well-controlled fabrication process, both in terms of targeted dimensional parameters and 3D element compositions (e.g., dopant distributions). Thus, there is a strong need for metrology techniques which enable characterization of these parameters with sufficient sensitivity [5]. Metrology concerns the development of measurement techniques of reference values and all sorts of parameters prioritizing the highest possible accuracy and precision. Measurement techniques can in many cases not be extended into the nano-regime without better understanding of the physical phenomena at the nanoscale. Therefore, it is necessary to introduce nanometrology as a sub-section of metrology [6]. Equipment and standards must be developed to allow reproducible production and support the increasing applications of nanotechnology. Additionally, at the nanoscale the different research fields from biology, chemistry, and material science meet. For research oriented nanometrology, the highest resolution possible should be achieved and as many parameters as possible need to be measured. The requirements for industrial nanometrology are quite different [7, p. 111]. For industrial nanometrology the measurement process needs to be effective with minimum time and cost. The measurement needs to produce reliable results even under the rough conditions that can be found in the production process. In general, it is not only important to measure and characterize nanostructures, but also to determine the uncertainties of the measured parameters

Typical analytical or dimensional techniques used in nanometrology are scanning and transmission electron microscopy (SEM, TEM) [8], [9], atomic force microscopy (AFM) [10], as well as techniques that also address elemental distributions such as secondary ion mass spectroscopy (SIMS) [11], atom probe tomography (APT) [12], and energy-dispersive X-ray spectroscopy (EDX) combined with scanning transmission electron microscopy (STEM) [13]. All these techniques have different advantages and disadvantages regarding sample preparation and consumption, achievable spatial resolution, required duration, and other

experimental parameters (e.g., tip sizes). Optical metrology based on light-matter interaction has a significant advantage in terms of measurements and usually does not require destructive sample preparation (like cutting lamella for TEM). There are two basic approaches for optical inspection: imaging, e.g., microscopy, which has the advantage of directly providing a 2D image of the specimen, and non-imaging, respective scattering measurements. The latter are not principally limited in resolution by the wavelength of the radiation used, as is the case for imaging microscopy. Instead, as an indirect measurement technique, they require solving an inverse problem. Optical scatterometry in semiconductor metrology is also known as optical critical dimension (OCD) metrology [14]. It is still used as a work horse and continuously improved, despite sensitivity limits that are hard to overcome. To keep pace with shrinking structures, intensive research is being carried out to reduce the wavelength (e.g., utilize X-rays) of the employed radiation. In X-ray scattering techniques, a distinction is made between measurements in transmission, also known as critical dimension small angle X-ray scattering (CDSAXS) [15], and reflection mode known as grazing incidence small angle X-ray scattering (GISAXS) [16], [17]. Both techniques have already shown that they allow for the dimensional reconstruction of nanostructures with an uncertainty in the sub-nm range [18], [19] and can statistically average over large areas, but they require either sample thinning (to achieve reasonable transmittance for CDSAXS) or large targets, which are required for the elongated footprints of the photon beam for GISAXS [20]. Usually, these techniques employ high-energy X-rays, which limits the optical contrast between the different materials within the investigated nanostructures. New approaches are using soft X-ray, deep ultraviolet (DUV) or extreme ultraviolet (EUV) radiation as a promising approach to limit the footprint and increase material sensitivity. In addition, studying the X-ray fluorescence (XRF) of a sample can provide analytical information about the sample.

The experimental basis of the method developed in this work is angle-dependent XRF measurements; a sample is illuminated by X-rays, which causes XRF photons to be generated by the atoms of the sample [21]. The energy of the fluorescence X-rays is characteristic of the fluorescing atom (e.g., the chemical element). The method to characterize nanostructures presented in this thesis will utilize grazing incidence X-ray fluorescence analysis (GIXRF) with incident soft X-rays (<1 keV). GIXRF takes advantage of the X-ray standing wave (XSW) field originating from the interference of the incoming and reflected beam [22] under sufficiently grazing incidence. GIXRF measurements were used for non-destructive analysis of atoms near the surface in [23]. Synchrotron radiation was employed to excite the fluorescence radiation, and the measurement results were successfully modeled with corresponding calculations. Additionally, GIXRF was used to locate impurity atoms below [24] and above a reflective surface [25]. The spatial position of the impurity atoms is encoded in the fluorescence measured for different angles of incidence. This can be explained with the XSW field effect, which can function as a nanoscale sensor. The XSW field inside the structure changes with different angle of incidence and the intensity modulation inside the XSW field significantly influences the X-ray fluorescence intensity of an element depending on its spatial position within the electromagnetic field distribution. In [26], thin layered samples were investigated with X-ray reflectivity (XRR) and angle dependent XRF, showing how these two methods work together and give complementary information about the sample. In [27], GIXRF measurements were applied to nanostruc-

tures (gratings with different thicknesses), showing that 2D structures can be analyzed with GIXRF measurements. In a later measurement, where not only the angle of incidence was changed, but additionally the 2D grating was rotated with respect to the incoming beam from parallel to perpendicular to the grating lines [28],they observed modulations that cannot be described with the calculation of a 1D XSW field. In [29], GIXRF measurements were used to reconstruct 2D nano-gratings. They used the adapted Sherman equation [30], [31] for GIXRF measurements and extended it to 2D systems, and showed that Parratt's [22] formalism to calculate the electric field is not capable of calculating the XSW field for 2D or even 3D structures. The shape of the nanostructure has an influence on the generated XSW field and therefore on the measured XRF. They used a finite element method (FEM) Maxwell solver for the calculation of the XSW field.

Similar FEM calculations were used in the computational part of the thesis. FEM is a general method to approximate the solutions of partial differential equations [32]. In [33], a fast time-harmonic FEM solver was presented and used for the calculation of photolithography masks. The FEM has the advantage that it can be applied for the reconstruction of nanostructures of almost any shape. For periodic nanostructures like line-gratings the computational domain can be reduced to a 2D problem, which reduces the computational demand significantly [34]. This enables successful FEM calculations for wavelengths that are significantly smaller than the computational domain [18]. The application of the FEM solver gives the possibility to reconstruct complex nanostructures from GIXRF measurements. It is important to use a reasonable parameterization of the nanostructures for an adequate performance of the reconstruction. Therefore, different parameterizations are investigated in this work. If the measurement process can be described mathematically, it is possible to predict the results of a measurement. This is called the forward problem. The inverse problem is inferring the parameters that describe the system from the result of a measurement [35]. Deriving the shape of a nanostructure from GIXRF measurements with the help of the FEM is an inverse problem. A solution of this problem can be found by applying an optimization method to the forward problem. Generally, the objective function has a complex shape with various local minima or maxima, which can throw off gradient based optimizers. Therefore, global optimization methods are needed. There is also a trade-off between using a realistic model capable of closely describing the real system and the increasing complexity of the virtual experiment resulting in a non-realistic computational effort. An optimization method well suited to this task is the Bayesian optimization (BO) [36]. The key of the BO is that it searches for the desired minimum (or maximum) without the need to derivate the objective function and a minimum of function evaluations [37]. The BO has been successfully combined with a FEM in [38]. To gain access to the uncertainties of the optimization results a huge amount of function evaluations can be necessary. Therefore, depending on the type of inverse problem it can even be feasible to generate a surrogate model from a finite number of computational expensive simulations [39]. A working and trustworthy surrogate can approximate the results of the simulation in a certain domain with significantly less computational time than the simulation itself.

The aim of this thesis is to develop a nanometrology method to characterize nanostructures. GIXRF measurements and a FEM Maxwell solver are utilized to retrieve the shape parameters. For an exact recon-

struction, reabsorption needs to be considered by applying the Beer-Lambert law for the generated fluorescence photon deep inside the structures. Using tabulated data obtained from measurements of bulk material or calculations are often not realistic for nanolayers and nanostructures [40]. Therefore, XRR measurements are part of this work to determine the optical constants. Different optimization methods are applied to the reconstruction process. To validate the results alternative nanometrology techniques like AFM, SEM and TEM cross sections are used. To reduce the optimization time and investigated the uncertainties a surrogate model is created.

2 Theoretical Foundations

An inverse problem uses actual measurement outputs to infer the values of parameters characterizing the system [35]. In contrast, in a forward problem, the results of some measurements can be predicted if the physical system is fully described. For example, if we have a nanostructure containing different layers and we know the optical constants (for the incident energy), the thickness, and the roughness of each layer, we can calculate the reflectivity for a given incident beam. This would be a forward problem or forward calculation. In most cases, it is the other way around: we have some measured reflectivity for a given incident beam and we want to infer the optical constants, the thickness, and the roughness of the structure. This is the first inverse problem discussed in this thesis. The second problem is the determination of the shape of a nanostructure that is the cause of the measured fluorescence.

In the following, I will give an overview of the interaction of X-rays with matter (Section 2.1) to give the physical background for the reflectivity and fluorescence. This is important to understand the forward calculations that will be described in Section 2.2. To infer solutions for the inverse problems, different optimization methods are used. In Section 2.3, the background for the optimization process is given and the optimizers used in this work are introduced.

2.1 Interaction of X-rays with matter

X-rays are electromagnetic waves with a wavelength even shorter than that of ultraviolet radiation, which were discovered by Wilhelm Conrad Röntgen in 1895 [41] . There are two main mechanisms for generating X-rays. The first is the radioactive decay of some isotopes. The other method is based on bremsstrahlung: the interaction of accelerated electrons with matter (X-ray tubes) or magnetic fields (accelerator or storage rings). The measurements used in this work were conducted at the storage ring BESSY II, which provided an incident beam with high brilliance, high polarization, and tunability of the wavelength.

Even before the discovery of X-rays, Pierre Bouguer described the attenuation of radiation when traveling through matter [42]. The resulting intensity of the radiation can be described with the Beer-Lambert law [43]:

$$I(E) = I_0(E) e^{\frac{-\mu_j(E)\rho d}{\sin\theta}} \tag{1}$$

Here, $I(E)$ is the transmitted part of the incident photons $I_0(E)$ depending on the energy E and the material j through a layer with a thickness d. Here, θ is the angle between the incident beam and the surface and ρ is the density of the layer. The mass attenuation coefficient or mass absorption coefficient $\mu_j(E)$ depends on the energy and the material. There are multiple ways for a photon to interact with an electron shell of the material it is passing through. The photon can be completely absorbed, scattered with no loss of kinetic energy (elastic scattering), or exchange some energy with the material (inelastic scattering).

In 1905, Albert Einstein described the so-called photoelectric effect [44]. To ionize an atom with a photon, only the wavelength of the photon is important, not the intensity of the light. The ionized atom is then in an excited state and can relax in two different ways: by photoemission (fluorescence) or the Auger effect.

Fluorescence means that an electron from an orbital with a higher energy relaxes into the one with a lower energy, from where an electron was ejected during the ionization process. The wavelength of the emitted photon is characteristic for the material and the main transitions are selected by the dipole selection rules. In the Auger process, the energy of the relaxing atom is not emitted as a photon but transferred to an electron with a lower binding energy, which is subsequently ejected from the electron shell. The resulting energy of this secondary electron is characteristic for the element. The Auger process dominates over fluorescence for atoms with low atomic number Z.

Scattering is the deflection of photons by a material. If the photon changes direction without losing energy after interacting with a tightly bound inner electron of an atom, this is called elastic or Rayleigh scattering. On the other hand, if the photon collides with a free electron or an outer shell electron that is not tightly bound, the photon may change direction and lose energy. This is called inelastic scattering or Compton scattering [21, Ch. 1.3.2]. Generally, the direction of the deflected photons is not pre-determined. Rayleigh scattering can have a fixed relation between the phases of the incident and scattered photons (coherent). Compton scattering is always incoherent. The total amount of energy and momentum needs to be conserved during inelastic scattering. The energy the photon keeps depends on the deflection angle and is independent of the material. The total attenuation of light passing through a material is the combination of the absorption and scattering processes.

In a homogeneous medium, X-rays can be described with plane waves. On the surface of the medium the X-rays can be partly reflected and refracted. They follow the law of reflection and Snell's law [21, Ch. 1.4]. This interaction can be described by the atomic scattering factor, if the individual atoms scatter independently. This is a good assumption for photon energies above 50 eV and outside the absorption edges [45]. The atomic scattering factor can be related to the optical constants of condensed matter by

$$n = 1 - \delta - i\beta = 1 - \frac{r_0}{2\pi}\lambda^2 \sum_q n_q f_q(0) \tag{2}$$

with r_0 the classical electron radius, λ the wavelength, n_q the number of atoms of type q, and $f_q(0) = f_{1q}(0) + i f_{2q}(0)$ the tabulated complex forward atomic scattering factor for atom q [45]. The attenuation of the material $\mu_j(\lambda)$ can be related to β with

$$\mu_j(\lambda)\rho = \frac{4\pi\beta}{\lambda} = 2r_0\lambda \sum_q n_q f_{2q}(0)\,. \tag{3}$$

For X-rays, most materials have a refractive index that is smaller than one (lower optical density than vacuum), therefore the refracted beam is deflected to the surface of the material. Refraction is only possible for angles above a critical angle α_{crit}. For incidence angles below the critical angle, the beam is reflected. This is called total reflection. The value for the critical angle can be retrieved from Snell's law for a surface in vacuum ($n_1 = 1$)

$$\cos\alpha_{crit} = \frac{n_2}{n_1}. \tag{4}$$

In a typical X-ray reflectometry profile, the critical angle can be approximated with

$$\alpha_{crit} \approx \sqrt{2\delta}. \tag{5}$$

The ratio between the intensity of the incident and reflected beam defines the reflectivity. This can be calculated with the Fresnel equations. They can be derived from Maxwells equation and depend on the incident angle as well as the refractive indices of the materials. For thin homogenous layers on a thick and flat substrate, the reflectivity of this system additionally depends on each layer's thickness. For grazing angle of incidence, interference effects can be observed for samples containing one or more thin layers. Reflection of X-rays on upper and lower boundaries of thin layers causes phase differences that determine constructive or destructive interference [21, Ch. 2.1]. Additionally, maxima and minima can be observed for the reflectivity depending on the incidence angle [46]. The period between the maxima (or minima) is inversely proportional to the layer thickness.

A total wave field is formed when the incident and reflected beams are superimposed. The resulting wave pattern may change with a certain velocity. It can also have a fixed pattern of nodes and anti-nodes, which is called a standing wave [21, Ch. 2.2].

Inside of the substrate, the intensity decreases exponentially; the penetration depth depends on the angle of incidence. In a layer system the standing wave field occurs in the layers. In 1955, Jacob Sherman published an equation correlating the fluorescence intensity with the composition of a material [31]. This equation can be adapted to calculate the fluorescence intensity $\Phi(\theta, E)$ of a specific fluorescent element for a given (X-ray) standing wave (XSW) field $I_{XSW}(\theta, E, t)$ with the incident energy E and incident angle θ [30]

$$\Phi(\theta, E) \propto \int_0^t P_{mat}(t', E) I_{XSW}(\theta, E, t') \exp(-\xi_{abs}(\theta, t'))\ dt' . \tag{6}$$

For the integration over the material profile depending on the depth t (from the surface) an absorption correction for the generated fluorescence photons is applied. The absorption correction $\xi_{abs}(\theta, t) = \sum_{mat} \rho(t, mat) \mu_{E_f}(t, mat) y_{dis}(\theta, t, mat)$ depends on the density ρ, mass attenuation coefficient μ_{E_f} for the fluorescence photon energy E_f, and the distance y_{dis} that the fluorescence photon travels through the different materials mat on its way to the detector. The parameters $P_{mat}(t, E) = W_i(t) \rho(t) \tau(E) \omega_k$ (mass fraction of the fluorescent element in the material W_i, photo-ionization cross section for the incident photon energy $\tau(E)$, and the fluorescence yield ω_k) depend on the distribution of the different materials in the layers system. The dependence of the field intensity on the angle of incidence θ_i (and therefore the dependency of the fluorescence signal on the angle of incidence) can be used for depth profiling of layered structures.

Equation 6 can now be generalized to 2D and 3D structures [29], [47] by integrating over the volume V of the sample for each fluorescent element

$$\Phi(\theta, E) \propto \iiint P_{mat}(V', E) I_{XSW}(\theta, E, V') \exp\big(-\xi_{abs}(\theta, V')\big)\, \mathrm{dV}' . \tag{7}$$

If we can calculate the $I_{XSW}(\theta, E, V)$ distribution within a structure for a known $P_{mat}(V, E)$, we can retrieve the expected fluorescence intensity $\Phi(\theta, E)$ for each fluorescent element of the sample. In the next section the calculation of $I_{XSW}(\theta, E, V)$ with a FEM Maxwell solver is introduced.

2.2 Forward calculations of the reflectivity and the fluorescence

The method developed in this thesis involves the forward computation of reflectivity and fluorescence. This is an essential part of the inverse problem of finding the nanostructure that produced the corresponding measured reflectivity or fluorescence intensity. In the following, I will show how to calculate the reflectivity for of a layered system (Section 2.2.1) or the fluorescence of a random structure (Section 2.2.2).

2.2.1 Calculation of the reflectivity with the TMM

If the complex refractive index n of a material is known, the expected reflectivity from the surface can be calculated as a function of the incidence angle θ. This is the forward calculation. To find the refractive index that can describe the measurements, we need to solve the inverse problem. In this work the Transfer-Matrix Method (TMM)method is adapted. The TMM provides a method to calculate the specular reflectivity and transmittance for a multi-layer system depending on the layer thickness and the roughness by using matrix multiplication [48, Ch. 3.2.1].

Assuming a Gaussian distribution of roughness and interdiffusion, the modified Fresnel coefficients $\tilde{r}_j$ and $\tilde{t}_j$ for each interface j can then be approximated [49] as

$$\begin{aligned} \tilde{r}_j &= r_j \exp\left(-2k_{z,j}\, k_{z,(j+1)} \tilde{\sigma}_j^2\right) \\ \tilde{t}_j &= t_j \exp\left(\frac{\left(k_{z,j} - k_{z,(j+1)}\right)^2 \tilde{\sigma}_j^2}{2}\right). \end{aligned} \tag{8}$$

The parameter $\tilde{\sigma}_j$ represents the mean square intermixing at the jth interface. The change of the wave vector component k_z at the j interface can be written as

$$k_{z,j} = \sqrt{\left(n_j k_0\right)^2 - \sin^2(\theta)\, k_0^2} \tag{9}$$

with k_0 as the incident wave vector.

This can easily be implemented in an algorithm to calculate the reflectivity of multi-layer system [48, Ch. 3.2.1], which can be solved by computers with a small computational effort[1]. The model parameters for this problem are the refractive indices, the thickness, the polarization, and the roughness for of the different layers. It can be feasible to include measurements of the reflectivity for different energies. This leads to an increase of the number of model parameters because the refractive indices are energy de-

[1] For the implementation used here, the calculation of the reflectivity of 3 layers on a substrate for one energy and 163 angles takes $199\mu s \pm 66\mu s$ on a standard computer with an Intel® Core™ i7-2600 CPU @ 3.4 GHz.

pendent. Additionally, the number of layers can be increased due to interdiffusion or contamination. Nevertheless, not every parameter needs to be optimized, for some parameters (e.g., if the refractive index is well known for a material) a tabulated value can be sufficient. The small computation time of a forward calculation allows to solve this inverse problem with many model parameters and using optimization methods, that need a lot of function evaluations (forward calculations).

2.2.2 Simulation of E-Field and calculation of the fluorescence

For the second inverse problem, it is necessary to calculate the fluorescence depending on different angles of incidence for a nanostructure. For a layered system, algorithms based on [22] or TMM from [48, Ch. 3.2.1] can be adapted to calculate the intensity of the XSW field. With the Sherman equation [31] the angular dependence of the fluorescence intensity can be calculated from the XSW field. For 2D (and 3D) nanostructures different non-analytical approaches are available. One possible approximation is the many-beam dynamical diffraction theory (MBDDT) also known as rigorous coupled wave analysis (RCWA) [50]–[52]. In [53], MBDDT was used to simulate the near and far field in the EUV regime. MBDDT approximates structures using box arrays, but corrections are needed for structures with side wall angles and surface oxidation. It's typically only used for s-polarized X-rays, and corner rounding or complex structures require a staircase approximation [54], increasing the computational effort. This work uses a more flexible tool to minimize approximations and decrease calculation errors. A FEM based Maxwell solver can calculate the complex electromagnetic field distribution for nearly any kind of periodic nanostructure [33]. Here the structure is divided into small triangles, which reduces the error from round features if the triangles are small enough. Additionally, the polarization of the incident beam can be chosen freely between s- and p- polarization. The more exact way to model the nanograting and the possibility to choose the right polarization led to the decision to use an FEM based Maxwell solver from JCMwave [55] even if this increases the computational effort.

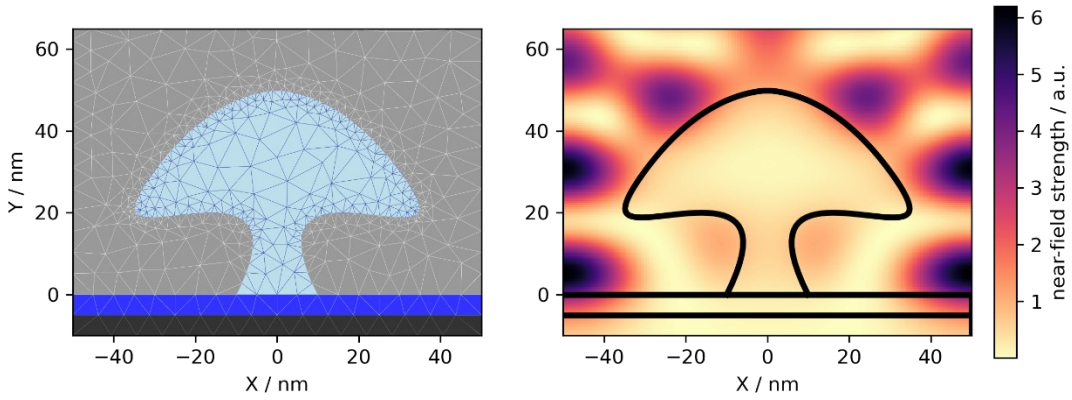


Figure 2.1 Left side show the FEM mesh for a nanostructure with periodic boundary condition in horizontal direction. The right picture shows the intensity of the XSW field for a given incident energy (680 eV) and angle (2.3°) of the same structure.

Figure 2.1 shows a cross section of a periodic nanostructure composed of finite elements. On the right side of the figure the corresponding electric field distribution is shown, which allows us to calculate the fluorescence intensity of any periodic nanostructure. The disadvantage of this method is, that it is time consuming to calculate the intensity of the XSW field depending on the size of the nanostructure and the wavelength of the incident beam.

To calculate the fluorescence $\Phi_{mat}(\theta, E)$ of a fluorescent material for a given incident energy E and incident angle θ_i equation 7 can be adapted e.g., for a 2D grating. The compositional and fundamental parameters of the respective material (mass fraction of the fluorescent element in the material W_i, photoionization cross section for the incident photon energy $\tau(E)$ and the fluorescence yield ω_k) are obtained from databases [56] or dedicated experiments [57]. The intensity of the XSW field $I_{XSW}(x,y) = \left|\tilde{E}(x,y)\right|^2$ is calculated from the electric field strength $\tilde{E}(x,y)$. The integration of the intensity is performed separately for the area A of each material, so that $P_{mat}(E) = W_i \rho \tau(E) \omega_k$ is similar at each position of the integration. The generated fluorescence photons can be reabsorbed on their way through the sample to the detector with a probability depending on $\xi_{abs}(x,y) = \sum_{mat} \rho(x,y,mat) \mu_{E_f}(x,y,mat) y_{dis}(x,y,mat)$. Here all materials mat between the origin of the fluorescence photon and the detector must be considered. The reabsorption correction $\xi_{abs}(x,y)$ depends on the mass attenuation coefficient μ_{E_f} for the fluorescence photon energy and the density ρ of the different materials. The distance $y_{dis}(x,y)$ is the total distance the photon travels through the material. The fluorescence $\Phi_{mat}(\theta, E)$ can be calculated with

$$\Phi_{mat}(\theta, E) = \frac{P_{mat}(E)}{\sum \mathrm{d}x} \cdot \sum_{x \epsilon A} \sum_{y \epsilon A} \left|\tilde{E}(x,y)\right|^2 \cdot \exp\left(-\xi_{abs}(x,y)\right) \mathrm{d}x \mathrm{d}y \,. \tag{10}$$

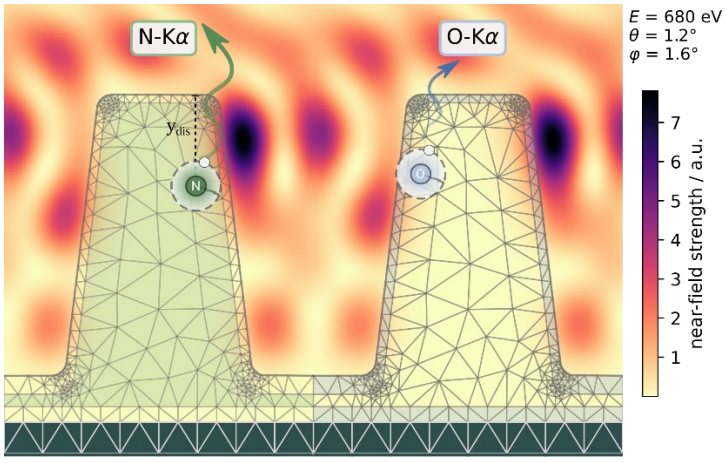


Figure 2.2 The calculated electric field strength inside and outside the structure is shown for $\theta = 1.2°$ and $\varphi = 1.6°$. For the nitrogen and oxygen fluorescence, the electric field strength is integrated, and the reabsorption is calculated with the distance y_{dis} the photon must travel to leave the structure. Taken and adapted from [58].

Figure 2.2 shows the finite elements and the calculated electric field strength for a 2D nanograting with a material containing nitrogen (green area) and a material containing oxygen (blue area) on a substrate

(dark green area). Oxygen and nitrogen fluorescence is generated with the incident energy. The software package used here from JCMwave [55] includes a FEM Maxwell solver and the ability to integrate the electric field strength. In this work, the reabsorption correction was implemented directly in the integration. For the reabsorption, the distance y_{dis} (see Figure 2.2) that the photons travel through the different materials is calculated for each point of the integration. The implementation of reabsorption led to a reduction in computational time and error because the electric field distribution does not need to be exported to a regular Cartesian grid to do the integration and reabsorption correction by re-importing the grid during an optimization loop.

Uncertainty quantification and global sensitivity analysis are only feasible with surrogate models when working with computationally expensive models or forward calculations [39]. In this thesis the polynomial chaos expansion (PCE) is applied as a surrogate to replace the calculation of the fluorescence intensity. One of the advantages of PCE is the possibility to analyze the global sensitivity for all model parameter combinations by using variance-based sensitivity analysis (or Sobol indices) [59]–[61]. The Sobol indices can help to decide which model parameters to include in the optimization and can be obtained analytically from the PCE coefficients [61]. In 1938 Norbert Wiener showed that any function has a representation as an infinite sum of Hermite polynomials, if it has a second moment with respect to a Gaussian measure [62]. This infinite sum must be approached with a finite subset of polynomials. There are some strategies for this approximation, the choice of strategy depends on the problem we want to model.

There are different ways to analyze the surrogate and test if it is a good approximation for the forward calculation. To calculate the error, the surrogate is not trained on the full dataset. Normally 10% of the data set is used as a test dataset (y_{test}). Here n_{test} is the amount of data points in the test data set. The surrogate is evaluated on the test data set and the result (pc_{test}) is compared with the expected values. E.g., the root-mean-square error (RMSE) ($error_{RMSE}$) can be used to evaluate the error of the surrogate:

$$error_{RMSE} = \sqrt{\frac{1}{n_{test}} \sum_{i=1}^{n_{test}} \left(y_{test,i} - pc_{test,i}\right)^2} \tag{11}$$

From the equations above the relative error $error_{RMSE\ relative}$ can be calculated.

$$error_{RMSE\ relative} = \sqrt{\frac{\sum_{i=1}^{n_{test}} \left(y_{test,i} - pc_{test,i}\right)^2}{\sum_{i=1}^{n_{test}} \left(y_{test,i}\right)^2}} \tag{12}$$

Additionally, for PCE the condition number of the Gram matrix can be investigated. This matrix contains the inner product of all polynomial basis functions of the PCE [63, Ch. 7.2]. The Gram matrix should be regular, close to identity [64] and well-conditioned to indicate a good approximation of the forward model by the surrogate. The matrix depends on the total number of forward calculations used to train the PCE and some weights.

2.3 Using optimization to find solutions of the inverse problems

The aim to reach optimality is a fundamental principle in many laws of physics [65]. Mathematical optimization is the concept of finding the best possible solution (of the objective function) by changing parameters, typically within given constraints. Often the objective function is optimized (minimized or maximized) during the optimization process. During optimization, forward calculations for different model parameters are computed and compared with the experimental data to solve the corresponding inverse problem. The next step is finding the model parameters or parameterization, which are quantities that can be varied and will influence the objective function. Additionally, it can be reasonable to apply some constraints on the model parameters. Constraints can influence the performance of the optimization process and help avoid unphysical results. The different local and heuristic global optimizers used in this thesis are described in Appendix A.

To find the most probable solutions of an inverse problem the likelihood function L can be used under the assumption that the error σ can be modeled through a Gaussian probability density function [66, Ch. 2.3]:

$$L\left(I_{exp}, I_{model}, \sigma\right) = \prod_{i}^{N} \frac{1}{\sigma_i \sqrt{2\pi}} \exp\left(-\frac{\left(I_{exp,i} - I_{model,i}\right)^2}{2\sigma_i^2}\right). \tag{13}$$

Here, I_{exp} are the results of the experiment and I_{model} are the results of the corresponding forward calculation with N the number of different entries in I_{exp}, I_{model}, or σ. If the likelihood L is close to a delta function (small variance), L can get numerical instable. For numerical stability, it is often better to use the log-likelihood $\mathcal{L}$ [67, Ch. 2.1]

$$\mathcal{L}\left(I_{exp}, I_{model}, \sigma\right) = \log\left(L\left(I_{exp}, I_{model}, \sigma\right)\right) = -\frac{1}{2}\sum_{i}^{N}\left[\log\left(2\pi\sigma_i^2\right) + \frac{\left(I_{exp,i} - I_{model,i}\right)^2}{\sigma_i^2}\right]. \tag{14}$$

Since logarithms are strictly increasing functions, maximizing the likelihood is the same as maximizing the log-likelihood.

For most of the cases σ is the uncertainties estimated from the experiment and doesn't change during the optimization. Therefore, the first part of the sum in equation 14 is constant and it suffices to minimize:

$$\chi^2 = \sum_{i}^{N} \frac{\left(I_{exp,i} - I_{model,i}\right)^2}{\sigma_i^2}. \tag{15}$$

The error σ behaves like a weight to the difference between the experimental data and the forward calculation. It has an impact on the behavior and the result of the optimization. The error can be a combination of the measurement error σ_{exp} and the error of the forward calculation (including the model error resulting from the imperfect approximation of the real world) or simulation σ_{sim}. The measurement error follows from the experimental set-up and the devices used during the measurement. Typically, the forward calculation cannot be done analytically, and numerical methods must be used. This results in an error of the simulation, which is captured by σ_{sim}. If the error σ can be approximated with an error model

(EM) depending on additional hyper-parameters, the hyper-parameters can be included in the optimization as well.

To find the optimum of the objective (e.g., χ^2 or $\mathcal{L}$) many different algorithms can be utilized. There are global and local optimization methods [68]. A local optimization routine cannot be expected to find a global extremum as it only tries to find parameters combinations which minimize the objective in a neighborhood around the initial point. A global optimizer on the other hand tries to find the global optimum of the objective. The inverse problem determines the choice of optimization method, which is not straightforward. Some optimization methods require a significant number of forward calculations to find the optimum, which can increase the total optimization time to an impractical level. The behavior of the optimization can be affected by the shape of the objective. For most of the cases, an optimizer which gave good results in an acceptable amount of time was chosen without searching for an optimal method. Testing different optimization methods can be a time-consuming task, depending on the forward calculation and is not the goal of this thesis, because here I was more interest in the result of the optimization process and not optimizing the reconstruction process.

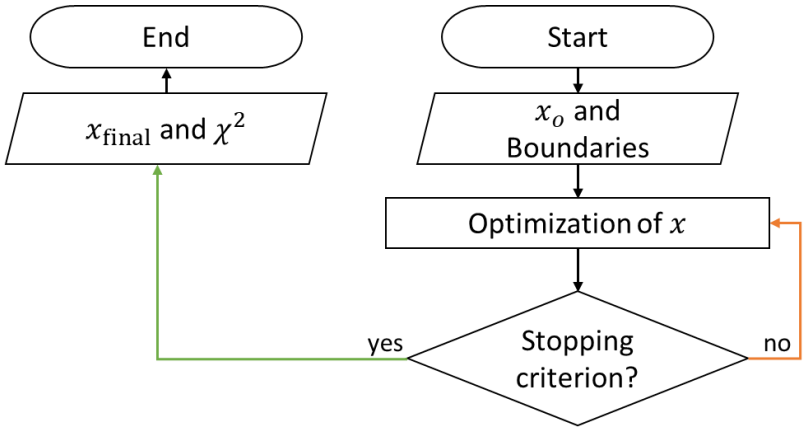


Figure 2.3 Flowchart of optimization setup A, where all parameters are fitted simultaneously with a global optimization method. The starting point x_0 contains the initial values of the optimization. The boundaries define the parameter space in which the model parameters are optimized. Hereχ^2 is minimized. At the end, the optimal value x_{final} and the final χ^2 are returned.

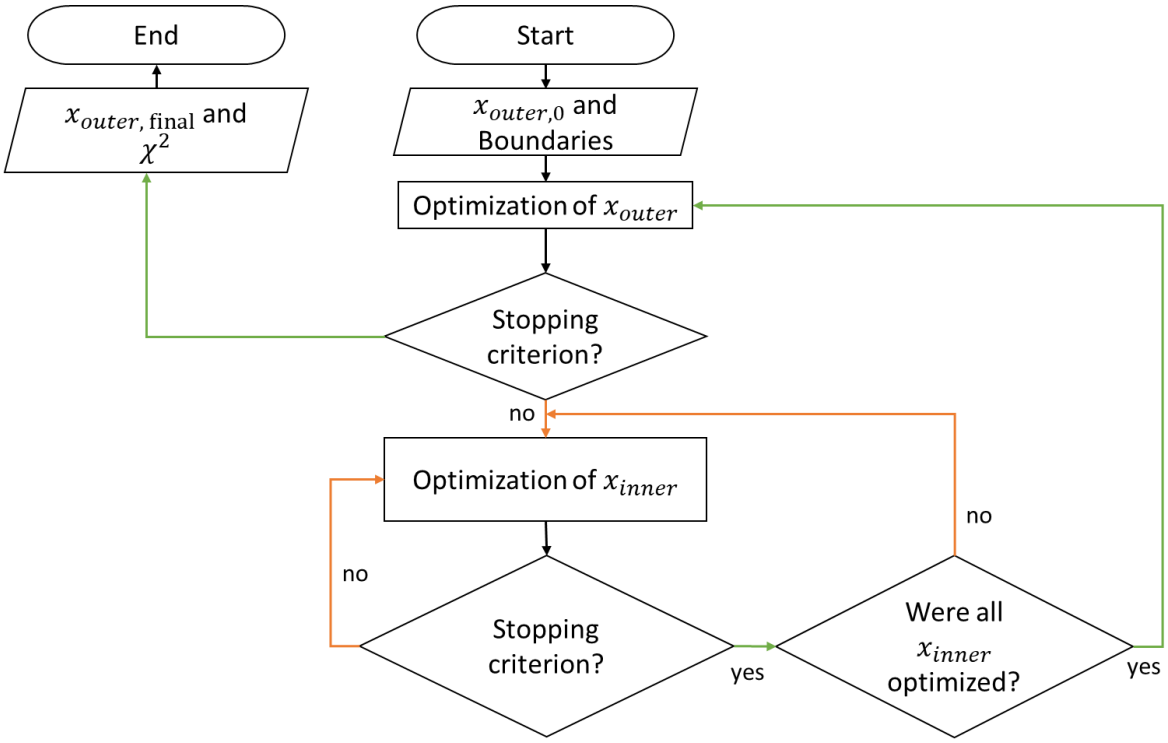


Figure 2.4 Flowchart of optimization setup B, where an outer and inner optimization for the respective model parameters is used. $x_{outer,0}$ contains the values for the starting point of the fit for the outer model parameters. The boundaries define the parameter space in which the model parameters are optimized. Here χ^2_{outer} is minimized. At the end, the optimal value $x_{outer,\mathrm{final}}$ and the final χ^2_{outer} are returned.

In this thesis I used two different optimization set-ups. A sketch of the set-ups can be found in Figure 2.3 for the set-up A and in Figure 2.4 for the setup B. In setup A all model parameters are optimized with a global method. For some methods a starting point x_0 for the reconstruction is given. Random values within the domain boundaries are taken as x_0 if there are no values from other optimizations or databases. The optimization method tests if a stopping criterion is reached and whether it is likely that a smaller value of the χ^2 can be achieved. A stopping criterion can be e.g., a predefined number of iterations or run time. There can be optimization set-ups where it makes sense to split the optimization into an outer and an inner optimization (set-up B). This is called bilevel optimization or bilevel programming [69]. These kinds of problems were first formulated by Heinrich Freiherr von Stackelberg in 1934 [70]. Some of the problems described in this thesis can be formulated as a bilevel optimization problem. E.g., if some of the optimization parameters depend on a quantity independent of other model parameters, such as incident energy. For both optimization schemes, a global optimization method is preferred because we want to find the global optimum of the objective problems. For some special cases, a local optimization method can be used, because local optimization methods can be faster and for convex problems, the global solution is given. For set-up B, the inner optimizer can be local if x_0 is close to the optimum.

After the successful optimization of the objective a probable solution for the inverse problem is found. Some optimization method provides not only a probable solution but the probability distribution of the solutions, from which the information about the confidence intervals of the model parameters $\vec{x}$ can be

retrieved. Another way to find the confidence intervals is to invert the Hessian matrix $\boldsymbol{H}$ of the objective function,

$$(\boldsymbol{H})_{\boldsymbol{kj}} = \frac{\partial^2 \chi^2}{\partial x_k \partial x_j} \tag{16}$$

at the minimum of χ^2 [71]. The Hessian matrix is used by some optimizers to find the optimum[2]. The inverse Hessian matrix close to the minimum of χ^2 is the estimated covariance matrix of standard errors of $\vec{x}$, if the errors are Gaussian distributed. The standard deviation or noise level at the global minimum of χ^2 is defined as $\mathrm{STD} = \sqrt{\frac{\chi^2}{\mathrm{DoF}}}$, where $\mathrm{DoF} = N - M$ is the difference between the length N of I_{exp} and M the length $\vec{x}$ (number of model parameters), and thus the degrees of freedom (DoF). The model parameter confidence interval can then be calculated as

$$\sigma_{\vec{x}} = \mathrm{STD}\sqrt{\mathrm{diag}(\boldsymbol{H})^{-1}}. \tag{17}$$

[2] For optimizers that do not compute the Hessian or the distribution of the solution, the confidence intervals are not estimated.

3 Samples, nanometrology and measurements

To test the method to characterize nanostructures different samples were produced. Additionally, different nanometrology tools are applied to provide a possibility to compare the results of the method to different measurements. In the last part of this section the measurement set-ups and the necessary post processing of the measurements are presented.

3.1 Samples

The samples used to test the method were produced by different sources. The sample should fulfill two requirements. On the one hand, the patterns should be close to real samples, but on the other hand, they should be simple in order not to increase the computational effort, which is unnecessary for a first test of the method. The samples should be regularly spaced over a certain area to be used for grazing measurement methods with a large footprint. Here, 2D line gratings with a pitch of 100 nm or less are used.

The first sample is a lithographically patterned silicon nitride grating on a silicon substrate. It was manufactured by means of electron beam lithography (EBL) at the Helmholtz-Zentrum Berlin (HZB). The nominal pitch of the grating is $p = 100\ \mathrm{nm}$, the nominal line height is $h = 90\ \mathrm{nm}$ and the nominal line width of the sample is $w = 50\ \mathrm{nm}$. For the manufacturing of the gratings, a silicon substrate with a 90 nm thick Si_3N_4 layer was used. ZEP520A, a positive resist (organic polymer), was spin coated on the substrate and structured with a Vistec EBPG5000+ e-beam writer, operated with an electron acceleration voltage of 100 kV. The grating was etched via reactive ion etching using CHF3 and to remove the remaining resist an oxygen plasma treatment was applied. The total structured area of the grating was 1 mm x 15 mm and the sample area outside the patterned region consists of the originally deposited unstructured Si_3N_4 layer. During fabrication, cross-section SEM images have been recorded on a witness sample. The results are presented in the Section 3.2.2. In the following the sample will be called Si_3N_4 sample.

The second sample set was produced at the Paul Scherrer Institute (PSI) with funding received from the NFFA (H2020 project). In this set samples with different shape (e.g., pitch and width) are fabricated. This set contains four different line gratings: one with a pitch of 100 nm (Sample 1), two with a pitch of 50 nm (Sample 2 and Sample 3), and one with a pitch of 60 nm (Sample 4). Witness samples were produced for TEM images. Additionally, some 3D nano pillars and a 1D layer with a length of 25 mm and a width of 1 mm made of the same materials were produced. To generate nanostructures with a small pitch a high-resolution negative photo resist was applied. The resist is made of Hydrogen Silsesquioxane (HSQ) [72], which is a high purity, silsesquioxane-based grade polymer. HSQ can be used to fabricate sub-10 nm half-pitch structures by EBL [73], [74]. In the production process HSQ is spin coated on a Si wafer. With EBL the nanograting's are written. After the removement of the undeveloped resist only the structure remains on the wafer. In the following the sample set will be referred to as HSQ sample set.

3.2 Microscopy characterization

To compare the results of the method developed in this thesis with the results from standard nanometrology tools, the samples were measured with different kinds of microscopes. Some of the samples are characterize by means of Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM) or Transmission Electron Microscopy (TEM). TEM and cross-section SEM measurements are destructive and were conducted on sister samples produced for this kind of measurements. The AFM and SEM measurements were made on the same sample used for the application and testing of the method (Section 4).

3.2.1 TEM

The first TEM was developed by Max Knoll and Ernst Ruska in 1931 [75]. In 1986, Ernst Ruska received the Nobel Prize in Physics. The basic principle of a TEM is the transmittance of a focused electron beam through a sample. The optics of electron microscopes can produce an image of the electron intensity coming from the sample [76]. For TEM measurments, a thin sample is required to allow the electrons to pass through the sample. TEM is based on measuring the transmittance of an electron beam through the sample and therefore requires a thin slice of the same. An imaging and a scanning variation exist; both require the use of electron lenses to guide the electron beam onto the sample and detector. Therefore, the focused ion beam technique (FIB) is ideal for the sample preparation [77] because with FIB thin foils from the sample can be produced.

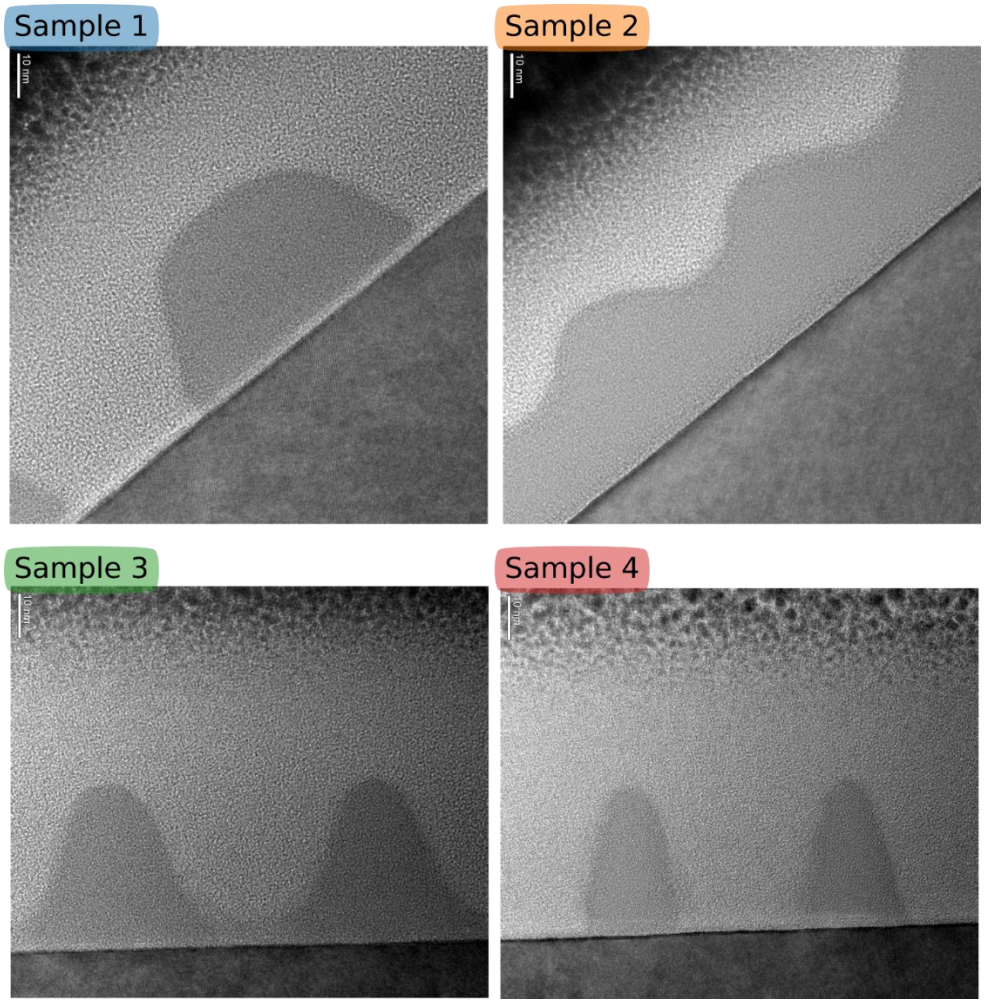


Figure 3.1 TEM images of all sister samples from the HSQ sample set are displayed. The cross section of the different gratings is shown. Images provided by CEA-Leti as part of the NFFA project (ID 685).

In Figure 3.1 the TEM images of 2D nanostructures from the HSQ sample set are shown. The structures were covered with a layer of SiOx to protect them from the FIB preparation. The preparation and the corresponding images were made at CEA-Leti as part of the NFFA project (ID 685). In the lower part the Si wafer can be seen. All samples have a smooth and round cross section. Nevertheless, the shape is different for all samples ranging from a trapezoid with rounded corners (Sample 1), a sinus wave (Sample 2), a gaussian bell (Sample 3) and a sugar loaf (Sample 4). For Sample 2 the grooves of the grating are filled with HSQ wich leads to more material and shallower grating compared to the other samples. The different shapes make this sample set a good test for the method developed in this work.

3.2.2 SEM

In a SEM a focused electron beam is scanned over the sample. The interaction of the electrons with the sample surface generates a variety of signals like secondary electrons, backscattered electrons, and X-ray radiation [78]. The difference between the secondary and backscattered electrons is their origin: the former are weakly bound valence or conduction band electrons that are ejected from the atom by the beam electrons, while the latter electrons are scattered or deflected by the electric field of the atoms in the

sample. Additionally, SEM can be used to investigate the material composition of the sample by analyzing the emitted X-rays, which are characteristic for the material [78]. In 1935, the concept of the SEM was introduced by Max Knoll [79], [80]. Two years later Manfred von Ardenne explained the principles of an SEM and produced the first scanning electron microscope [81].

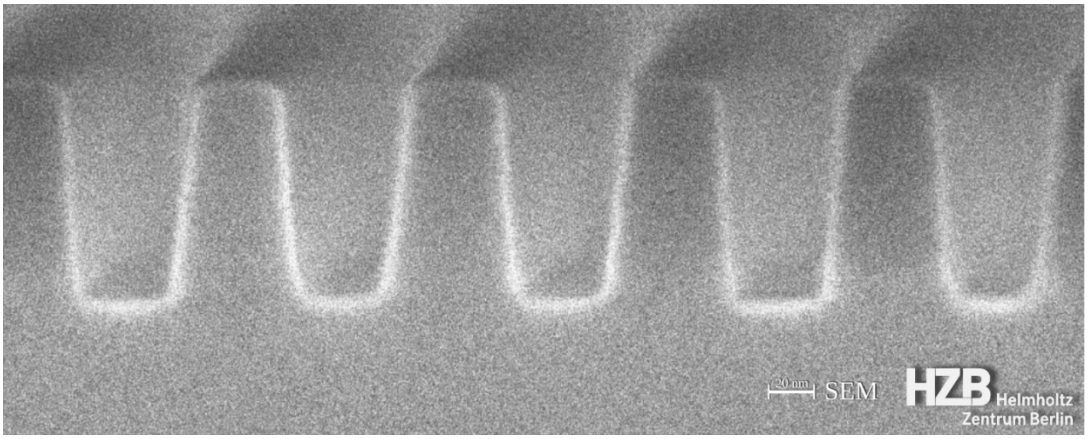


Figure 3.2 Cross section SEM of the Si_3N_4 sample made from a sister sample is shown. Image provided by the HZB.

Figure 3.2 shows a cross section SEM image of a sister sample of the Si_3N_4 sample. The structure of the 2D nanograting can be seen. The shape can be described with a symmetric trapezoid with rounded corners at the top and in the groove. The brighter part at the surface of the grating can results from boundary effects during the SEM measurements [78, Ch. 3.4].

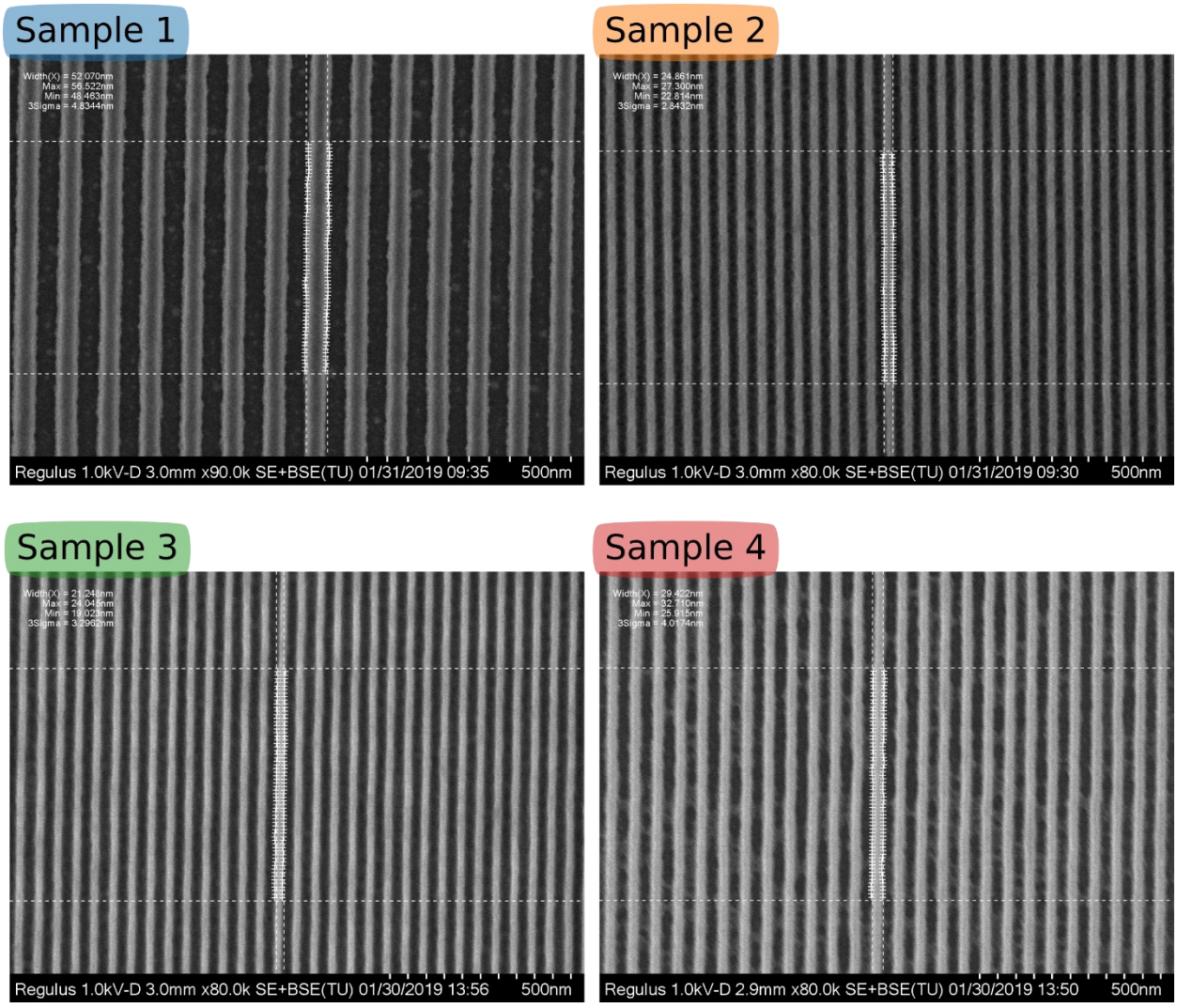


Figure 3.3 SEM image as a top view on the samples from the HSQ sample set. The width of the structure is retrieved using the firmware software. Images provided by PSI as part of the NFFA project (ID 685).

A SEM can be used to image the surface or top view of the sample. In Figure 3.3 the different top views of the samples from the HSQ sample set are shown. The images were made at PSI as part of the NFFA project (ID 685). Additionally, the width of the gratings is estimated with the help of a tool include in the SEM software.

Table 1 Overview over the estimated width w from the SEM image for the different samples including the pitch p, the maximum width w_{max}, the minimum width w_{min}, and the 3σ standard deviation.

	p / nm	w / nm	w_{max} / nm	w_{min} / nm	3σ / nm
Sample 1	100	52	57	48	5
Sample 2	50	25	27	23	3
Sample 3	50	21	24	19	3
Sample 4	60	29	33	26	4

In Table 1 the widths from the SEM images are summarized. The widths of the gratings match the target width of the EBL process. The gratings appear to be rough along the lines indicated by the standard deviation of the measurement and the images themselves. For Sample 1 and Sample 4, there appears to be some material left in the grooves. Nevertheless, from this top view the height of the structure and the real width is difficult to retrieve without destroying the sample.

3.2.3 AFM

The predecessor of the AFM was the scanning tunneling microscope (STM) [82] which is able to image single atoms but needs conductive samples to measure the tunneling current [83]. That led to the invention of the AFM [84], [85], which doesn't require conductive samples.

The AFM measures the force between the sample and the tip, and can operate in static or dynamic mode [86]. In the static mode, the deflection of the cantilever (a flat spring) with a tip at its end is used to measure the force between the sample surface and the tip, which is kept constant while the sample is scanned. In the dynamic mode the cantilever oscillates. Close to the surface the interaction between the tip and the sample changes the resonance frequency. This leads to a change of the amplitude, which is used as the feedback signal to regulate the distance between sample and tip.

The following AFM measurements were performed at the Helmholtz-Zentrum Berlin (HZB) with a Nanosurf Nanite 25 x 25. The sample was measured under tapping mode condition and a standard pyramidal shaped silicon probe with a tip radius <10 nm was used, as it is commonly applied for the inspection of diffraction gratings [87]. The AFM probe is characterized by a resonance frequency of 190 kHz and force constant of 48 Nm. The inspected sample area was 500 x 500 nm^2 in size covering about five grating lines. The Si_3N_4 sample and the HSQ sample set were measured with an AFM.

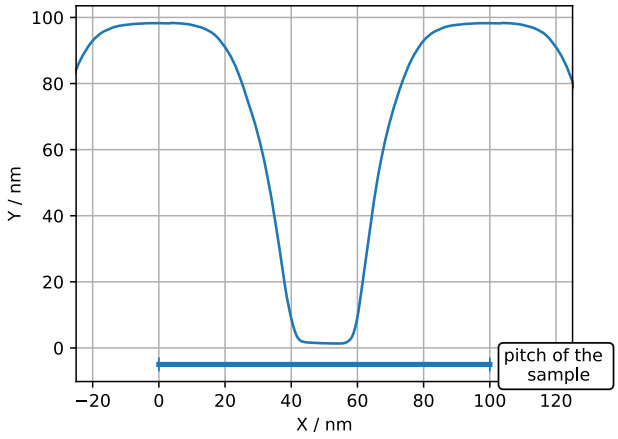


Figure 3.4 The mean of the AFM measurements over the measured area and different gratings line for the Si_3N_4 sample.

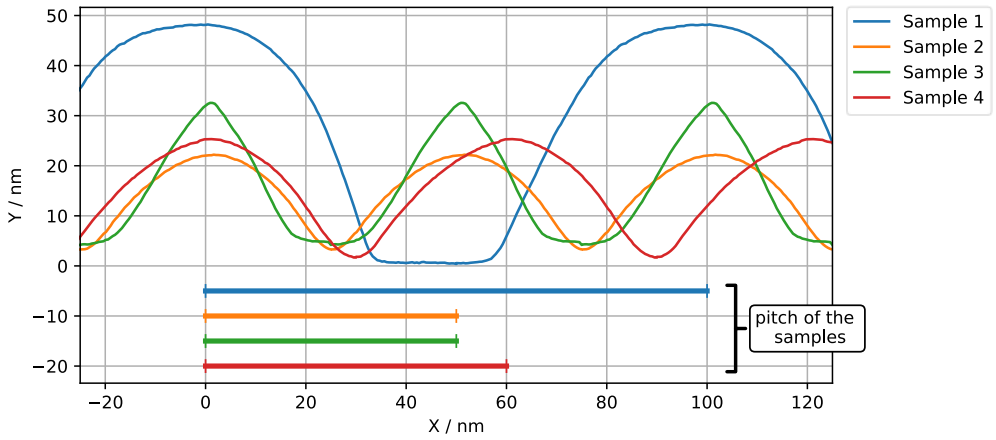


Figure 3.5 The mean of the AFM measurements over the measured area and different gratings line for the HSQ sample set.

In Figure 3.4 and Figure 3.5 the results of the AFM measurements are displayed. Because of the periodicity of the sample the measurements along the lines and over the different periods is taken to generate an average cross section of the samples. From the measurement it is difficult to get the width of the structure, because the shape of the tip has an influence on the results [86, Ch. 8.1]. If the sample has a larger aspect ratio than the tip it is not imaged correctly. The result is a convolution of the tip shape with the high aspect ratio structure of the sample. Nevertheless, the height of lines can be retrieved if the tip has reached the bottom of the groove during the measurement, which can be the case if a flat line in the groove is observed. This seems to be the case for the Si_3N_4 sample, Sample 1, and Sample 3. For the other samples, it is difficult to discuss because the shape of the sample does not include a flat line in the groove because the samples are not fully etched, or the groove is too small.

3.3 Measurements of soft X-ray reflectivity and grazing incidence X-ray fluorescence

To find the shape of an unknown nanostructure, measurements that contain this information need to be conducted. Soft X-ray reflectivity (XRR) and fluorescence (XRF) are measured in the PTB laboratory at BESSYII for the method described in this thesis.

One part of the soft XRR measurements was conducted at the X-ray radiometry (SX700) beamline [88]. The beamline covers the photon energy range from 45 eV to 1800 eV and is designed to produce a beam with low divergence (<1 mrad) with minimal halo. The energy resolution of this beamline is $\frac{E}{\Delta E} = 1000$ [88]. The samples were mounted on a 6-axis goniometer in the ellipso-scatterometer under ultrahigh vacuum (UHV) conditions. A GaAsP photodiode is mounted on a moveable detector arm inside the vacuum chamber to measure the incident and reflected beam. The angle of incidence θ_i is defined as the angle between the X-ray beam and the sample surface and was aligned with respect to the beam with an uncertainty of 0.01°. The angle of incidence can be varied between 0° and 90° in s-polarization direction from the sample normal. The measured range of incident angles was adapted to the different photon energies to effectively cover the relevant angular range for the different samples.

The other soft XRR and GIXRF measurements were performed at the plane-grating monochromator (PGM) beamline [89], [90] for undulator radiation, which covers an energy range from 40 eV to 1860 eV. The sample was mounted in an UHV measurement chamber [91], where a 9-axis manipulator allows for an accurate sample alignment with respect to the direction of incident X-ray beam. For the measurements on the structures not only θ is aligned but additionally the azimuthal angle φ. This angle is defined as the angle between the incident beam and a plane, which is normal to the sample surface and parallel to the direction of the grating lines such that $\varphi = 0°$ is defined as the orientation parallel to the plane of incidence. Both sample rotation axes can be aligned with an uncertainty below 0.01°. As we have employed radiometrically calibrated X-ray fluorescence (XRF) instrumentation, we can perform reference-free GIXRF [92] and gain quantitative access to the elemental mass depositions present on the sample [93]. At each angular position for θ and φ, a fluorescence spectrum is recorded with a calibrated silicon drift detector (SDD) [94] and the incident photon flux is monitored by means of a calibrated photodiode.

3.3.1 X-ray reflectivity experiments

The measurement of the XRR is rather simple: the sample and the detector need to be able to rotate and a monochromatic X-ray source is needed. At a synchrotron it is impractical to rotate the source, therefore, the sample and the detector need to be rotated to measure the reflectivity for different θ. This leads to a θ-2θ geometry, where the detector is moved by 2θ to measure the specular reflectance I_R of the sample, which is rotated with θ. The intensity of the incident beam I_0 is measured by placing the detector in the direct beam without a sample for each incident E. Intensities are determined by measuring the current of the diode. Therefore, the measured currents need to be corrected by the dark current I_{DC} of the multimeter and diode. The dark current depends on the measurement range and is determined before the experiment. The resulting reflectance is calculated with

$$R(\theta, E) = \frac{I_R(\theta, E) - I_{DC}(I_R)}{I_0(E) - I_{DC}(I_0)}. \tag{18}$$

The corresponding relative uncertainty $u_{R,rel}$ of the measurements is calculated with

$$u_{R,rel} = \sqrt{stab^2 + noise^2 + \left(\frac{noise_{Diode}}{I_R}\right)^2}. \tag{19}$$

with $stab = \begin{cases} 1 - \frac{I_{0,before}}{I_{0,after}} & 1 - \frac{I_{0,before}}{I_{0,after}} > 1e-3 \\ 1e-3 & 1 - \frac{I_{0,before}}{I_{0,after}} \leq 1e-3 \end{cases}$, $noise = 1e-3$, and $noise_{Diode} = 0.2\,pA$. The equation was provided by Christian Buchholz (PTB)

To obtain good results it is important that rotation axis for θ is at the surface of the sample. Additionally, each measurement point is measured multiple times to account for systematic errors from the diode or parts of the multimeter. This is important for measuring low currents at the limit of the set-up. Additionally, I_0 can be measured before and after the measurement and multiple times to be able to estimate the drift of this quantity.

3.3.2 X-ray fluorescence experiments

The measurement of the fluorescence intensity involves more steps compared to the XRR measurements. Here as an energy dispersive detector a silicon drift detector (SDD) is utilized. For the angle dependent XRF measurements the detector is kept perpendicular to the incident monochromatic beam with an energy E and the sample is rotated with the incident angle θ.

To retrieve the emitted normalized fluorescence intensity $\Phi(\theta, E)$, the detection efficiency ϵ_{E_f} for the corresponding fluorescence photon energy E_f, the incident photon flux N_0, effective solid angle of detection $\frac{\Omega}{4\pi}$ needs to be known:

$$\Phi(\theta, E) = \frac{F(\theta, E)}{N_0 \epsilon_{E_f}} \frac{4\pi \sin\theta}{\Omega}. \tag{20}$$

Additionally, the detected count rate $F(\theta, E)$ needs to be calculated from the measured spectra. With the reference free set-up of the PTB it is possible to quantify the absolute mass deposition through the absolute fluorescence intensity [93]. Here, the fundamental parameters need to be known and the instrumental quantities need to be calibrated to be able to work without calibration samples. The incoming photon flux N_0 can be calculated from the photocurrent measured with a calibrated photodiode [95]–[98]. The effective solid angle of detection $\frac{\Omega}{4\pi}$ depends on the geometry of the measurement set-up and measurement chamber and can be calculated [99]. The solid angle is needed to calculate the complete fluorescence intensity emitted by the sample based on the detected fluorescence in the field of view of the detector. In grazing geometry, the footprint of the beam on the sample is elongated. For the calculation of the effective solid angle of detection the profile of the incident beam needs to be known or measured. The angle dependent projection of the incident beam on the sample needs to be convoluted with the

position dependent solid angle [100]. The detection efficiency ϵ_{E_f} can be determined experimentally [101], [102].

For the detected count rate $F(\theta, E)$ the spectra measured with the SDD need to be deconvoluted. Therefore, the energy dependent response function [94], [103] for each fluorescence line in the spectra is fitted to the measured spectra. Additionally, the relevant background contributions like bremsstrahlung or resonant Raman scattering are modeled [104]. The spectra in this work were deconvoluted with a routine provided by Philipp Hönicke (PTB).

The relative experimental error $\sigma_\Phi(\theta, E)$ for the emitted fluorescence intensity $\Phi(\theta, E)$can be retrieved from the estimation of the error for the effective solid angle of detection σ_Ω, and the contribution from the counting statistics $\frac{\sqrt{F}}{F}$ for the respective fluorescence lines [92]:

$$\sigma_\Phi(\theta, E) = \sqrt{\left(\frac{\sqrt{F(\theta, E)}}{F(\theta, E)}\right)^2 + \sigma_\Omega(\Omega)^2} \tag{21}$$

with

$$\sigma_\Omega(\Omega) = \left|\frac{\Omega}{\max(\Omega)} - 1\right|^{10} * 0.1 + 0.015\,. \tag{22}$$

The contribution of the error of the incident photon flux and the detector efficiency are not included here because they are angle independent.

4 A method to characterize nanostructures by solving inverse problems

The method developed in this thesis solves an inverse problem to characterize nanostructures. The method can be applied to 1D, 2D or even 3D periodic nanostructures and can be used as a metrological tool to reconstruct such structures from experimental GIXRF data. Shape and elemental composition are obtained in a non-destructive manner by combining X-ray fluorescence measurements with FEM simulations. The simulation requires the optical constants of the materials which are obtained dedicated XRR measurements and corresponding modeling.

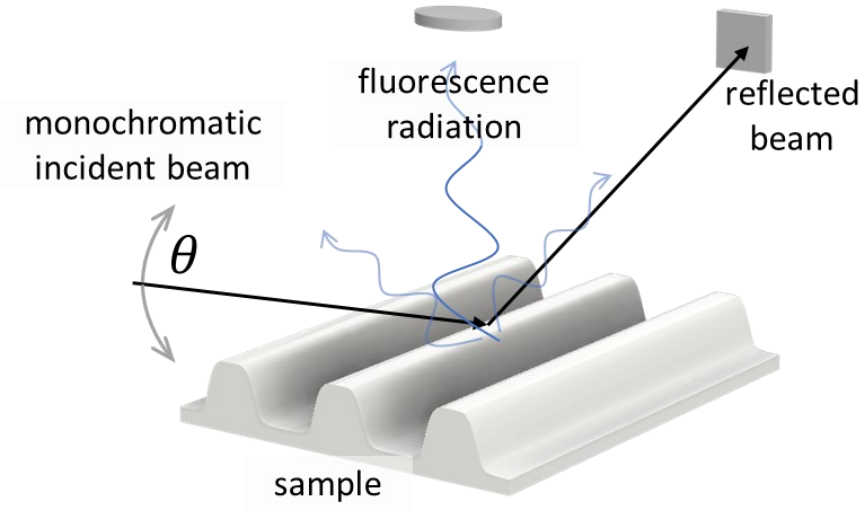


Figure 4.1 Sketch of the measurement principle (GIXRF and XRR) used in this thesis, with θ the grazing angle of incidence. For XRR measurements the sample is not a grating but a layered structure.

Figure 4.1 shows a sketch of the two measurement principles utilized in this work: angle dependent GIXRF and angle dependent XRR. Detailed information regarding the measurement set-ups for the measurements can be obtained from Section 3.3. The method comprises multiple computational components. The inverse problem of determining the optical constants from XRR measurements can be solved using the TMM and an appropriate optimization algorithm. Specifics of the forward calculations can be found in Section 2.2.1. Determining the shape of the nanostructures entails a more complex forward computation, whereby a model parameterization must be identified for performing the FEM Maxwell solver calculation corresponding to the structure under consideration. The fluorescence intensity can be retrieved using the 2D Sherman equation and the computed electric field distribution from the Maxwell solver. The fundamentals of this forward calculation are outlined in Section 2.2.2. The essence of this method lies in the reconstruction process of the shape describing the fluorescence intensity measurement. The basics of the optimization are introduced in Section 2.3. Various optimization algorithms can be employed to ascertain the model parameters, as elucidated in Appendix A. Upon successful reconstruction, the shape of the

nanostructures can be determined. The uncertainties of the outcomes can be obtained based on the optimization algorithm used.

In the following section, the method is applied to different nanostructures: the Si_3N_4 sample and the HSQ sample set. The samples are introduced in Section 3.1. They are 2D nanograting's fabricated to be used as test samples for this method. The Si_3N_4 sample contains a surface oxide layer on top of the structure, which allows the method to be tested for finding the elemental distribution of different materials. The HSQ sample set contains different 2D nanograting's with different pitch and shape. This gives the opportunity to discuss different parameterizations of the shapes and the sensitivity to smaller structures.

For this method, accurate optical constants of the material for the incident energy must be known, because the interaction (reflection and fluorescence) of the X-rays with the sample depends on the optical constants of the materials. Therefore, for the most accurate calculation of the FEM Maxwell solver, accurate optical constants for the incident energy are of paramount importance. For bulk materials without an absorption edge in the vicinity of the incident energy[3], optical constants retrieved from a database may be sufficient. However, for nanostructures and thin films, the density may be reduced, resulting in a change in the electron density and thus a change in the optical constants of the nanostructure or thin film. Section 2.2.1 describes how the optical constants of the samples are determined. First, the optical constants of a quartz crystal are obtained from the XRR measurement covering the Si-L edges and the O-K edge. The optical constants show the anisotropy of the quartz crystal. In addition, a deviation between a database and the measured optical constants is observed at the edges. This motivates a closer look at the optical constants of the material from which the nanostructures are made. For this reason, XRR measurements are performed on layers made of the same materials from which the nanostructures are made.

In Section 4.2, the results of the GIXRF measurements for the different samples are presented. For each sample, the angle-dependent fluorescence intensity is calculated from the measured spectra, from which preliminary information such as the elemental composition of the sample can be obtained.

For the reconstruction of the shape (Section 4.3) the fluorescence intensity for the same incident energy and angles is calculated with the FEM Maxwell solver and equation 10. This requires parameterizing the shape of the nanostructure, which also defines the model parameters used for the optimization (Section 4.3.1). Different optimization methods can be utilized to solve the inverse problem of inferring the shape of the nanostructure from the GIXRF measurements. There are optimizers that use intelligent sampling to achieve a solution with less computational time. This is investigated in a part of Section 4.3.2. In addition to experimental data, the reconstruction process is tested with artificially generated data. A possible way to reduce the computational effort is the application of different surrogate models. In Section 4.3.3 a surrogate of the forward calculation is created and used for solving the inverse problem.

The results of the reconstruction are compared to AFM, SEM or TEM measurements in Section 4.4.

[3] At the absorption edge there can be a sharp change in the optical constants. In addition, the optical constants depend on the fine structure of the material.

Section 4.5 investigates the influence of an additional carbon contamination layer, which is not accounted for in the model but can be observed in the measured spectra.

4.1 Characterization of the optical constants of nanolayers and quartz

As mentioned before, the determination of accurate optical constants is a key factor for modeling the light-matter interaction. There are several ways to determine or verify the optical constants experimentally. One option is to measure the transmission of the sample. From transmission measurements the imaginary part of the optical constants can be obtained directly. The Kramers-Kronig relation relates the imaginary part of the optical constants to the real part by integrating over the full energy range. Using transmission measurements and the dispersion relation has drawbacks. If the absorption is high, the samples must be extremely thin to measure any transmission. To apply the Kramers-Kronig analysis from a limited energy range of measurements, the range must be extrapolated, which can be a source of uncertainty [105].

Another option is to measure the angle dependent reflectance. This is an inverse problem investigated in this section. The setup must be able to measure the reflectivity for different angles of incidence, which is introduced in Section 3.3.1. This type of measurement can provide not only the optical constants, but also information about layer thicknesses, roughness, and interfaces [106]. It is non-destructive and can be used on samples with a thick substrate. The measurement can be extended to different energies, providing additional information about the sample (e.g., the fine structure of an absorption edge).

To solve this inverse problem different optimization setups are used depending on the total number of model parameters. The different approaches and optimizers are introduced in Section 2.3. For a small number of model parameters (<10) optimization setup A (see Figure 2.3) is used, where all model parameters are optimized at the same time. Here, MCMC simulations can be a good choice because this allows to estimate the full posterior distribution of the model parameters and, when combined with an error model of the inverse problem, the error of the input such as the measurement and the forward calculation.

For many model parameters (>>10) a different approach is used. For example, the number of measured photon energies increases the number of model parameters, because the optical constants of the different layers depend on the photon energy. Most meta-heuristic approaches such as DE are very computationally expensive with several hundreds of model parameters. The optimization process can be divided into two parts: an external optimization for the energy independent parameters such as the geometrical parameters (layer thickness, roughness, or angle offset) and an internal optimization for the photon energy dependent parameters like the optical constants. This is typically referred to as bilevel optimization or bilevel programming (setup B, see Figure 2.4).

First, the sensitivity of reflectivity measurement and optimization methods are tested with the measurement of the anisotropy of the quartz crystal. Then the optical constants of a Si_3N_4 layer with an oxide layer and a developed HSQ layer are investigated around the oxygen-K edge. This pre-characterization is

necessary to obtain good optimization results in Section 4.3. HSQ is a photoresist to produce nanostructures with a critical dimension in the nanometer regime [73], [74]. Normally the resulting structures are not made of this material, because it is removed after the pattern transfer with e.g., reactive ion etching [107]. Nevertheless, the optical constants are of interest and needed for the reconstruction of nanostructures. For the investigation of the uncertainties of the results MCMC simulations are conducted for the incident energy of the fluorescence measurements (680 eV).

4.1.1 Anisotropy and optical constants of quartz crystal

Quartz is the crystalline form of silicon dioxide (SiO_2). Quartz glass is used in a wide variety of applications like simple laboratory glassware and optics or semiconductor manufacturing and lithography. In optoelectronics and micro-electro-mechanical systems, quartz is used as a carrier material for mirrors, nanostructures, and other functional surface features. For the development of new optical components with pre-defined properties (e.g., maximum reflectivity for a given wavelength) reliable material knowledge is important. Knowing the optical constants is a key factor in modeling light-matter interactions. Different measurements of the optical constants of amorphous SiO_2 are published elsewhere [105], [108]–[110]. For a material like quartz, the optical constants or permittivity depend not only on the wavelength but also on the crystal orientation, which is called optical anisotropy. This dependence is also called spatial dispersion or optical anisotropy and can occur, for example, in a perfect cubic crystal that should be completely isotropic. The optical anisotropy effect near absorption edges was postulated in [111]. Near a strong absorption edge, the dielectric function and thus the refractive index increase strongly. These anisotropies have been observed in the past in various materials [112]–[114] and have also been confirmed for quartz at the O K edge by near-edge X-ray absorption spectroscopy [115]. This section is based on the publication [116] where the anisotropy of y-cut quartz crystal is shown.

The reflectance was measured for different angle of incidence between 0° and total reflectance in the s-polarization direction from the sample normal for the two different crystal orientations. The range of incident angles was adapted to the different photon energies to effectively cover the relevant angle range. At low reflectance, the measured signal is below the detection limit of the experimental setup, so it is unnecessary to measure in this regime. The photon energy covered the range from 45 eV to 620 eV, which includes the Si-L edges and the O-K edge. The optical constants were compared to ab initio Bethe-Salpeter calculations for the O-K edge made by Philipp Hönicke (PTB) and John Vinson (NIST).

This data set significantly extends the existing data on the optical constants of quartz in the near edge regions. Due to the large number of energies and thus fitting parameters, it was not possible to use MCMC. Optimization setup B is used with DE as a global optimization and an internal fit with the Limited-memory-Broyden–Fletcher–Goldfarb–Shanno-with-Bounds algorithm (L-BFGS-B). The layer model used includes a contaminant layer and the substrate. The sample was handled under normal laboratory conditions, which can lead to an ultrathin carbon-like layer with unknown optical and dimensional parameters due to adsorption of volatile organic materials. This was confirmed by the ab initio Bethe-Salpeter calculation. The anisotropy reconstructed without the contamination layer deviates significantly and can no longer be correlated with the calculation.

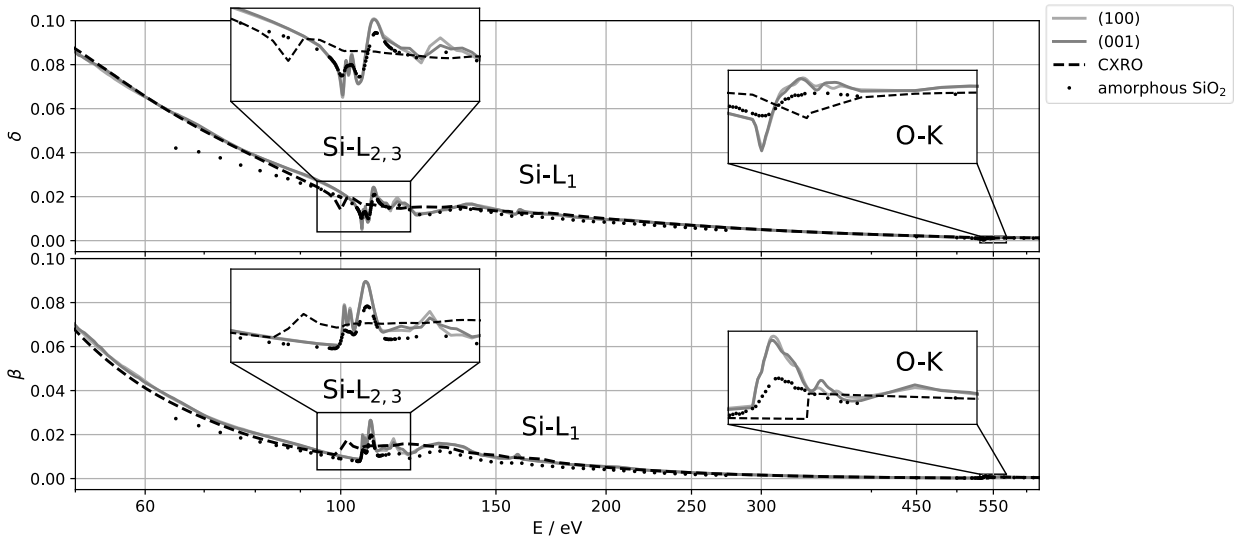


Figure 4.2 Reconstructed refractive index of a quartz crystal. Shown are the $1-\delta$ and β parts for the ordinary (001) (dark grey line) and extraordinary (100) (grey line) orientations of the crystal compared with the Center for X-ray Optics (CXRO) database values for SiO_2 (black dashed line) [45] and with the optical constants from amorphous quartz (black points) [105]. The insets show a magnified view of the reconstructed anisotropy in the absorption edge areas. The data are available online [117].

Figure 4.2 compares the results with the existing database [45] for an expected density of $2.65\,\frac{\mathrm{g}}{\mathrm{cm}^3}$. Away from the absorption edges, the tabulated values agree well. In the insets of Figure 4.2 the refractive index around the absorption edges is highlighted. Here the anisotropy of the quartz can be seen as the difference between the results for the two orientations ((001) vs. (100)). The anisotropy is more pronounced at the oxygen absorption edge. The position of the absorption edges for our results and the amorphous quartz [105] are energetically shifted compared to the tabulated data, which are mostly calculated with the atomic scattering factor (see equation 2). This assumption is only valid outside of absorption edges. Near the absorption edges the specific chemical state must be considered. This can be seen by the lack of fine structure and the shifted position of the absorption edge in the tabulated values.

To explain not only the fine structure but also the dependence of the fine structure on the orientation, the local environment of the absorbing atom needs to be considered. The quartz crystal structure consists of connected SiO_4 tetrahedra in a trigonal lattice. The unoccupied Si d orbitals are split by the tetrahedral symmetry of the neighboring oxygen atoms into doubly degenerate e and triply degenerate t_2 orbitals, with the broad absorption around 115 eV made up of transitions into the e: d_{z^2} and $d_{x^2-y^2}$. These orbitals are split locally by distortion in the oxygen tetrahedra and, through hybridization, by the long-range anisotropy of quartz. Similarly, at the O-K edge, the features from $545\ \mathrm{eV}$ to $550\ \mathrm{eV}$ are due to hybridization with these same Si d orbitals [118].

This shows that the values taken from existing databases of a material in the edge region are often in exact. Using reflectivity measurements to calculate the optical constants in the edge regions can be more precise.

4.1.2 Analyzing the optical constants of the Si_3N_4 sample and developed HSQ around the oxygen edge

To reconstruct the nanostructures described in Section 4.3, the optical constants and the density of the material under investigation must be known. The method applied to the quartz crystal can be used to obtain the optical constants of a (multi-)layered sample. Here, the optical constants of the unstructured area of the Si_3N_4 sample and the 1D sample of the HSQ sample set are obtained. In [40], [119] it was shown that the optical constants of a material can depend on the thickness of the layers. Additionally, the density of a nanostructure can be lower than the bulk density. To use database values for a material containing different atoms, the stoichiometry must be known. For HSQ, the stoichiometry varies depending on the exact development process and can therefore only be estimated. HSQ before development contains cubic-like [$HSiO_{3/2}$] nanoclusters, which are transformed into a network structure during development by losing hydrogen and forming chemical bonds with neighbors [120]. Therefore the optical constants should be comparable to amorphous silica [121] with a reduced density. Since the exact stoichiometry of the developed HSQ is variable, the use of a database may result in an inaccurate optical constant, which can be solved by utilizing values obtained from soft XRR measurements.

4.1.2.1 Reflectivity measurements

The reflectivity of the Si_3N_4 sample (Figure 4.3 left) and the 1D HSQ sample (Figure 4.3 right) were measured for different incident energies around the oxygen edge. In Table 2 the expected layers and the settings of the corresponding reflectivity measurements are summarized.

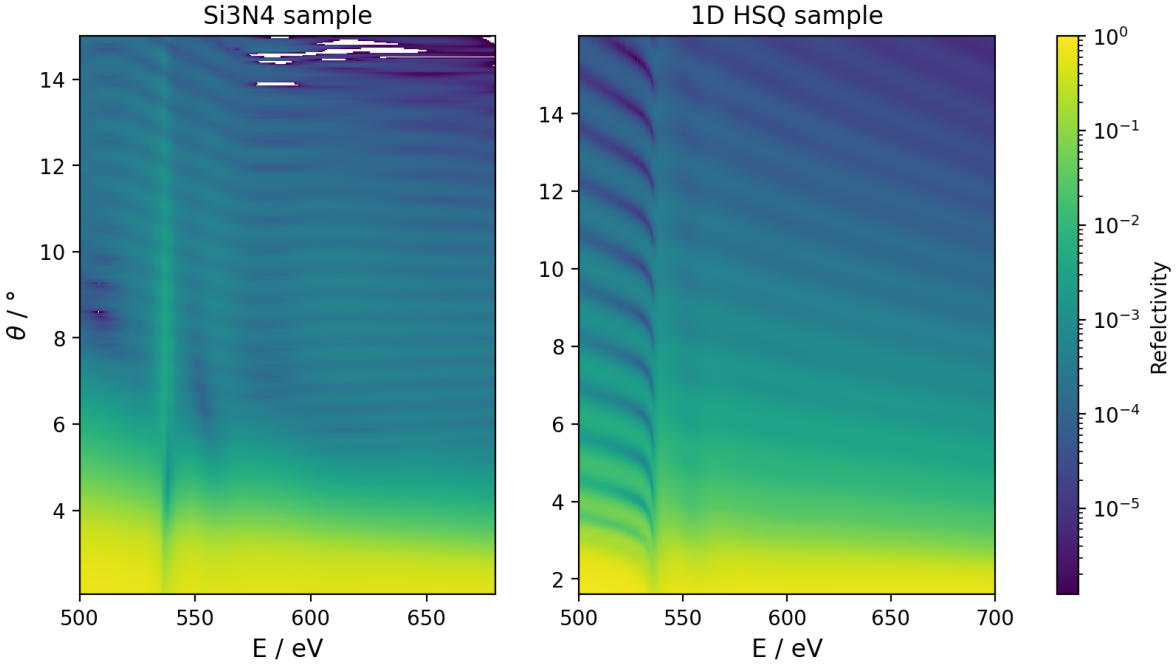
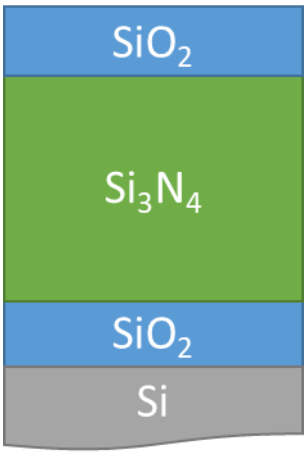


Figure 4.3 Reflectivity of the Si_3N_4 sample (left side) and 1D HSQ sample (right side) for different energies E around the oxygen absorption edge (seen around 530 eV) and grazing angle of incidence θ.

Table 2 Overview over the expected layers and the settings of the measurements for the different sample.

	Si_3N_4 sample	**1D HSQ sample**
Layers		
Energy range with varying step size	500 eV to 680 eV	500 to 700 eV
Polarization	p-polarization	s-polarization

The substrate of both samples is a Si wafer that may have a native oxide layer. The layer system of the unstructured area of the Si_3N_4 sample above the substrate (with oxide layer) consists of a Si_3N_4 layer with an oxidized surface due to the removal of the photoresist. The high frequency and the low frequency oscillations visible in the XRR shown in the left side of Figure 4.3 are a clear indication of a multilayer system. It is reasonable to expect the optical constants of the unstructured region to be the same as the structured region since they are made of the same material. The 1D sample from the HSQ sample set contains a layer of developed HSQ on the substrate (with oxide layer). For both samples, the oxygen absorption edge is visible in the reflectivity around 530 eV. The optical constants were modeled for all measured energies.

From the expected layer system, the model parameters for the optimization are selected. They can be divided into energy independent model parameters (e.g., the thickness and the roughness of the different layers, a theta offset, and a dark current) and energy dependent parameters (e.g., the optical constants of the layers). For the substrate, the optical constants were not modeled but taken from the database [45], because the substrate can be approximated as a bulk material and the measurement was not close to a Si absorption edge. A contamination layer has been added to both samples because the evaluation of the quartz dataset showed that this is necessary when modeling the optical constants in the region of an edge. This additional layer consists of various organic compounds. At the oxygen edge, the optical contrast changes extremely with high sensitivity to different chemical samples [122]. The contamination can occur due to storage in ambient conditions.

4.1.2.2 Optimization results

To obtain the optical constants for the entire energy range of both data sets, the optimization setup B (see Figure 2.4) was used. DE was chosen as the outer optimization method for both samples. In the following the optimization results of the Si_3N_4 sample are discussed. For this sample, the Truncated Newton Constrained (TNC) method was chosen for the inner minimization loop, because it allowed the global optimization to converge to a better minimum $\chi^2_{TNC}/DoF = 2.93$ compared to L-BFGS-B ($\chi^2_{L-BFGS-B}/DoF = 9.84$) with $DoF = 8370$. TNC and L-BFGS-B differ in their approach to calculating the search direction. TNC uses an elaborate variable-cost iteration with partial second derivative information, while L-BFGS-B uses a fixed low-cost formula that requires no additional derivative information [123].

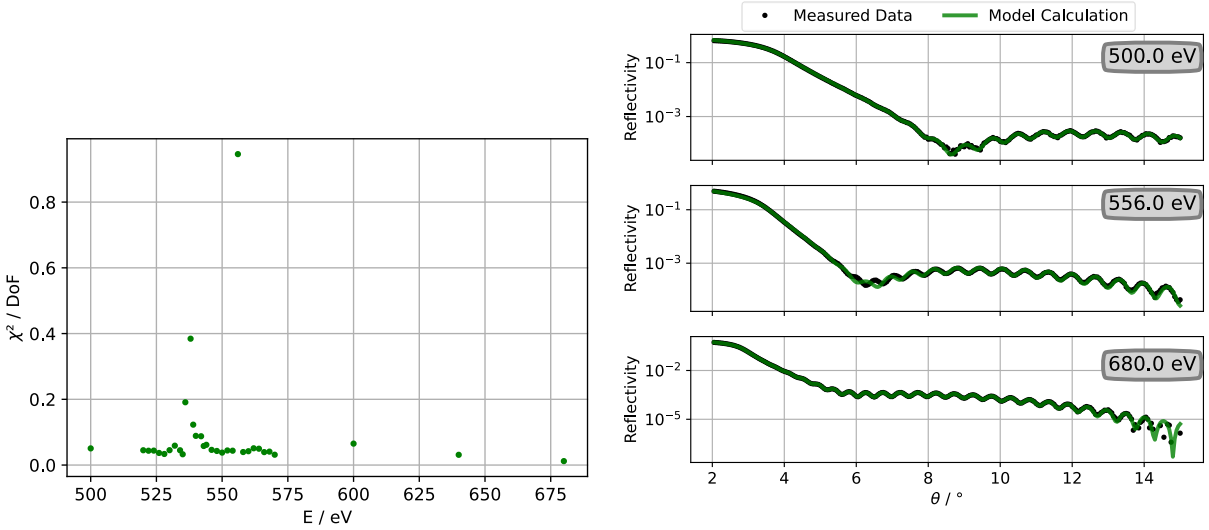


Figure 4.4 The left side shows the $\chi^2(E)/\ DoF$ depending on the different energies E and summed over θ of the successful optimization for the Si_3N_4 sample. On the right, the comparison between the model calculation and the measured reflectivity can be seen for three different energies.

In the left side of Figure 4.4 the $\chi^2(E)/DoF$ of the Si_3N_4 sample for the best optimization results is shown. For most energies E, the model describes the experimental data well. The oxygen absorption edge is clearly visible at 530 eV. The maximum $\chi^2(E)/DoF$ is located at $E = 556.0\text{eV}$ which is located above the edge. At the right side of Figure 4.4 the measured data of the Si_3N_4 sample and the corresponding model calculation for three energies are displayed. For $E = 500.0$ eV (below the oxygen edge) and $E = 680.0\text{eV}$ (above the oxygen edge) a good agreement between the experimental data and the calculation is observed. For $E = 556.0$ eV, the model calculation shows a deviation from experimental data around $\theta = 6°$. At the oxygen K-edge, the reflectivity profile is extremely sensitive to oxygen [122]. Due to the chemical shift, there can be a resonance energy at $E = 556.0\text{eV}$ of some compounds incorporating oxygen present at the surface of the sample.

In an additional measurement the reflectivity of the Si_3N_4 sample is measured at the PGM beamline and in p-polarization, because the higher flux allows the measurement of the reflectivity at higher angles of

incidence. These results are published in [58]. The same settings (incident energy of 680 eV and p-polarization) were used for the reflectivity measurements as for the fluorescence measurements. Because of the reasonable number of degrees of freedom, a MCMC method was applied as a global optimization (optimization set-up A). Parameters such as layer thickness, roughness, and optical constants were modeled. A three-layer model is used for the simulation of the XRR. Here the model was simplified compared to the measurement of the soft XRR around the oxygen edge by removing the contamination layer, following the Occam's razor philosophy [124] of using the simplest possible model. The modeled contamination thickness for the previous optimization was 0.1 nm with a roughness like that of the oxide layer (0.55 nm). At 680 eV, the measurement is no longer highly sensitive to oxygen compounds in a thin contamination layer, because this energy is far from the oxygen K-edge. Therefore, outside the absorption edge the influence of such a small contamination layer is negligibly small. The three layers model describes the experimental data very well, which supports the decision to reduce the number of model parameters (see Figure 4.5).

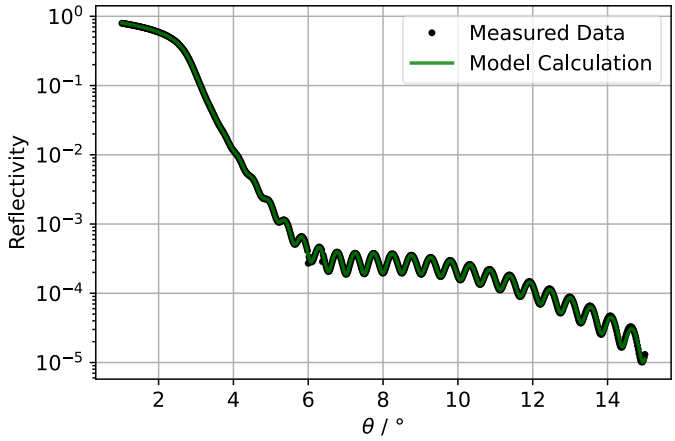


Figure 4.5 The result of the MCMC simulation and the corresponding measurement data from the PGM beamline for the Si_3N_4 thin film sample for different angle of incidence for an energy of 680 eV is shown.

In the following, the optimization results for the 1D HSQ sample are presented. In [121], the near edge X-ray absorption fine structure of exposed and unexposed HSQ was studied at the oxygen K-edge. To find the shifted oxygen K edge position of the HSQ sample investigated here, the total number of measured energies was increased compared to the Si_3N_4 sample. To obtain a stable fit, the optical constants of the substrate oxide layer were interpolated from the results of the Si_3N_4 sample around the oxygen edge (from 500 eV to 570 eV) or taken from the database for $E \geq 570$ eV. The oxide layer of the substrate is expected to be thin (<2 nm) and below the HSQ layer. For small angles below the critical angle, the thin oxide layer does not contribute to the reflected signal. Therefore, the internal optimization is not sensitive to the optical constants of this layer, and the performance of the overall optimization can be significantly improved by not including it in the optimization.

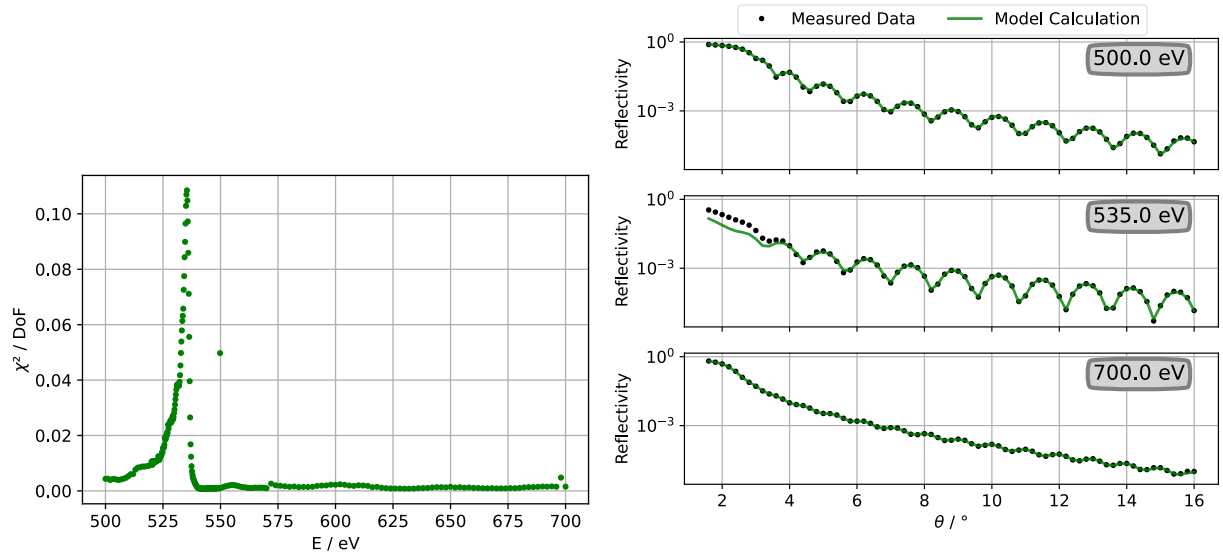


Figure 4.6 The left side shows the $\chi^2(E)/DoF$ depending on the different energies E and summed over θ of the optimization. On the right the comparison between the model calculation (green line) and the measured reflectivity (black dots) can be seen for three different energies.

For the 1D HSQ sample optimization set-up B is combined with either-TNC or L-BFGS-B as an inner optimization method because both lead to similar minima of the χ^2 optimization. The overall $\chi^2(E)/DoF$ with $DoF = 17656$ for the HSQ sample is shown in Figure 4.6 (left side). For energies above 550 eV, $\chi^2(E))/DoF < 0.1$ for most incidence angles. This indicates that the modeled reflectivity matches the measured reflectivity. Figure 4.6 (right side) the measured data and the calculated reflectivity are displayed for three different energies. At the oxygen edge, the model was not able to describe the measured data, much like for the Si_3N_4 sample. The extreme sensitivity to oxygen as a chemical species, the atomic, electronic structures and chemical bonds [122] is even more pronounced. In the HSQ sample, oxygen is not only present in a small layer but in all layers above the substrate. Due to chemical shifts, there can be different resonance energies for some compounds with oxygen. In addition, exposed HSQ contains different chemical bonds between the silicon, oxygen, and hydrogen atoms. The distribution of these chemical bonds can differ throughout the structure, because the surface can have an influence on the cross-linking process during the exposure[121].

For the HSQ sample, an MCMC simulation at 680 eV was performed with a two-layer (no contamination) model to estimate the density of the HSQ layer. The optical constants for HSQ were taken from the database [45] for a $H_8Si_8O_{12}$ material with varying densities. The two-layer model is closer to the model used for the reconstruction of the fluorescence measurement since no contamination layer is added. The estimated density is $\rho_{HSQ} = 2.12 \pm 0.04 \frac{\mathrm{g}}{\mathrm{cm}^3}$.

The sample is approximated by discrete ideal layers (here TMM). The interface imperfections are estimated by damping factors. In the sample, there may be defects, especially between different layers, or inhomogeneities in the density profile of a layer. Far from the oxygen edge, traces of oxygen-containing compounds are uncritical for the model, but at absorption edges of oxygen the model becomes deficient [125], [126]. To model interdiffusion or density profiles, one can add more layers to the model. This greatly

increases the computational effort and is not necessary here, because the four-layer model worked well for most energies.

4.1.2.3 Discussion of the determined optical constants

The experimentally determined optical constants for both samples are compared in different figures with tabulated CXRO data [45] and the values retrieved around the oxygen edge form the quartz crystal.

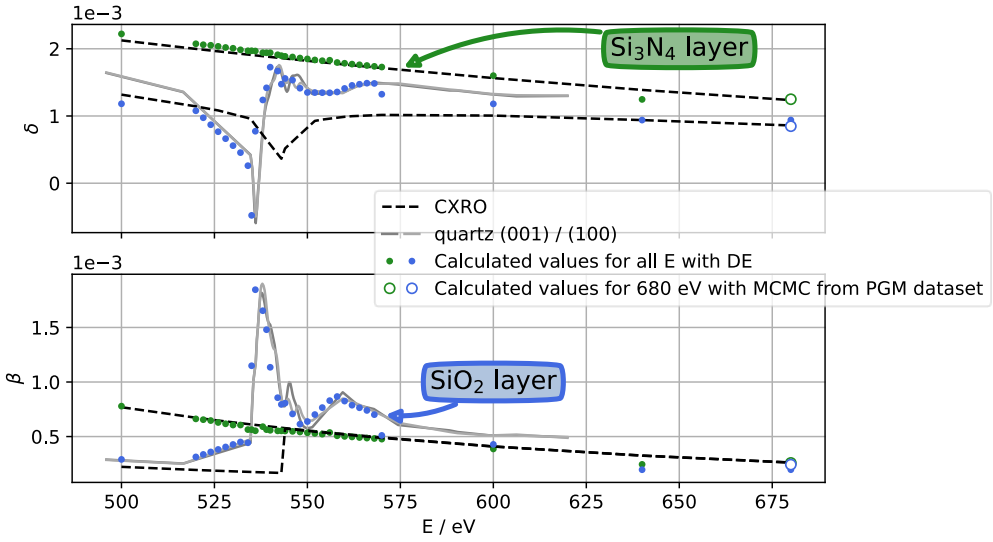


Figure 4.7 Reconstructed refractive index of the Si_3N_4 sample. Shown are the δ and β parts for the Si_3N_4 layer (green) and SiO_2 layer (blue) compared with the values from the CXRO database values (black dashed line) [45] with the reduced densities and the quartz results from the previous section (grey). The points are connected to the reconstructed values from the PGM dataset.

The results of all the modeled optical constants (dotted lines) for the different layers of the Si3N4 sample are shown in Figure 4.7. The values are compared to the values from the CXRO database (black lines)[4] [45] and the reconstructed optical constants of the quartz crystal (grey lines). Relative densities of (0.86 ± 0.01) and (0.79 ± 0.01) were found for Si_3N_4 and SiO_2, respectively. The densities are estimated by comparison of the results of the MCMC simulation to the tabulated data. In [29], the observed material density reduction is already discussed. The results of the MCMC simulation from the PGM dataset is visible as circles. In Figure 4.7 an overall good agreement is seen for the values of the nitride layer. At $680\ \mathrm{eV}$, the optical constants of the nitride layer (obtained from the PGM dataset) agree with the results of the energy dependent measurement. For the oxygen layer, an energy shift at the edge and a more complex fine structure can be observed compared to the database. Comparing the results of the optical constants of the oxide layer with the quartz data, it is noted that the fine structures visible in the quartz data are not visible for the oxide layer, because this layer is amorphous. The oxygen K-edge for the oxide layer seems to be shifted to lower energies compared to the quartz indicating a slightly different chemical shift. In the amorphous silicon oxide layer, there may be different specimens of silicon oxide present. For the oxide

[4] Here the reduced density retrieved from the MCMC simulation of the PGM dataset is used.

layer a difference between the result of the MCMC simulation and the results from the DE optimization can be observed. This difference is larger than expected from the uncertainties of the MCMC simulation, which can be explained by the difference in the model used for optimization (with and without contamination) and the different measurement setup and time between measurements. For a complete comparison of the two different optimization results, the uncertainties of the energy-dependent measurement results must be calculated. Due to high computational effort caused by the large number of different energies, this was not investigated here, but is an interesting topic for future studies.

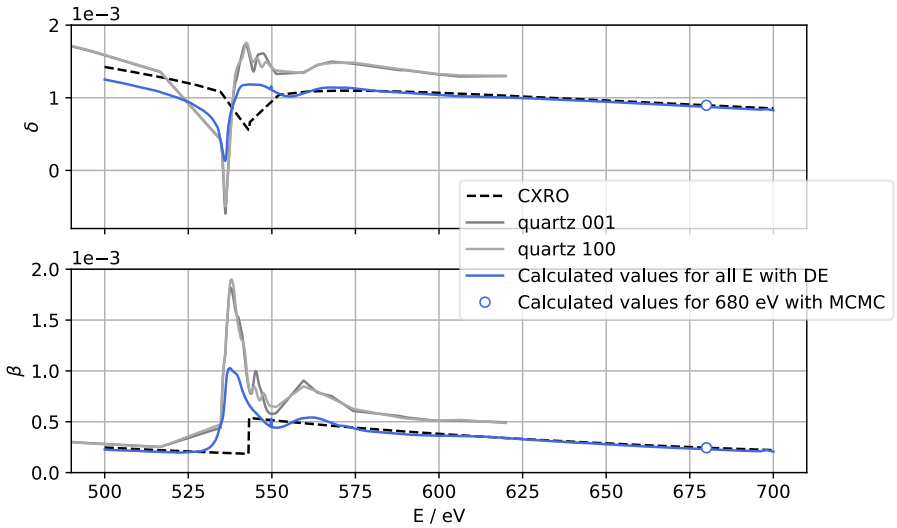


Figure 4.8 Comparison of the δ and β values for the HSQ layer of the samples for different energies (blue line) compared to the results from the CXRO database [45] (black dotted line) and the quartz crystal (grey lines). The circle shows the results of the MCMC simulation.

The results for the optical constants of the HSQ layer are summarized in Figure 4.8. The blue circles show the results of the MCMC simulation, and the black dotted line are the values taken from the database for $H_8Si_8O_{12}$ with the density obtained from the MCMC simulation. The reconstructed values for the complete energy range are shown in blue and, for comparison, the quartz results are added to the plot in grey. The position of the oxygen K edge is clearly shifted compared to the values from the database but matches the position from the quartz data. Like the oxide layer of the Si_3N_4 sample the optical constants of the HSQ layer are missing the fine structure present in the quartz, because HSQ behaves not like a crystal but more like amorphous SiO_2. The height of the peak of the HSQ at the edge is lower compared to the quartz, indicating that in the HSQ layer different chemical species of oxygen are present compared to the quartz, which leads to a broadening of the peak. An additional difference is the shift to lower δ or β compared to the quartz, indicating a reduced density of the HSQ layer [40], [119].

In summary, studying the reflectivity around an absorption edge can provide detailed information about the chemical composition of the sample. At these edges, database values are often not applicable, because the values are mostly determined by calculations based on the atomic scattering factor [45] (see

Section 2.1). In addition, the influence of surface contamination on the optimization process of the optical constants needs to be further investigated for a better interpretation of the results. Nevertheless, the method provides reasonable optical constants for the materials studied, which can be used in the next section.

4.2 Grazing incidence fluorescence measurements

For the characterization of regular ordered nanostructures, I employ GIXRF measurements. In the angle dependent fluorescence measurements, the shape of the structure is encoded. In this section the results of the measurements for the different samples are presented. Details of the experimental set up can be found in Section 3.3.2. The measurement and post-processing of the fluorescence data involves more steps than the reflectivity measurement described in the previous section. The fluorescence spectra are measured for different angles of incidence with an incident energy of $680\ \mathrm{eV}$, which allows the measurement of fluorescence photons from oxygen, carbon, or nitrogen. All these elements are present in the Si_3N_4 sample. In the HSQ sample set, only carbon and oxygen can be observed.

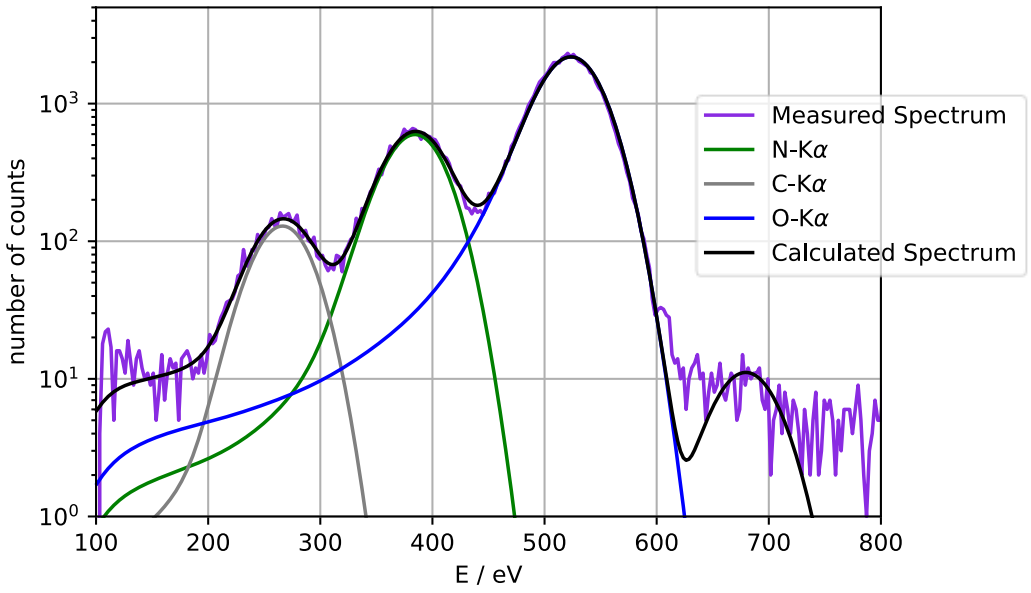


Figure 4.9 Example spectrum of the Si_3N_4 sample (purple) with the calculated spectrum (black) and the constituent spectra: carbon in grey, oxygen in blue and nitrogen in green.

In Figure 4.9 an example spectrum of the Si_3N_4 sample is shown. The peaks of the oxygen, carbon and nitrogen fluorescence are visible. The calculated spectrum is the sum of the different parts of the response function, which were modeled to the measured spectrum. This process was applied to every measured spectrum. In the following, only the postprocessed measurements are shown. The postprocess not only includes the deconvolution of the spectra, but also the correction to the detector efficiency, the solid angle of detection, the angle of the incident beam to the surface of the sample, and the incident photon

flux (see equation 20). Considering these factors allows for calculating the fluorescence intensity per incident photon.

4.2.1 Measurements of two different materials on a sample

The first sample is the Si_3N_4 sample, which is a 2D nanograting with an oxide layer originated from an oxygen plasma treatment to remove the remaining photo resist after the etching of the structure. The detected carbon fluorescence is caused by a contamination layer that is present on the structure because of the storage and previous experiments of the sample. Nevertheless, the carbon fluorescence intensity is small (thickness below 1nm estimated by the XRR characterization). For the reconstruction only the intensity of the oxygen and nitrogen fluorescence is used. Measuring the angle dependent fluorescence of more than one element gives additional information about the distribution of the different elements inside the structure, because the XSW field acts like a nanoscale depth sensor.

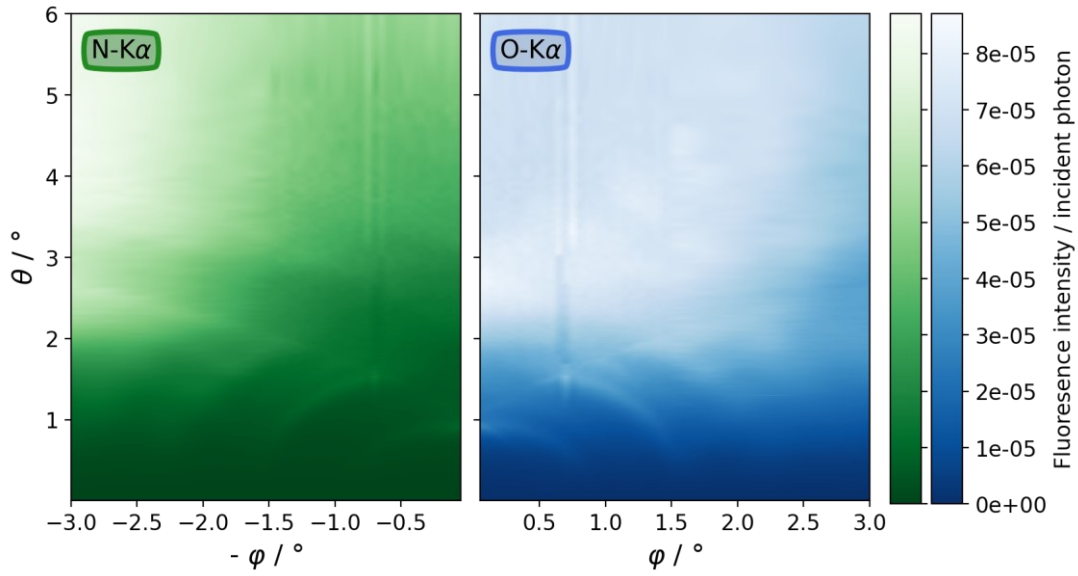


Figure 4.10 Plot of the fluorescence signal for oxygen (blue) and nitrogen (green) for different angles of incidence θ and azimuthal angle φ for an incident energy of 680 eV.

Figure 4.10 shows the fluorescence intensity for different azimuthal angles φ from 0° (grating parallel to the incident beam) to 3° and angles of incidence θ from 0° to 15° for an incident energy of 680 eV (~1.83 nm). For azimuthal rotation of the sample a change in the measured fluorescence signal for oxygen and nitrogen can be observed. This modulation cannot be observed in 1D structures like layered systems. Nevertheless, for the reconstruction only the fluorescence for the minimal φ was utilized to reduce the computational effort. The fluorescence for the minimum azimuthal angle was measured first and therefore has the least influence of any (carbon) contamination, as the contamination increases during the measurements.

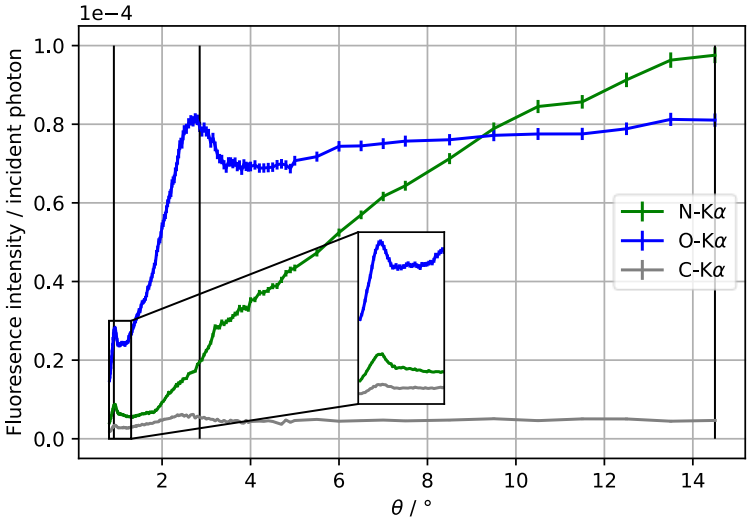


Figure 4.11 Fluorescence intensity per incident photon depending on the grazing angle of incidence of a 2D nanograting made from Si_3N_4 with an SiO_2 oxide layer. The inset shows the small angle regime. The black lines indicate the angles at which the electric field distribution is shown in Figure 4.12.

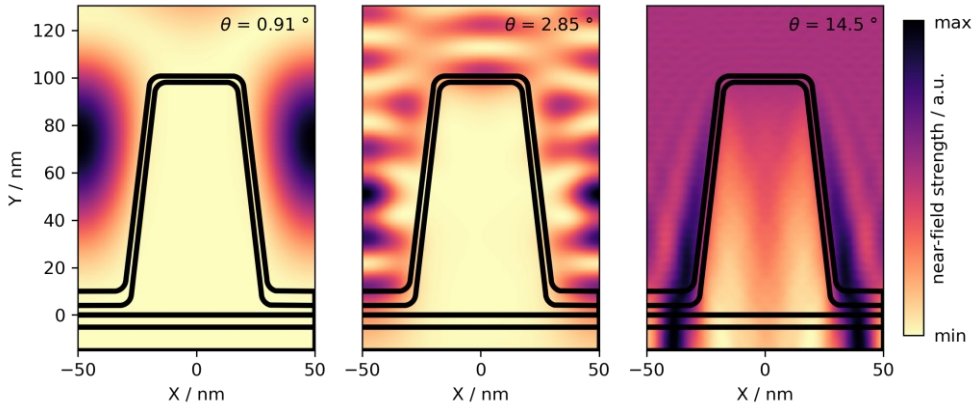


Figure 4.12 The electric near-field distribution for a given incident energy (680 eV) and three angles of incidence θ of a possible shape of the Si_3N_4 sample outlined by the black lines.

In Figure 4.11 the fluorescence intensity per incident photon depending on the grazing angle of incidence for the minimal φ is shown for the Si_3N_4 sample. For small angles (<7°) the oxygen signal dominates the measured fluorescence which indicates an oxidation of the Si_3N_4. The penetration depth of the XSW field depends on the angle of incidence. At grazing angles, only the atoms near the surface contribute to the fluorescence. For both fluorescence lines, first local maxima (FLM) around 1° can be observed. The location of this peak depends on the width of the groove of the nanograting, because here the first mode of the XSW field fills the groove (see Figure 4.12 left). As the angle of incidence is increased, the penetration depth of the XSW field increases and atoms deeper in the structure begin to emit fluorescent photons. For the maximum of the measured oxygen fluorescence around $\theta = 2.85°$ the near field distribution is shown in Figure 4.12 in the middle. At this angle, one mode penetrates the oxide layer at the top of the structure, and one penetrates the oxide layer at the bottom of the groove. For higher angles (>8°) the

penetration depth increases, which leads to an increase in nitrogen fluorescence. For angles above the critical angle, the beam is diffracted into the structure. At high angles of incidence, the fluorescence intensity levels off. Figure 4.12 right side shows the near field distribution for $\theta = 14.5°$. No near-field oscillation is seen above the structure and nearly the entire sample volume contributes to the fluorescence. For higher angles, the self-absorption increases, especially for the nitrogen fluorescence, because fluorescence photons are generated deep inside the structure. The fluorescence photons travel through a significant amount of material where they can be reabsorbed before reaching the detector.

4.2.2 Measurements of samples with different shape

The second set of samples used to develop the method contains four different HSQ samples with different shapes and pitches. With an incident energy of $680\ \mathrm{eV}$, oxygen and carbon fluorescence can be observed.

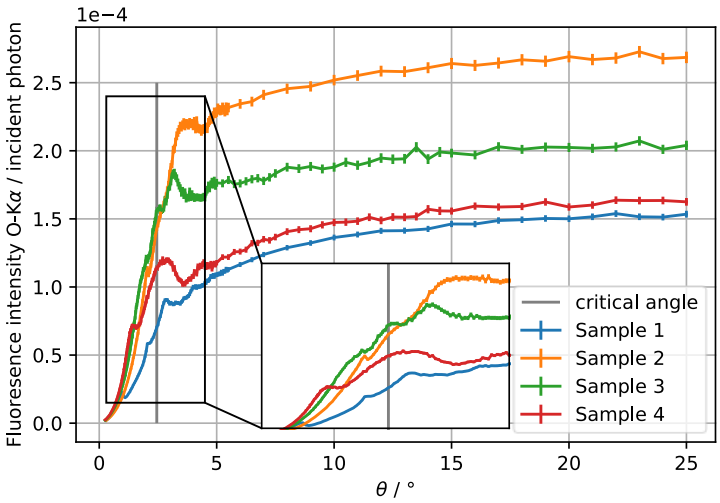


Figure 4.13 Oxygen fluorescence intensity per incident photon depending on the grazing angle of incidence for four different samples. The grating was oriented parallel to the incident beam. The inset shows the small angle regime. The grey line shows the position of the critical angle of a single HSQ layer.

Figure 4.13 shows the oxygen fluorescence for the grazing angle of incidence for all four samples with the grating oriented parallel to the incident beam. In contrast to the Si_3N_4 sample, only fluorescence from the oxygen K edge can be observed. In the inset of Figure 4.13 the small angle regime is displayed. The location of the FLM depends on the pitch of sample. For Sample 1 with a pitch of 100 nm, the FLM occurs around 1° and for Sample 2 and 3 with a pitch of 50 nm around 2°. Sample 4 has a pitch of 60 nm and the FLM is between 1° and 2°. The critical angle (see Equation 5[5]) can be correlated to a local maximum for Sample 3 and 4. For Sample 1 and 2 it is not clear. At the critical angle, the first mode of the XSW field is in the grooves of the nanograting, and the field starts to penetrate deeper into the structure. Depending on the depth of the groove, the position of the second mode can be located at the top of the structure or in the

[5] Calculated from the optical constant of the HSQ layer.

groove. Alternatively, the absence of features at this angle may indicate the absence of material in the groove or the presence of overlapping fluorescence from the top of the structure. For high angles ($\theta > 20°$), the fluorescence intensity levels off, indicating that all fluorescent atoms are contributing to the measured intensity. By taking the mean of the fluorescence intensity $\Phi_{\mathrm{mean}}(680\ \mathrm{eV})$ (retrieved with equation 20, Section 3.3.2) for angles between 20 ° and 25° the area density $\frac{m_O}{F_i}$ can be calculated:

$$\frac{m_O}{F_i} = \Phi_{\mathrm{mean}}(680\ \mathrm{eV})\frac{1}{\omega_k \tau(680\ \mathrm{eV})} \tag{23}$$

Here $\tau(680\ \mathrm{eV}) = 11356.9\frac{\mathrm{cm}^2}{\mathrm{g}}$ is the photo-ionization cross section of the oxygen K-shell for the incident photon energy of 680 eV taken from a database [56]. The fluorescence yield $\omega_k = 0.00688$ is taken from dedicated experiments [57].

Table 3 The area density $\frac{m_O}{F_i}$ for the different HSQ samples with the corresponding standard deviation (STD).

	$\left(\frac{m_O}{F_i}\right) / \left(\frac{kg}{m^2}\right)$	$STD\ /\ \left(\frac{kg}{m^2}\right)$
Sample 1	1.95E-05	1E-07
Sample 2	3.44E-05	3E-07
Sample 3	2.07E-05	2E-07
Sample 4	2.60E-05	3E-07

Table 3 summarizes the area density of the different samples. Sample 2 has the highest area density, indicating that the total number of oxygen atoms in a unit cell of the cross section of the grating is the highest. Due to the fabrication process, the height of the HSQ grating on the substrate should be similar for all samples. The difference in area density of Sample 2 compared to the other samples indicates that for this sample a significant amount of material is present in the groove. This corresponds to the TEM image of the sample compared to the other samples (see Figure 3.1).

4.3 Reconstruction of nanostructures from GIXRF measurements

In this section, a procedure for solving the inverse problem of inferring the shape of a nanostructure from GIXRF measurements is described. For a successful reconstruction it is important to parameterize the shape. Different options are presented in Section 4.3.1. The model parameters resulting from the parameterization are optimized with different global optimization methods (Section 4.3.2). To find a good solution for the inverse problem, where the calculated fluorescence intensity matches the experimental one as closely as possible, different optimizer are compared regarding their efficiency and the best solution achieved by the optimizer. The Bayesian Optimizer (BO) shows the best performance for this inverse problem. Additionally, the BO is analyzed with the help of a virtual experiment to evaluate the reconstructed model parameters with their confidence intervals. Then the method is applied to the measured fluorescence intensity. The performance of two different parametrizations is compared. A surrogate is trained

to replace the time intensive forward calculation with the FEM based Maxwell solver (Section 4.3.3), which allows for uncertainty quantification and a reduction in optimization time.

4.3.1 Parametrization of the samples

To solve the inverse problem, it is essential to find a parametrization of a system to describe the model [35]. These model parameters are optimized during the reconstruction and can have an influence on the convergence of the optimization e.g., a higher number of model parameter increases the computational effort of the optimizer and can increase the calculation time. For a grating, the basic parameters are the linewidth w, the line height h, and the pitch p. Often a box is a too simple model, because during the etch process a grating with tilted side walls can be created [127]. Therefore, a trapezoid can be used as a model. Due to the symmetry of the structure, it is sufficient to add one side-wall angle swa as a model parameter. The width w is defined at half height. The corners are not sharp, but rather smooth, so a corner rounding with a given radius R is added to the model. An etch offset h_g is added as a model parameter because the grating may not be fully etched to the etch stop (Si wafer). The oxide layer of a structure can be modeled with a thickness for the grating t_t and a different thickness in the grove t_g. The oxide layer is created by removing the resist after etching. The resist is only present on the top of the structure. Therefore, there was no resist in the groove during oxygen plasma cleaning, which can result in a thicker oxide layer in the groove compared to the top of the structure. The oxide layer of the substrate is defined with a thickness t_s. The oxide layer is expected to be thin (<2 nm) and located below the silicon nitride layer. Its contribution is small compared to the surface oxide layer. Therefore, I anticipate low sensitivity to this model parameter. Including model parameters with low sensitivity in the optimization process results in a high uncertainty for this parameter. To reduce the numerical effort reference values from independent measurements can be used. For the value of the thickness of oxide layer of the substrate t_s the thickness from the XRR reconstruction was used. The samples are produced with an E-beam writer. During the lithography process the resulting width of the structure can vary due to the interaction of the electrons with the resist. The variation in the pitch of the sample is insignificant compared to the variation in the width. The width is determined not only by the E-Beam exposure, but additionally by the etching process, while the pitch depends only on the E-Beam, which is precisely controlled by E-fields. Therefore, the pitch was not modeled during the reconstruction. This type of parameterization was used for the Si_3N_4 sample, because in the SEM image a trapezoid with rounded corner can be observed. In Figure 4.14 the model parameters (red) and the constant parameters (black) needed for the parameterization of the shape of the Si_3N_4 sample are shown. By comparing the shape with the expected shape obtained from the SEM image, this type of parametrization seems reasonable.

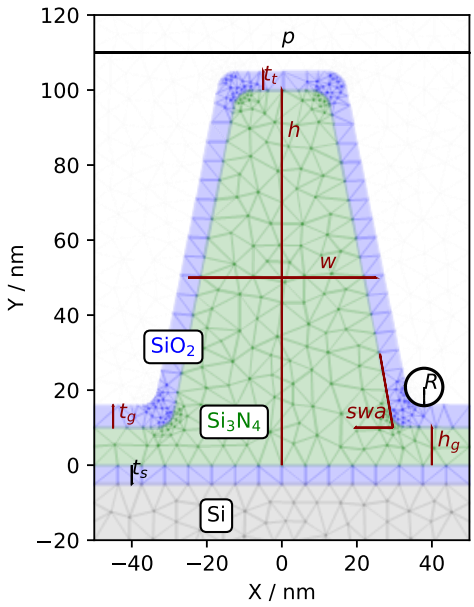


Figure 4.14 Parameterization of the Si_3N_4 sample. The parameters are width w, the height h, the pitch p, radius of the corner R, side-wall angle swa, etch offset h_g, thickness of the oxide t_t and thickness of the oxide in the grove t_g. The model parameters are shown in red, while the fixed parameters are shown in black.

For the HSQ sample set, this type of parameterization is not feasible. From the TEM images presented in Section 3.2.1, a more rounded shape of the structure is expected compared to the Si_3N_4 sample. Two different parameterization methods were investigated and compared for the HSQ sample set. The different parametrization methods were chosen by analyzing the TEM images and the corresponding outline of the samples.

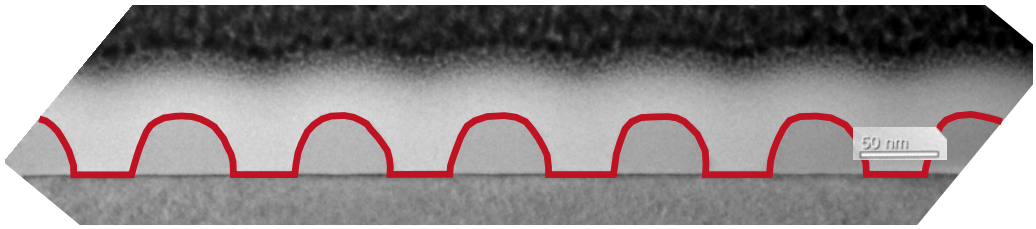


Figure 4.15 TEM image of Sample 1 with the traced outline in red.

The outline of the TEM image is obtained by tracing the surface for some grating lines for each sample. An example of the result can be seen in Figure 4.15 (red line) for Sample 1. I determined the scale of the pixels using the known pitch. I then identified the positions where the outline repeats to calculate the size of the pitch in pixels. By comparing the pitch in pixel to the pitch in nanometers, I was able to determine the shape of the structure at the nanometer scale. Calculating the mean of the traced grating lines gives the X and Y values for a unit cell and the corresponding standard deviation. I assume that the samples are symmetrical, so the average of the right and left sides of the unit cell is taken. The results are shown in Figure 4.16.

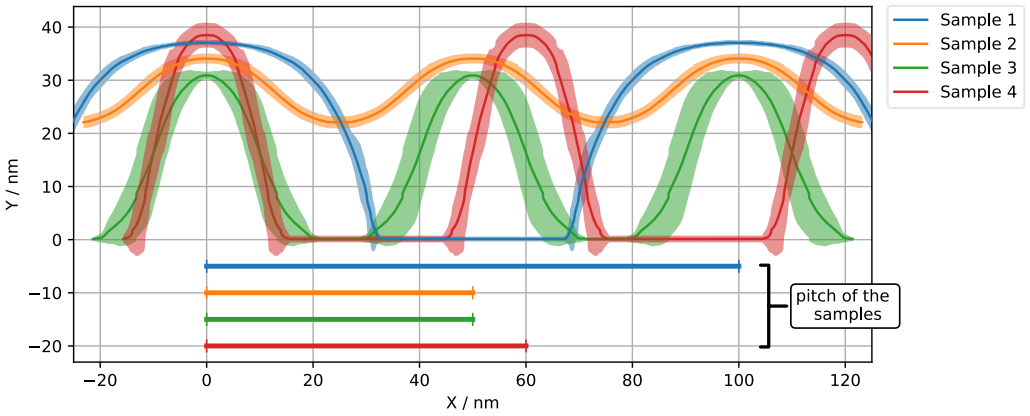


Figure 4.16 The mean shape of the samples retrieved from the TEM image with standard deviation and the pitch of the samples.

This shape can now be parameterized with two different methods: Gaussian shape and B-splines. At first a Gaussian like function was applied

$$Y_{Gaussian}(X) = \begin{cases} h\, e^{-\frac{X^2}{2w^2}} + t_t, & h\, e^{-\frac{X^2}{2w^2}} + t_t < 0 \\ 0, & h\, e^{-\frac{X^2}{2w^2}} + t_t \geq 0 \end{cases}. \tag{24}$$

The gaussian parametrization has the advantage, that it only utilizes 3 model parameters (height h, the offset t_t and the standard deviation or width w) to describe the shape. The offset t_t can have negative values, which may result in negative values for $Y_{Gaussian}(X)$. To ensure that all shape values remain non-negative, any negative values of $Y_{Gaussian}(X)$ are set to zero. For the parameterization, the peak of the Gaussian can be sufficient.

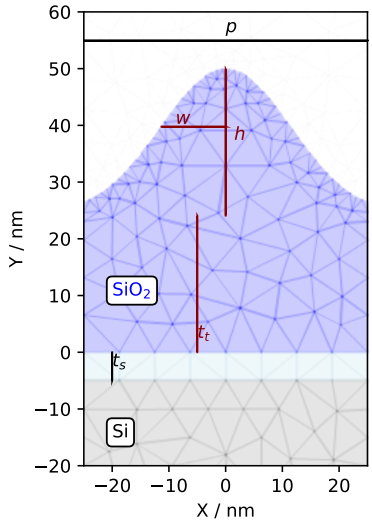


Figure 4.17 A possible parametrization of a shape with a Gaussian like profile. The width w of the shape is the standard deviation of the gauss function. This is scaled with the height h and an offset t_t was added as a model parameter. In parameters in black (the pitch p and the thickness of the oxide layer of the substrate t_s) are not modeled during the optimization.

In Figure 4.17 the model parameter for the gaussian parameterization (red), the constant parameter (black) and the resulting shape are shown. The outline from the TEM images of the HSQ samples (see Figure 4.16) set can now be modeled by using equation 24. Here the global optimizer DE is used to minimize the $\chi^2 = \sum \frac{\left(Y_{TEM}(X) - Y_{Gaussian}(X)\right)^2}{\sigma^2_{TEM}(X)}$ with the mean shape $Y_{TEM}(X)$ and the standard deviation $\sigma_{TEM}(X)$ from the TEM images. For all samples of the HSQ sample set the process is repeated. Here, a global optimizer was used because the boundaries of the model parameters were chosen to cover a huge search space to find the best model parameters. With the chosen boundaries a local optimizer can get stuck in a local minimum resulting in inexact model parameters for the shape.

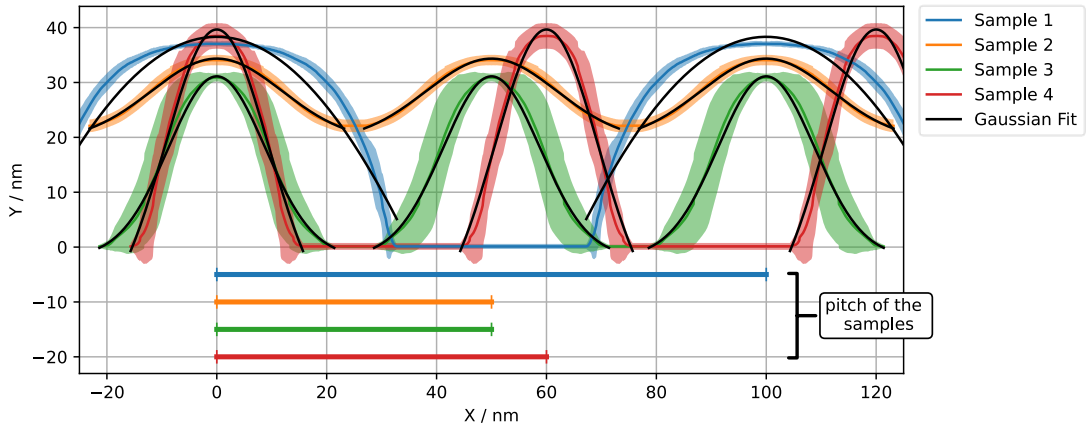


Figure 4.18 Results of the modeled outlines from the TEM image of the HSQ sample set (blue, orange, green, and red lines) with the gaussian like parameterization (black lines).

In Figure 4.18 the result of the minimization can be seen. For all samples except Sample 1 a good parametrization of the shape with the gaussian like function was obtained. For Sample 1 a difference between the fit result and the shape from the TEM image can be seen. I was not able to describe this shape with a gaussian like function. Using the Gaussian profile has the advantage of having a small number of model parameters, which reduces the computational effort required for the reconstruction process. Employing a parameterization method that cannot describe all samples in the set can serve as a test for the shape sensitivity of the method. If the Gaussian parameterization is used to reconstruct the shape from the fluorescence measurements, the reconstruction results for Sample 1 should exhibit a significant difference from the experimental data.

Therefore, another way to parameterize this shape was investigated. For this type of smooth curve, B-splines can be used not only for interpolation, but also as a parameterization method [128]. This is a kind of model-free parametrization because B-splines do not have standard model parameters such as height or width. B-splines are described by control points, and each control point has an x and y value. For this parameterization, a distance Δx is modeled between the different points. If all points were inside the unit cell (computational domain), the outermost point was moved to the edge of the unit cell x_0 to ensure that reasonable structures were obtained. The height h was modeled without constraints. The value of the other control points p_{0-4} was fitted as a percentage c_{0-4} of the h and an offset h_o to allow the points to be hidden in the substrate. In addition, the points were not allowed to be more than 1% higher than the point closer to the center:

$$p_i = (h - h_o)c_i + h_o. \tag{25}$$

From this follows the vector of x position of the control points

$$\overrightarrow{cp_x} = \begin{cases} \Delta x * [-5,-4,-3,-2,-1,0,1,2,3,4,5], & 5\Delta x \geq \frac{p}{2} \\ \left[-\frac{p}{2}, -4\Delta x, -3\Delta x, -2\Delta x, -\Delta x, 0, \Delta x, 2\Delta x, 3\Delta x, 4\Delta x, \frac{p}{2}\right], & 5\Delta x < \frac{p}{2} \end{cases} \tag{26}$$

and for the y positions $\overrightarrow{cp_y} = [p_0, p_1, p_2, p_3, p_4, h, p_4, p_3, p_2, p_1, p_0]$.

From $\overrightarrow{cp_x}$ and $\overrightarrow{cp_y}$ the spline representation of the shape is computed by the scipy function $splrep$ [129] based on the algorithms from P. Dierckx [130]–[132]. With the spline presentation the shape can be calculated for any point X in the computational domain defined by the pitch p with the scipy function $splev$

$$Y_{B-spline}(X) = splev(X, tck) \tag{27}$$

with $tck = splrep(\overrightarrow{cp_x}, \overrightarrow{cp_y})$. The pitch p and the thickness of the oxide layer t_s of the substrate were kept constant during the optimization. The structure is assumed to be symmetric, so a total of 7 model parameters are needed for this parameterization.

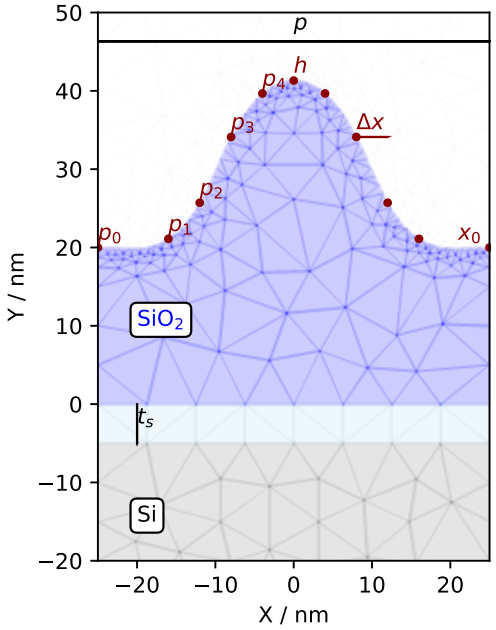


Figure 4.19 A possible parametrization of a shape with a B-splines. Here in red the modeled parameters: Δx is the distance between the different control points, x_0 is the x-value of the outer most control points, and p_{0-4}, h describe the height of each control point. In black the pitch p and the thickness of the oxide layer of the substrate t_s which are not optimized.

In Figure 4.19 the parameterization of the shape with B-Splines is shown. With B-Splines different shapes between a layer with different height and smooth and differentiable curves in between can be obtained. The outline from the TEM images of the HSQ samples (see Figure 4.16) set can now be modeled by using equation 27. Here the global optimizer DE is used to minimize the $\chi^2 = \sum \frac{\left(Y_{TEM}(X) - Y_{B-spline}(X)\right)^2}{\sigma^2_{TEM}(X)}$ with the mean shape $Y_{TEM}(X)$ and the standard deviation $\sigma_{TEM}(X)$ from the TEM images. The boundaries allowed the control points to have negative values and be positioned outside the expected shape from the TEM. The shape $Y_{B-spline}(X)$ was only evaluated at the X values from the TEM shape, where $Y_{TEM}(X) > 0$. Again, this kind of parametrization was used to model the TEM images for all 4 samples.

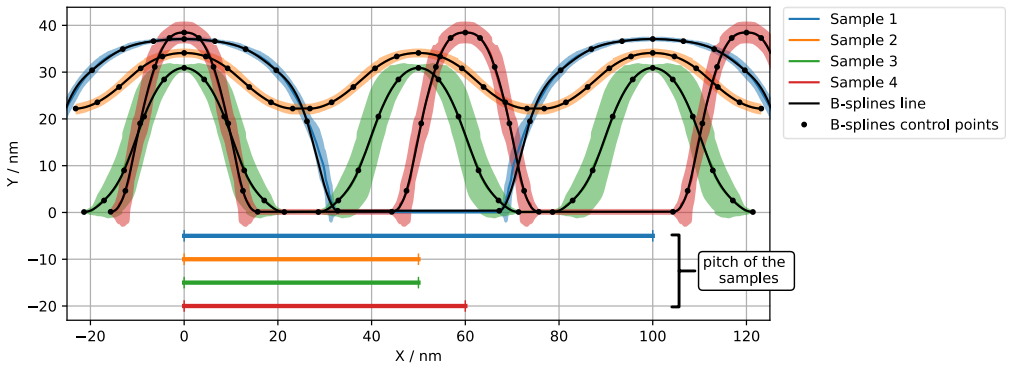


Figure 4.20 Results of the modeled outlines from the TEM image (blue, orange, green, and red lines) with the B-splines (black lines) and control points (black circle).

For all samples from the HSQ sample set, a close to perfect description of outline from the TEM can be seen (Figure 4.20). For all samples, certain control points have been shifted outside the shape, where their influence on the calculated shape is minimal or non-existent. This suggests that the total number of control points can be reduced without compromising the results. With B-splines, the total number of model parameters can be adjusted to fit the shape. This demonstrates the flexibility of the B-spline parameterization.

Table 4 Comparison of the Gaussian profile and the B-spline parameterization in terms of their advantages and disadvantages

	Gaussian profile	**B-splines**
Number of model parameters	3	≥ 3 depending on the number of control points
Advantages	Small number of model parameters	High flexibility to describe different shapes
Disadvantages	Only applicable to Gaussian like shapes	Can increase the total computational effort

Table 4 summarizes the advantages and disadvantages of the Gaussian profile and the B-splines parameterization. The Gaussian profile typically requires a smaller number of model parameters, leading to reduced computational effort during the reconstruction process. The Gaussian profile may not be able to capture complex or non-Gaussian shapes accurately (see Figure 4.18 Sample 1), potentially resulting in less accurate reconstructions. Compared to the Gaussian profile, the B-spline parameterization may require a larger number of model parameters, leading to increased computational effort during the reconstruction process. Nevertheless, B-splines offer greater flexibility in representing a wide range of shapes. With the B-splines the shape of all samples was successfully described (see Figure 4.20). The influence of the parameterization on the reconstruction is investigated in detail in Section 4.3.2.4.

4.3.2 Optimization with a global optimization method

In this section, the optimization process and the reconstruction results are investigated. First, different optimization methods (BO and PSO) are compared in terms of optimization time and obtained optimum. In the next section, a virtual experiment is performed to analyze the effect of including the fluorescence of more than one element on the confidence intervals of the results. The reconstruction is then applied to experimental data from the Si_3N_4 sample and the HSQ sample set. For the HSQ sample set, the different parameterization methods and their influence on the reconstruction are compared.

4.3.2.1 Deciding on the ideal optimization approach

To solve the inverse problem of finding the shape that describes the experimental data, different optimization methods can be used. Here, the two global optimizers PSO and BO are analyzed. Details of the optimization methods can be found in Appendix A. This section is based on the publication [133].

Table 5 Overview of the two global methods: PSO and BO

	PSO	**BO**
Type	stochastic global optimization method	Global optimization method based on Bayesian Inference
Description	The particles of the swarm are moving randomly through the predefined parameter space. The velocity of each particle is guided by the particle's best seen position as well as the swarm's best seen position.	BO uses a stochastic model (here Gaussian process (GP)) to update the prior distribution of the objective based on function evaluations. GP is commonly used to estimate the objective value without a parameterized function. The next sampling point is chosen using the expected improvement (EI) as an acquisition function, balancing global and local search for the optimum.

Table 5 gives an overview of the two optimizers compared in this section. Since BO takes all previous function evaluations into account it can be more efficient than other heuristic global optimization strategies [38] and local optimization strategies [134]. For example, PSO computes the next sampling position solely based on the current position and velocity of one swarm member as well as based on the best seen position of the member and the swarm. In this comparison, the hyperparameters of the optimization methods were not tuned, even though that could have improved the optimization speed and results.

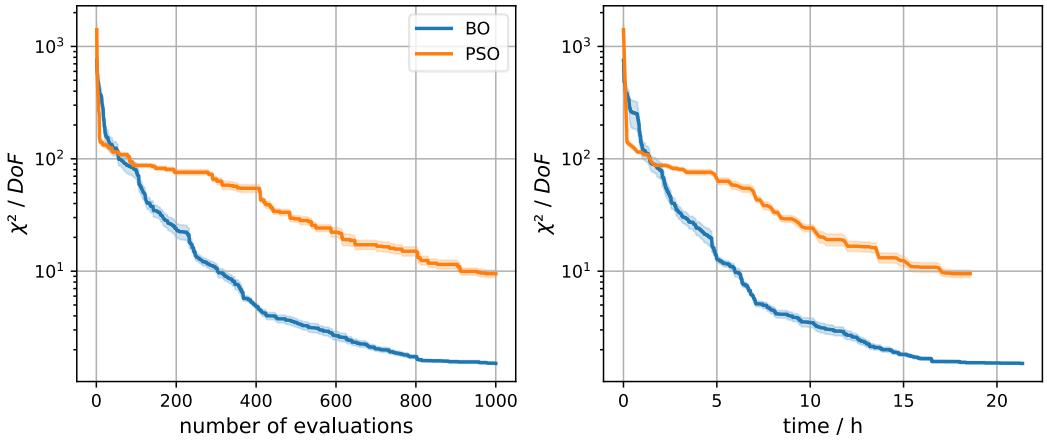


Figure 4.21 Here the minimal χ^2/DoF value during the optimization process as a function of the number of evaluations (left side) or computational time (right side) of the optimization averaged over six independent optimization runs of BO (blue) and PSO (orange), respectively. The shading indicates the uncertainty of the average.

Both methods try to minimize the χ^2/DoF by varying the geometrical parameters of the same model. For the benchmark, the optimization was repeated 6 times. The average values of the χ^2/DoF with $DoF = 318$ are shown in Figure 4.21. Clearly, BO requires significantly fewer function evaluations than PSO to reach good results but needs more time to calculate all the steps. During the 1000 evaluation the PSO was not able to reach the same minimum found by the BO, which indicates that the PSO was not able to find the global minimum in the given number of steps. Tuning the PSO hyper parameters, like start velocities and weighting of the global best and local best solutions, could give better results. Nevertheless, with the BO outperforming the PSO, the standard optimization method for the reconstruction was chosen to be the BO.

4.3.2.2 Virtual Experiment

To assess the performance of Bayesian optimization (BO) results independent of unaccounted model errors such as contamination or experimental error, an artificial dataset was created for two distinct incidence energies: one before the oxygen edge and one after. This dataset allows testing the advantages of incorporating fluorescence signals from multiple materials (silicon nitride and silicon oxide). This is part of the publication [58]. In previous work [29], measurement data at $E = 520$ eV were analyzed and an indirect sensitivity to the surface oxide layer was found, although no fluorescence signal originating from it was used for the reconstruction. The indirect sensitivity was only due to the attenuation behavior of the oxide layer, and thus I expect an increased sensitivity if a direct signal originating from the oxide layer is also used. For this purpose, I generated artificial experimental data by calculating model curves using the parameterization and optical constants of the Si_3N_4 sample, resulting in a structure made of Si_3N_4 covered by a SiO_2 layer. The model curves were calculated for the two incident photon energies $E = 680$ eV and $E = 520$ eV. To mimic the experimental noise, a Gaussian noise with a width of 3% is applied Table 6. By reconstructing the artificial datasets, I now analyze the benefit of the increased incident photon energy

on the reconstruction results and their confidence intervals without the influence of any experimental error contributions.

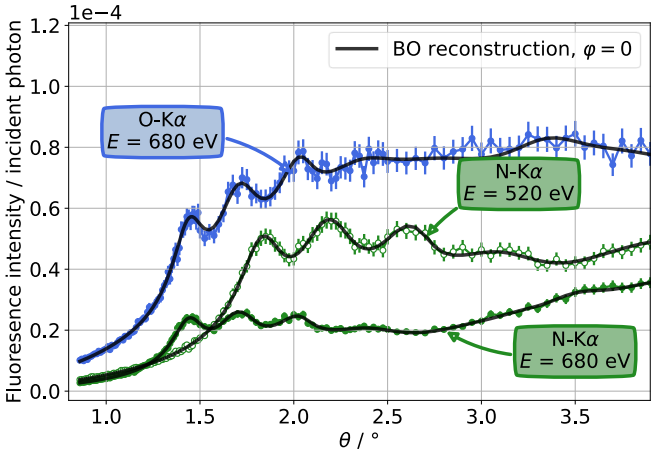


Figure 4.22 Comparison of the expected artificial disturbed simulated N-Kα (green points) or O-Kα (blue points) fluorescence intensities for the different excitation energies and the corresponding BO reconstruction (black lines).

In Figure 4.22 the corresponding artificial GIXRF curves and reconstruction results are displayed. Clearly visible is how the first peak in the nitrogen signal shifts with increasing energy to smaller incident angles due to the decrease in wavelength of the incident beam. They match the artificial data well for both photon energies and characteristic features, e.g., the first local maxima are very well retrieved. From the BO reconstruction, I can calculate the confidence intervals (equation 17) of the reconstructed parameters and, as shown in Table 6, they agree well with the initial parameters within the derived confidence interval.

Table 6 The values of the geometrical parameters of the synthetic data from the GIXRF BO reconstruction with the model parameter confidence intervals for $E = 520$ eV (Configuration A, only the nitrogen signal is modeled) and $E = 680$ eV (Configuration B, both nitrogen and oxygen signals are modeled). The pitch was set to $p = 50$ nm. The model parameters are the height h, width w, sidewall angle swa, oxide layer in the groove t_g, oxide layer on the grating line t_t, and modeling error for nitrogen ϵ_N and oxygen ϵ_O.

Parameter Name	Initial Value	Config. A	Config. B	Conf. Interval Ration B/A
h/nm	90	89.4(6)	90.3(4)	0.67
w/nm	25	24.87(7)	25.08(6)	0.85
$swa/°$	88	88.1(1)	87.9(1)	1.0
t_t/nm	3	3.13(6)	3.02(5)	0.83
t_g/nm	5	5.3(6)	5.0(1)	0.17
ϵ_N	1	1.02(1)	1.00(1)	1
ϵ_O	1		0.99(1)	

From a comparison of the determined confidence intervals for the two different configurations (Configuration A, $E = 520$ eV, only the nitrogen signal is modeled and Configuration B, $E = 680$ eV, both nitrogen and oxygen signals are modeled, see Table 6), the positive influence of also considering the oxygen fluorescence signal is obvious. Especially for the height and the groove oxide thickness, the achieved confidence intervals are significantly smaller for configuration B. Nevertheless, also for configuration A, the angle-dependent nitrogen fluorescence contains all relevant information about the dimensional properties of the nanostructure and even the surface oxide layer, as already pointed out in an earlier work [29]. This indicates a high sensitivity to surface layers, even if no direct fluorescence signal is measured or calculated from these layers.

4.3.2.3 Reconstruction of the Si_3N_4 grating from experimental data

Using the same model and methodologies, the real experimental data, measured according to configuration B, were modeled. This was published in [58] but here an additional model parameter, the height of the nitride layer in the groove h_o was added, because it is possible that the structure was not fully etched to the substrate. The results of the reconstruction can be found in Table 7 and Figure 4.23. Again, the confidence intervals of the structure parameters derived from the BO (equation 17) are like those determined in the virtual experiment.

Table 7 The values of the geometrical parameters of the nanostructures from the GIXRF BO reconstruction with the model parameter confidence intervals (one sigma). The parameters are height h, height of the nitride in the groove h_o, width w, sidewall angle swa, oxide layer in the groove t_g, oxide layer on the grating line t_t, modeling error for nitrogen ϵ_N and oxygen ϵ_O.

Parameter Name	**Reconstructed Value**
$\boldsymbol{h/\mathrm{nm}}$	97.8(6)
$\boldsymbol{w/\mathrm{nm}}$	49.79(8)
$\boldsymbol{swa/°}$	83.5(1)
$\boldsymbol{t_t/\mathrm{nm}}$	2.65(2)
$\boldsymbol{h_o/\mathrm{nm}}$	0.000(3)
$\boldsymbol{t_g/\mathrm{nm}}$	5.9(1)
$\boldsymbol{\epsilon_N}$	0.926(7)
$\boldsymbol{\epsilon_O}$	0.891(6)

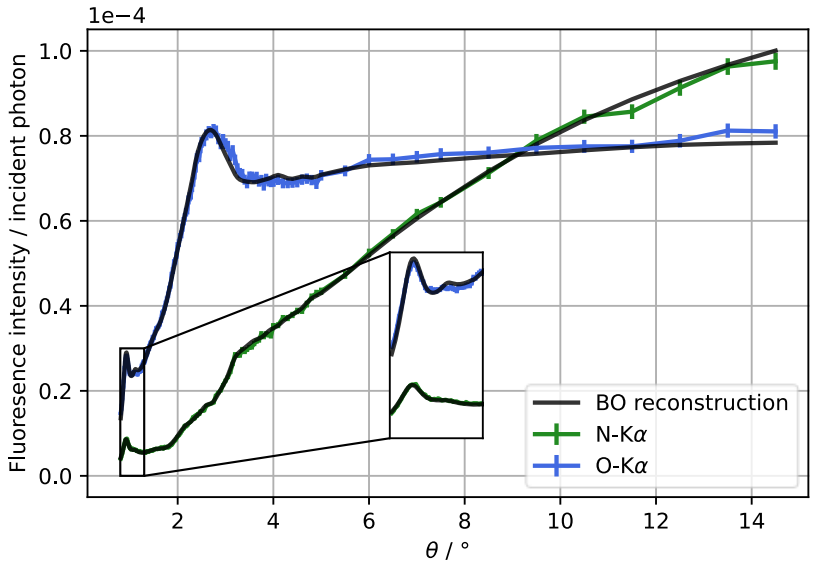


Figure 4.23 Comparison of the experimental N-Kα (green) or O-Kα (blue points) fluorescence intensities for $\varphi = 0°$ to the reconstructions from the Bayesian optimization (black lines).

Like the result from the virtual experiment, a higher sensitivity for the line width is observed as compared to the height. The line height of the nanostructure has a smaller influence on the XSW field and therefore the X-ray fluorescence compared to the line width. Thus, the nitrogen fluorescence radiation from the bottom of the grating line does not contribute significantly to the overall observed signal. Nevertheless,

all calculated confidence intervals are in the sub-nm regime and match the result from the virtual experiment.

It should also be noted that the two modeling error parameters ϵ_N and ϵ_O deviate from 1 within a range of 10%. There are multiple reasons for the deviation, which can all contribute to these modeling errors. They are in the same magnitude that one would expect the relative uncertainty of the employed fluorescence production cross-sections (product of fluorescence yield and photo ionization cross section) to be in. In addition, the modeling errors can correct for density variations in the nanostructure, because the density is estimated from the optical constants with a certain uncertainty. The parameters ϵ_N and ϵ_O may indicate a model error: roughness or contamination is present on the sample but not included in the model for simulation. The influence of such model imperfections is discussed in detail in Section 4.5.

4.3.2.4 Influence of different parameterizations on the reconstruction

This section focuses on the reconstruction of sample shapes from the HSQ sample set. The impact of parameterization on the reconstruction process and its outcomes is discussed extensively in this section. The parameterization of the sample can significantly affect the convergence of the global optimization method, which relies on the total number of model parameters and the shape of the minimum, as well as the overall space of the χ^2 function. All these aspects are influenced by the chosen parameterization of the model. In Section 4.3.1, two different approaches for parameterizing the HSQ sample set, namely B-splines and Gaussian profile, were applied to the shape extracted from TEM images. The results showed that the Gaussian profile may not be sufficient to describe all the samples. However, the Gaussian profile has significantly fewer model parameters compared to the B-splines (3 model parameters compared to 7 model parameters). In addition, the modeling error ϵ_O was added as a model parameter for both parameterizations. For the Gaussian profile, the optimization method A (see Figure 2.3) was adapted and the modeling error parameter ϵ_O was optimized along with the other model parameters. Here, the optimization method employed was BO.

To address the issue of slow convergence associated with B-spline parameterization, the reconstruction process utilized optimization method B (see Figure 2.4) with BO as the outer optimization method. This method effectively reduces the number of outer model parameters. For each shape, an optimal modeling error ϵ_O exists, which can be obtained through scalar minimization (inner optimization). This factor allows for scaling the calculated fluorescence intensity to the measured values. Incorporating this approach enhances the efficiency of the reconstruction process, as the computational effort required for the inner minimization is significantly lower compared to calculating similar shapes twice with different ϵ_O, as would be the case with optimization method A. The combination of inner optimization and the inclusion of constraints, such as ensuring the highest value is near the center of the shape, contribute to a stable convergence in the reconstruction process.

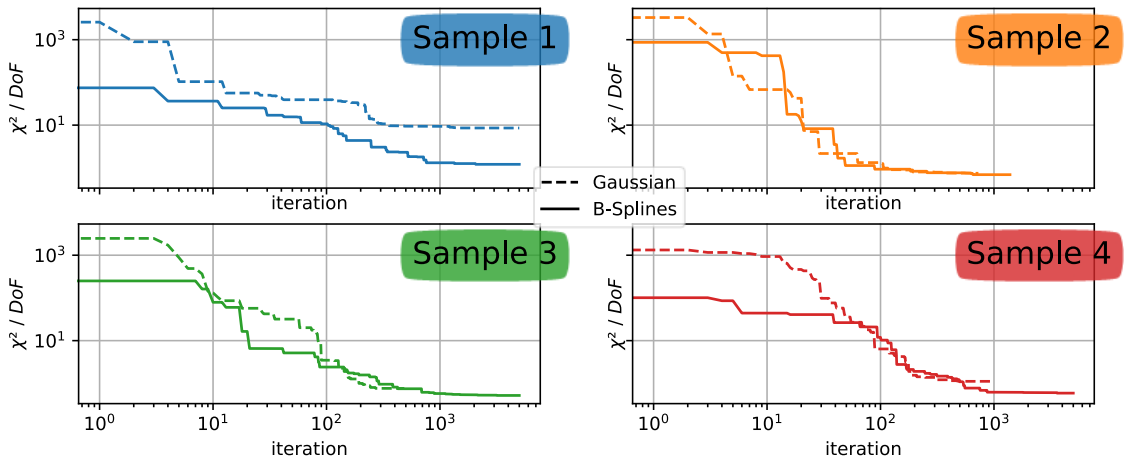


Figure 4.24 Comparison of the cumulative minimum of the χ^2/DoF of the BO with the Gaussian (dotted lines) versus the B-splines parameterization (solid lines) for all HSQ samples.

Figure 4.24 illustrates the comparison of the cumulative minimum χ^2/DoF achieved by the Bayesian Optimization (BO) method using Gaussian and B-spline parameterizations. Two stopping criteria were employed for the optimization process: 1) 5000 iterations to ensure reasonable computational effort and time, or 2) the condition where the probability of improvement (PI) for the last five function evaluations falls below 2e-12. The PI provides insight into the likelihood of obtaining a better solution (e.g., a smaller χ^2) with the next sampling point. Except for Sample 1, the second stopping criterion was met in fewer than 1000 steps for the Gaussian profile. Sample 2 also satisfied this stopping criterion when using the B-spline parameterization. For all other optimizations, the first stopping criterion was employed.

Table 8 Comparison of the stopping criteria and the χ^2/DoF with DoF the degree of freedom of all samples from the HSQ samples for the Gaussian profile and B-splines parameterizations.

	Sample 1		**Sample 2**		**Sample 3**		**Sample 4**	
	Gaussian	B-spline	Gaussian	B-spline	Gaussian	B-spline	Gaussian	B-spline
Stopping criterion	5000 iterations	5000 iterations	PI < 2e-12	PI < 2e-12	PI < 2e-12	5000 iterations	PI < 2e-12	5000 iterations
$\boldsymbol{\chi^2/DoF}$	8.54	1.23	0.75	0.69	0.76	0.54	1.10	0.62
$\boldsymbol{DoF}$	112	108	112	108	217	213	217	213

Table 8 provides a summary of the reconstruction results, including the achieved minimal χ^2/DoF and the corresponding stopping criteria. The B-spline approximation yields minimum χ^2/DoF values around 1 for all samples, indicating a reasonable fit. When χ^2/DoF is close to one, the calculated fluorescence adequately represents the variability observed in the experimental data. For most samples (excluding Sample 1), the χ^2/DoF values are below one, suggesting that the experimental error is larger than expected based on the difference between the results and the measurements. This indicates that the model

may capture additional variability beyond the noise in the data. In the case of Gaussian shape, the minimum χ^2/DoF values were higher for all samples compared to the B-spline approximation. The difference in χ^2/DoF was most significant for Sample 1 and least pronounced for Sample 2. However, apart from Sample 1, the minimum χ^2/DoF values for all samples using the Gaussian shape remain within an acceptable range, close to one.

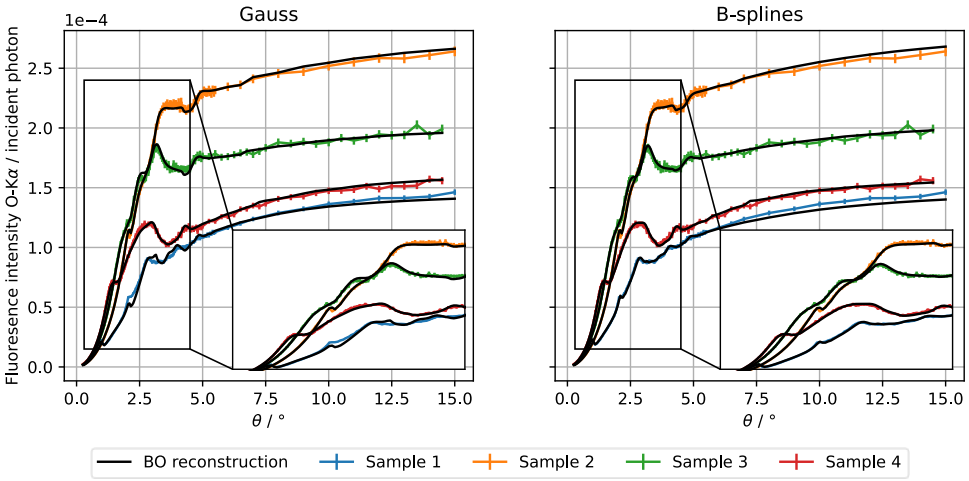


Figure 4.25 Here, the results of the reconstruction of the angle-dependent fluorescence intensities of all samples are shown for the Gaussian parameterization (left) and the B-spline parameterization (right). The black lines are the reconstruction result, and the colored line is the experimental data for the different samples.

Figure 4.25 shows the results for the different reconstructions with the different parameterizations. On the left are the results for the Gaussian profile and on the right are the results for the B-spline approximation. For the B-splines, the result of the optimization describes the experimental data within the measurement error for all four samples. For the Gaussian shape a good agreement is observed, except for Sample 1. This corresponds to the evaluation of the minimum of χ^2/DoF (see Table 8). The Gaussian profile is unable to accurately describe the shape of Sample 1, as previously observed in the TEM profiles. Specifically, when applying the Gaussian profile to Sample 1 in the angular range of 2.2° to 5°, a noticeable discrepancy between the experimental data and the reconstruction becomes apparent. While the position of the FLM aligns well, the intensity of the reconstructed shapes falls below the corresponding experimental data at the second local maximum. Moreover, between 2.5° and 5°, the reconstruction exhibits additional features not visible in the experimental data. This deviation underlines the method's sensitivity to the shape of nanostructures. If the chosen parameterization fails to accurately represent the shape, discrepancies between the calculated and measured fluorescence intensity arise. For all other samples, the BO reconstruction results show all features that are present in the experimental data.

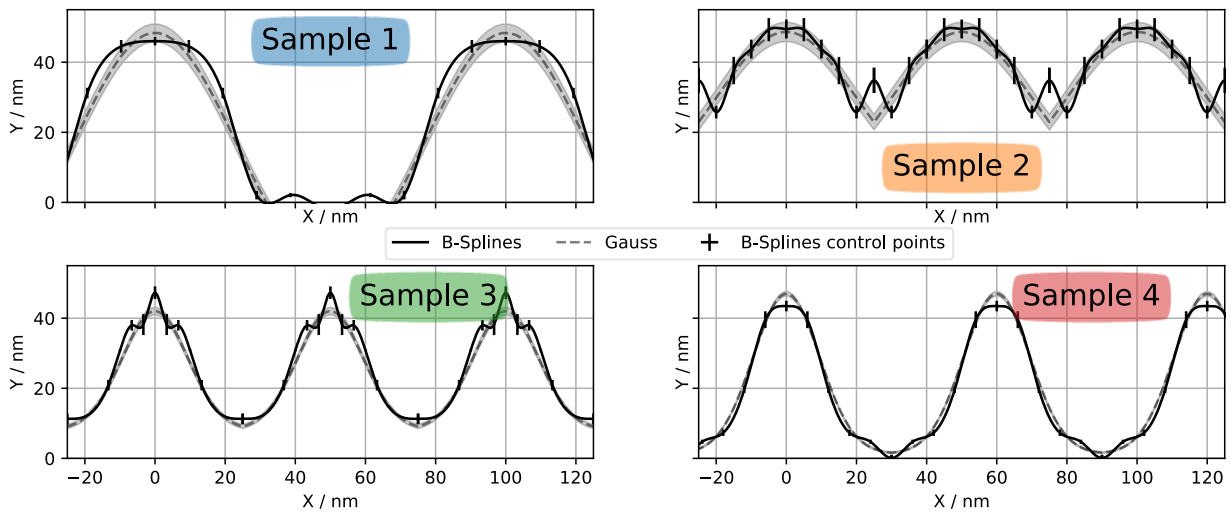


Figure 4.26 Comparison between the reconstruction results of the B-splines parameterization (black line) and the Gaussian parameterization (grey dashed lines) with confidence intervals for the HSQ sample set.

To better interpret the result, the different shapes are plotted in Figure 4.26. The overall shape is similar for all four samples. This is mainly seen in the width of the structures and the area under the line. There are some differences in the height of the structures (e.g., Sample 3 and Sample 4). In Sample 1, the shape differs in the upper part of the shape. The Gaussian profile is unable to describe the flattened top of the structure. For Sample 2 and Sample 3, a sharp feature (width less than 10 nm) can be observed with the B-spline parameterization. These features are unrealistic because similar features are not visible in the TEM images (Section 3.2.1). During fabrication, HSQ is spin-coated on the substrate and then exposed with an E-beam at different doses. This determines the possible height and width of the structure. During fabrication, the E-beam setting was chosen for a target grating width of 20 to 25 nm. During fabrication, each line is exposed to the e-beam once. This process makes a sharp feature unlikely because the achievable resolution of E-beam lithography depends on the chosen settings. The resolution is related to spot size, electron scattering, secondary electron range, resist development, and mechanical stability [135]. It is interesting to note that the width of the sharp features is related to the distance between two control points of the B-splines. These features are most likely artifacts of the parameterization. The high flexibility of the B-splines allows the optimizer to account for features in the angular fluorescence that can be noise or connected to an unaccounted model error. The real structure may have a deviation in the overall height and material in the groove that was probably not removed after E-beam lithography. This type of roughness is not added to the model.

Table 9 Comparison of the Gaussian profile and the B-spline parameterization in terms of their results of the reconstruction

	Gaussian profile	**B-splines**
Number of model parameters (excluding the modeling error)	3	7
Advantage	Stable and fast convergence	Results describes the experimental data very well
Disadvantage	Not able to describe all shapes	allows unrealistic features in the shape

Table 9 summarizes the advantages and disadvantages of both parameterizations. The flexibility of the B-spline allowed to include pointed features even if they are not realistic. This can be a disadvantage of this parameterization. The feature can result from the position of the control points of the B splines in the X direction. Here, the constraint that one control point must be in the center of the structure can lead to the observed artifacts. For some examples, reducing the total number of control points or including the X position of the control points as model parameter could improve the result. With the Gaussian, the measured fluorescence of Sample 2 and Sample 3 was described almost perfectly. The advantages of the Gaussian parametrization are the small number of model parameters and the realistic results. However, this parameterization was not flexible enough to describe the measured fluorescence of all samples.

In this section, it was shown that the parameterization has an influence on the results of the reconstruction process. For the flexible B-spline profile the parameterization needs to be given reasonable constraints and the total number of control points needs to be adapted to the sample making this parameterization more complex. Nevertheless, if the parameterization is not able to describe the shape, this can be seen in the results of the optimization. This underlines the high sensitivity of the method to the shape of the nanostructure.

4.3.3 Applying a surrogate

The previous section showed the positive effect of including the fluorescence of another material (e.g., oxygen) on the confidence intervals of the result. In addition, the parameterization of the model has an influence on the reconstruction results. To better understand the results, it may be interesting to investigate the posterior distribution of the model parameters. This can be done by MCMC simulation and allows to obtain correlations, multi modalities and a better estimation of the uncertainties of the reconstruction process. One of the drawbacks of MCMC is that it often requires more than a hundred thousand evaluations of the forward problem. This is not feasible for shape reconstruction with the finite element Maxwell solver due to the numerical effort. To solve this, the forward problem is replaced with a Polynomial Chaos Expansion (PCE) [62], [136] surrogate (see Section 2.2.2), which reduces the computational effort but introduces a small error. The PCE allows for a global sensitivity analysis of the model parameters with the

Sobol coefficients. For the training data set, the fluorescence is calculated for combinations of model parameters with a uniform distribution and specified bounds of the parameters. A percentage of the overall data set is used to test the accuracy of the surrogate.

For the Si_3N_4 sample, a data set of more than 11000 samples was generated. The boundaries of the different model parameters were set to the values from the BO reconstruction and the model parameters were uniformly distributed. The PCE was trained on 90% of the data set. The remaining 10% was used as a test data set to evaluate the error of the surrogate. For Si_3N_4 samples, two approximations are computed, one for the nitrogen and one for the oxygen fluorescence. Both approximations use the same set of model parameters. The error for the two surrogates is calculated using equation 12. The error for the nitrogen surrogate was found to be $error_{RMSE\ relative,N} = 0.8\%$ and for the oxygen surrogate $error_{RMSE\ relative,O} = 0.7\%$, which is a good approximation of the results from the FEM Maxwell solver. Additionally, the condition number of the Gram matrix (see Section 2.2.2) can be calculated, which should be close to 1 if the Gram matrix is regular and well-conditioned. The condition number is 4.7 for the nitrogen surrogate and 4.4 for the oxygen surrogate. The small $error_{RMSE\ relative}$ and condition number of the Gram matrix indicate that a good approximation of the forward calculation is being generated.

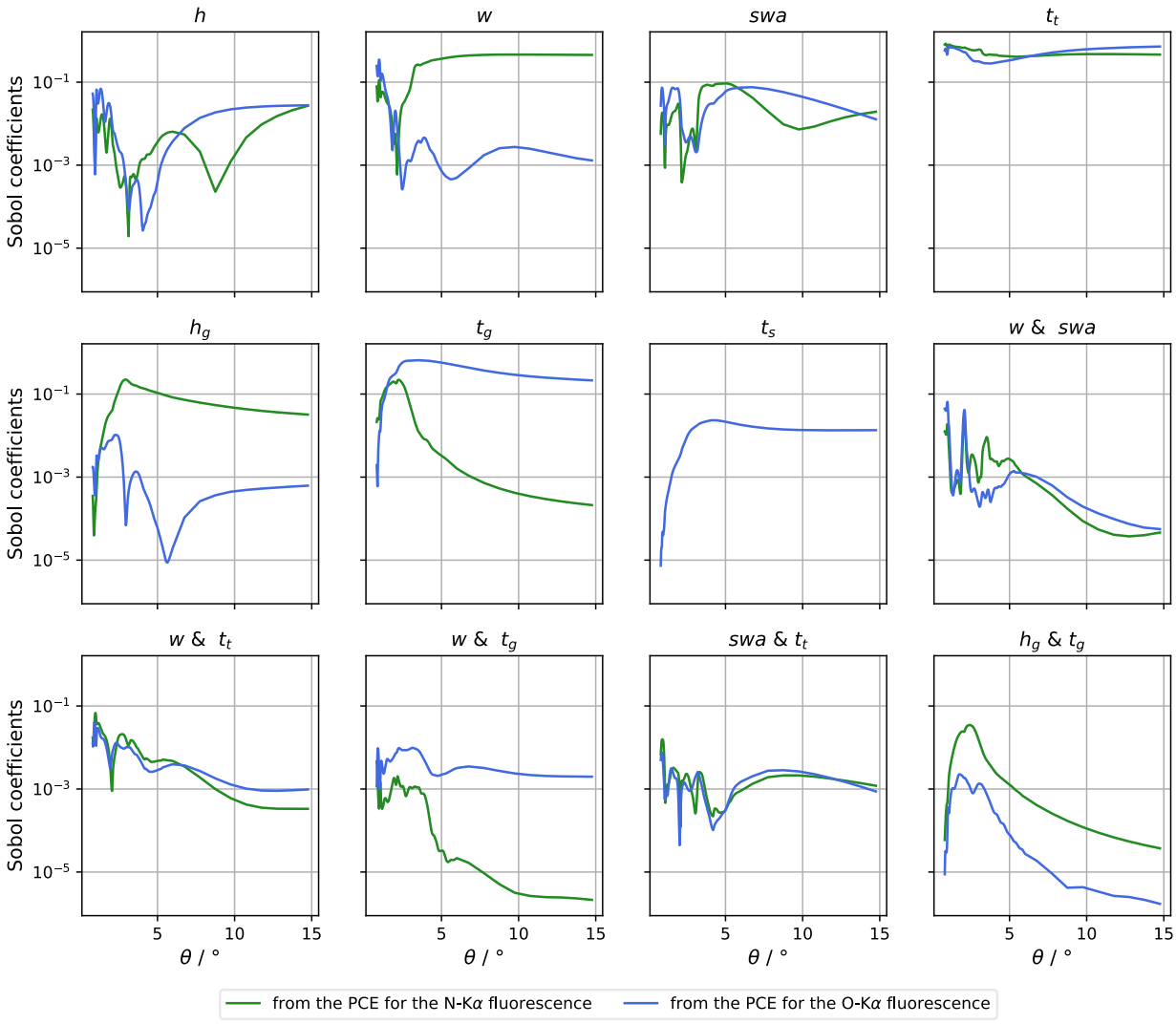


Figure 4.27 Sobol coefficients of the surrogate for the nitrogen (green) and oxygen (blue) fluorescence for the different grazing angle of incidence θ. The Sobol coefficients display the sensitivity of the surrogate to the different model parameters: line width w, line height h, side-wall angle swa, etch offset h_g, thickness of the oxide t_t, thickness of the oxide of the substrate t_s, and thickness of the oxide in the grove t_g.

The Sobol coefficients provide information about the sensitivity of the model parameters and their combinations. The higher the Sobol coefficient, the higher the sensitivity. They can be used to decide which model parameters to include in the optimization. Figure 4.27 shows the 12 Sobol coefficients with the highest values for the nitrogen (green) and oxygen (blue) fluorescence surrogates. The surrogates are most sensitive to the thickness of the top oxide layer t_t for all grazing angles of incidence. This underlines the high sensitivity of these measurements to surface effects. For grazing angles of incidence, the electric field intensity only penetrates into the surface of the structure. In addition, this parameter has an influence on the total measured fluorescence of the oxygen and on the shape of the groove of the structure.

Not surprisingly, the oxygen surrogate is more sensitive to the oxide layer in the groove t_g and the oxide layer of the substrate t_s than the nitrogen surrogate. For t_s the nitrogen surrogate is not sensitive at all. The oxide layer of the substrate is located below the structure made of silicon nitrite and contributes only

for higher angles of incidence (seen in the low sensitivity for small angles). Therefore, it has no influence on the E-field distribution inside the grating. This supports the decision to keep this parameter constant during optimization, since even the sensitivity of the oxygen surrogate to this parameter is low.

The sensitivity of the nitrogen surrogate is high for the line width w and the etch offset h_g for incidence angles above 3°. At these angles, the E-field starts to penetrate deeper into the structure starting at the top and in the bottom of the grooves. The line width w (with constant swa) determines the width of the upper part of the grating. If the structure is wider more fluorescence photons can be generated. For a higher etch offset h_g, more material is located in the groove of the structure, which may be fluorescent. For higher angle of incidence, the E-field intensity and therefore the fluorescence is stronger in the grooves (see Figure 4.12). This explains the high sensitivity to this model parameter compared to e.g., the side-wall-angle swa or the line height h.

In addition, there is a sensitivity to the combination of different parameters. Here the maximum of the Sobol indices is located below 5°. All the model parameter combinations have an influence on the shape of the groove, which influences the position of the first local maxima. Therefore, a high sensitivity to the parameter combination that determines whether the first mode is squeezed into the groove is expected. To sum up, the Sobol coefficients show a good sensitivity to all model parameters that are optimized and some Sobol indices for the nitrogen and oxygen surrogates complement each other. This supports the previous statement that it is useful to include oxygen fluorescence in the reconstruction to reduce the uncertainties of the reconstruction results.

With the surrogate, an MCMC simulation was performed to obtain the posterior distribution of the model parameters. The prior distribution for the MCMC was the same uniform distribution of model parameters used in the training data set. The MCMC simulation gave comparable results to the reconstruction with the BO (see Section 4.3.2.3) with small deviations. To evaluate the posterior distributions of the model parameters, a corner plot for all parameters is shown in Figure 4.28. No multi-modalities or strong correlations between the parameters are observed. All parameters are Gaussian-like distributed.

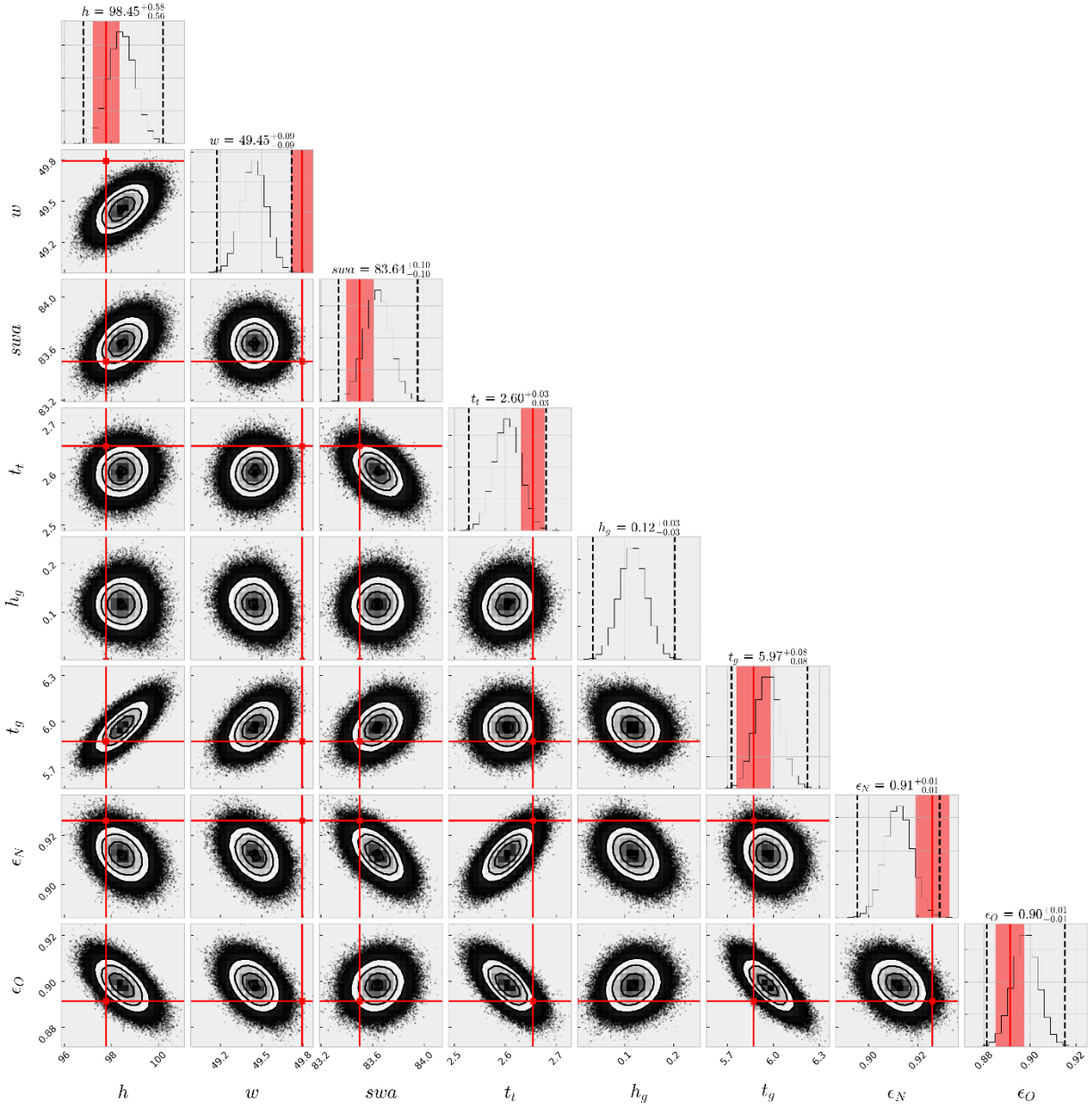


Figure 4.28 Corner plot of the MCMC chains for the different model parameters. The red dots are the results from the BO reconstruction with the corresponding confidence intervals. The rightmost plots show the 1D projection of the the model parameters. The model parameters are line width w, the line height h, side-wall angle swa, etch offset h_g, thickness of the oxide t_t, thickness of the oxide of the substrate t_s, thickness of the oxide in the grove t_g, modeling error ϵ_N and ϵ_O.

For comparison, the red dots show the results of the BO reconstruction. For all parameters except w and h_g, these results are in the 3σ regime of the posterior distribution of the MCMC. For the width of the nanostructure, the results differ more than expected. This can be explained by the error of the surrogates, which is only an approximation of the calculation with the FEM solver. This deviation to the FEM calculation is angle dependent and not included in the MCMC simulation. In combination with the angle-dependent sensitivity of the parameters, it can lead to the observed deviation between the results of the MCMC simulation and the BO reconstruction. There is a correlation between the height h and the thickness of

the oxide in the grove t_g. As t_g increases, so does the height. Therefore, the total depth of the groove remains the same. This indicates that the shape of the groove is significant for the measured fluorescence because it has an influence on the XSW field distribution inside the groove. The thickness of the oxide t_t is correlated with the modeling error ϵ_N for nitrogen fluorescence and anticorrelated with the modeling error ϵ_O for oxygen fluorescence. For nitrogen fluorescence, a thicker oxide layer attenuates the fluorescence. On the other hand, a thicker oxide layer increases the oxygen fluorescence. These effects can be counteracted with the corresponding modeling errors. Overall, a PCE has been successfully trained and applied to the reconstruction. The deviations to the results of the BO reconstruction need to be investigated further in future studies. A possible solution is to include the error of the surrogate or an error model[6] into the MCMC simulation. Nevertheless, a working surrogate allows for time-efficient reconstruction of similar samples or measurements.

To investigate the stability of the measurement, other measurements of the fluorescence from a different position on the Si_3N_4 sample were repeated 9 times on the sample with the same angular regime. The PCE surrogate was used to perform an MCMC simulation for all 9 measurements. From a MCMC, the model parameters and their uncertainties can be retrieved and compared for all measurements.

[6] The parameters describing the error model can be included as model parameters in the MCMC simulation and their posterior distribution can be calculated and analyzed.

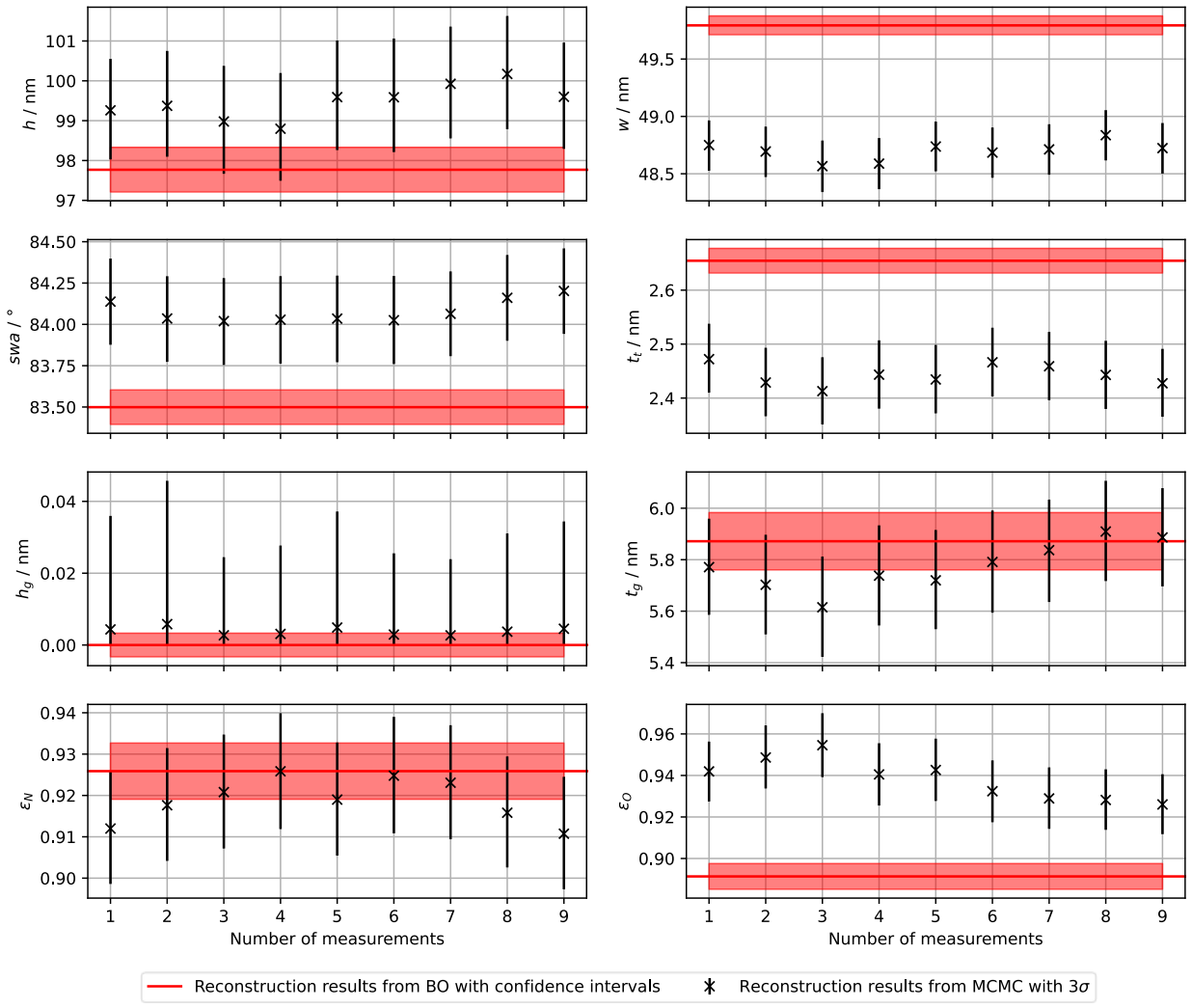


Figure 4.29 Comparison of the reconstruction results (different measurements and different position on the sample) from the BO (red) to the results from the MCMC for the repeated measurements (black) with the error bars indicating the 3σ standard deviation of the posterior distribution. The model parameters are line width w, the line height h, side-wall angle swa, etch offset h_g, thickness of the oxide t_t, thickness of the oxide in the grove t_g, modeling errors ϵ_N and ϵ_O.

Figure 4.29 shows all model parameters (black x with error bars) for the repeated measurement reconstructed with the PCE and the MCMC. In red, the results of the BO reconstruction (see Section 4.3.2.3 different measurement and different position on the sample) are shown. The results of the repeated measurements overlap with the 3σ standard deviation of their posterior distribution. This indicates that the error obtained from the MCMC simulation reflects the variation in the reconstruction results that can arise from the measurement and the measurement uncertainties. Comparing the results with the reconstructed values for the original measurements (red lines), a deviation is observed for the width w, the thickness of the oxide t_t, and the modeling error ϵ_O. For the other model parameters, there is an overlap with the BO reconstruction results for at least some of the reconstructed measurements. As discussed before, the error of the surrogate can have an influence on the reconstruction results. In addition, some time (about 2 years) elapsed between the original measurements analyzed with the BO and the repeated measurements. During this time, the Si_3N_4 sample was measured at various locations on measurement

setups. In addition, the sample was stored under ambient conditions. The storage and the measurements can cause a change and increase in the surface contamination layer. The influence of contamination is studied in detail in Section 4.5. The measured position on the grating was not the same, so inhomogeneities of the sample (e.g., in width) can explain the difference to the older measurement.

4.4 Comparison to alternative measurement methods

To validate the results, they can be compared with different methods that can provide information about the shape of the sample. For the Si_3N_4 sample, an SEM image was taken from a reference sample. For the HSQ sample, TEM images were taken on samples prepared with the same parameters. In addition, AFM measurements were performed on all samples. All the results are presented in Section 3.2.

Since AFM is a well-established method for mapping the surface topology of nanostructures with sensitivities down to the nm range, we can use an AFM to verify the GIXRF reconstructed total line heights (difference between the top surface and the groove surface) of the Si_3N_4 sample. The results are part of the publication [58]. From the AFM data, we determined a total line height of 98.4 nm. This agrees with the GIXRF reconstructions within the confidence intervals. It should be noted that the AFM measurement is only representative of a small area of the grating where it was performed, while the GIXRF result is averaged over a much larger area due to the elongated beam footprint. Due to the interplay between tip shape and nanostructure, other dimensional parameters such as line width or sidewall angle are not easily derived [137].

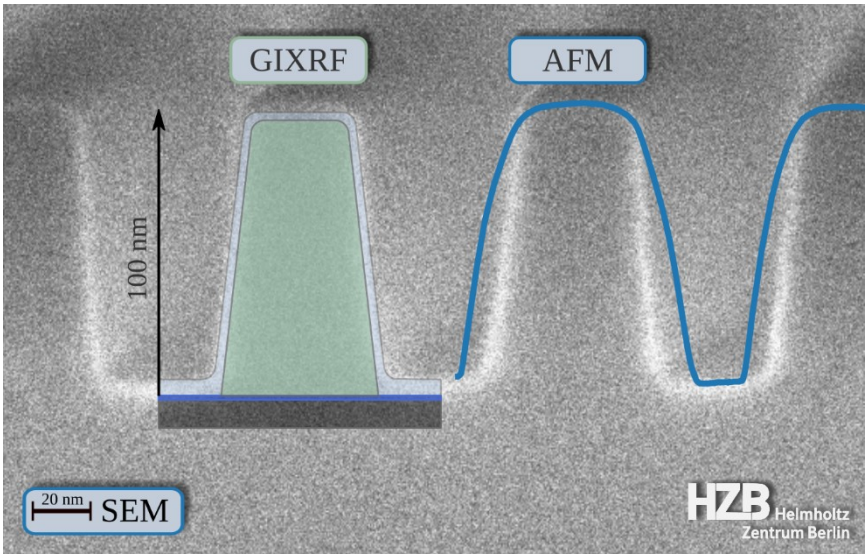


Figure 4.30 Comparison between the AFM measurement, the SEM image, and the reconstruction results of the Si_3N_4 sample. Image taken from [58]

In Figure 4.30, the GIXRF reconstructed shape of the sample and the AFM profile obtained by averaging two adjacent AFM line profiles are superimposed on the SEM cross section. Therefore, the GIXRF line profile and the AFM were scaled to the SEM image to match the line pitch while keeping the aspect ratios

constant. As far as the low contrast allows, the agreement in line height, sidewall angle and line width are quite good. The difference is greater than would be expected from the confidence intervals, but as noted before, the SEM image was taken from a witness sample. Therefore, an exact match is unlikely due to inaccuracies in the deposition of the Si_3N_4 layer and in the overall manufacturing process.

For the second HSQ sample set TEM images were taken from witness samples. To better verify the results, AFM measurements were performed on the same sample used for the GIXRF measurements. Details of the measurements are given in Section 3.2.

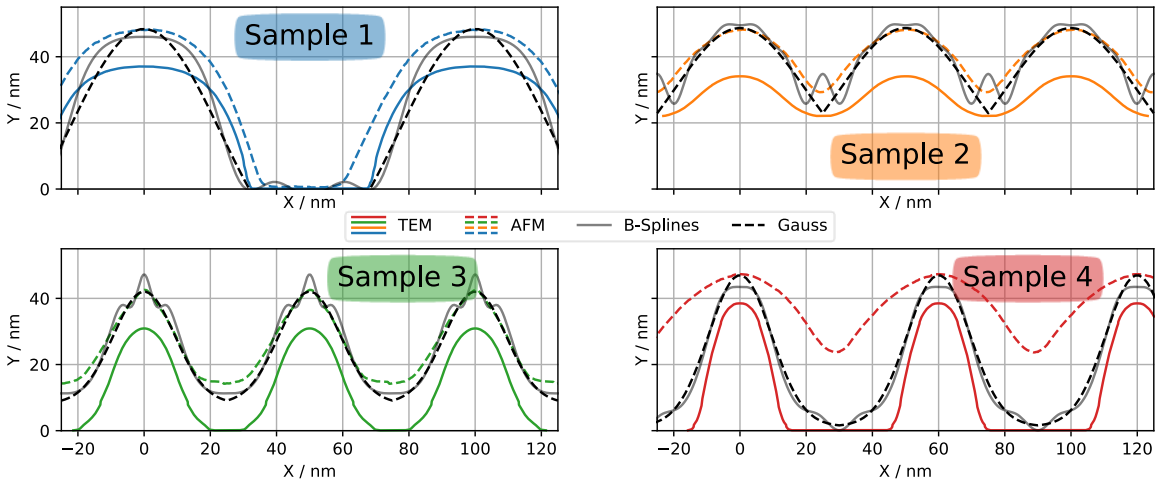


Figure 4.31 Comparison of the shape from the reconstruction with B-splines (grey line) and Gaussian parametrization (black dashed line) to the AFM (colored dashed lines) and TEM measurements (colored lines).

Figure 4.31 compares the results of the BO reconstruction and the Gaussian and B-splines parameterization with the AFM and TEM measurements. For most of the sample, the AFM measurements agree with the shape from the reconstruction. For structures of the same size as the AFM tip it is challenging to get the exact profile as mentioned above. Deviations in width and sidewall angle between the AFM and GIXRF measurements (e.g., Sample 1) can be explained by the tip. In addition, with a pitch size below 100 nm, it can be difficult to ensure that the tip has reached the bottom of the sample. Therefore, the height cannot be retrieved exactly (e.g., Sample 4). However, for Sample 2 and Sample 3, the AFM profile is consistent with the reconstruction results except for the point feature seen in the B-spline parameterized result. The AFM results indicate that the feature introduced by the B-spline parameterization, such as the top of Sample 3, is not real, as assumed and discussed previously (Section 4.3.2.4).

The deviation between the TEM images and the reconstruction results are smallest for Sample 3 and Sample 4, except for the offset in the Y direction, which was not seen in the TEM image. For Sample 1 and Sample 2 the deviation is significant. In addition, the AFM measurement does not fit the TEM image, indicating that the TEM image cannot be used for direct validation of the sample shape. For better validation, TEM images of the sample used for the GIXRF measurement could provide information, but this would require the samples to be destroyed, which was not done here. The samples were not produced in

the same run and the reproducible of samples produced with HSQ is temperature and ambient dependent [138]–[140].

4.5 The influence of model imperfection

As seen in the XRR measurements, the deviation between the numerical model and the actual physical structure can affect the reconstruction results if there is a sensitive of the measurement to this model imperfection.

The measurement is sensitive to very thin layers on the surface of the nanostructure for grazing angles of incidence. As an example, I will discuss the influence of contamination on the reconstruction of the shape of the nanostructure from the GIXRF measurements. This contamination is not accounted for in the model. Surprisingly, carbon K alpha fluorescence can always be observed in the measured spectra. Experimentally, it was found that carbon contamination can be caused by storage in air as well as by the measurement process itself. Here two samples are analyzed, a Si_3N_4 grating with a contamination line of unknown origin and one sample of the HSQ sample set. The Si_3N_4 sample had been measured with various measurement setups at different locations and was hence transported over long distances in an uncontrolled ambiance. For the sample of the HSQ sample set, the same GIXRF measurements were performed repeatedly over a 48 h period and included a scan along the grating lines to investigate the change in the fluorescence spectra during the measurements.

4.5.1 Influence of a contamination layer covering a part of the sample

By analyzing the Si_3N_4 sample with an optical microscope, I observed a discoloration in the center of the sample (see Figure 4.32). By scanning the sample (see Figure 4.33), I can associate the dark line with a significant increase in carbon fluorescence. Therefore, this sample offers the possibility to analyze the influence of a contamination on the reconstruction, because I was able to measure the contaminated and the uncontaminated part on the same sample during the same measurement time.

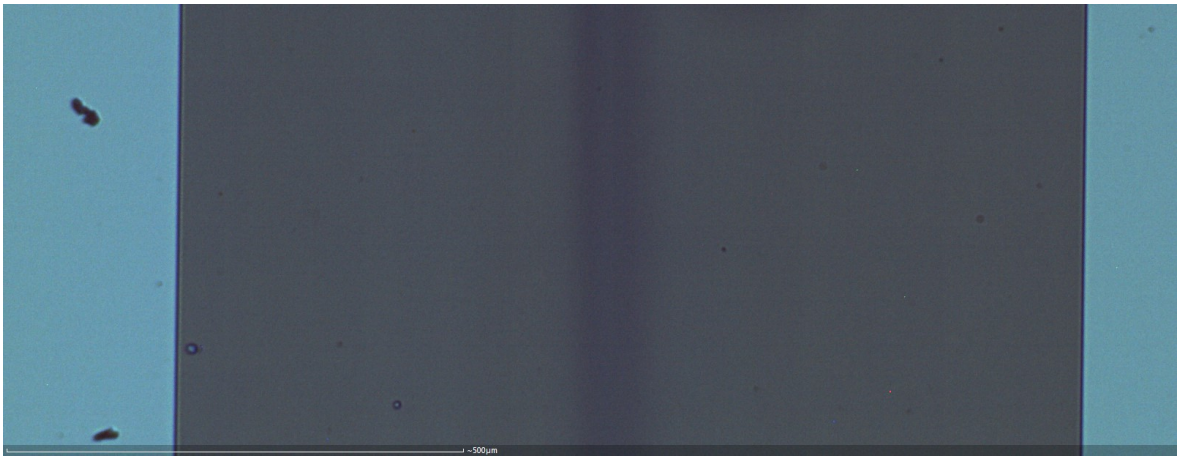


Figure 4.32 Microscope image of the sample with a dark area in the middle of the sample.

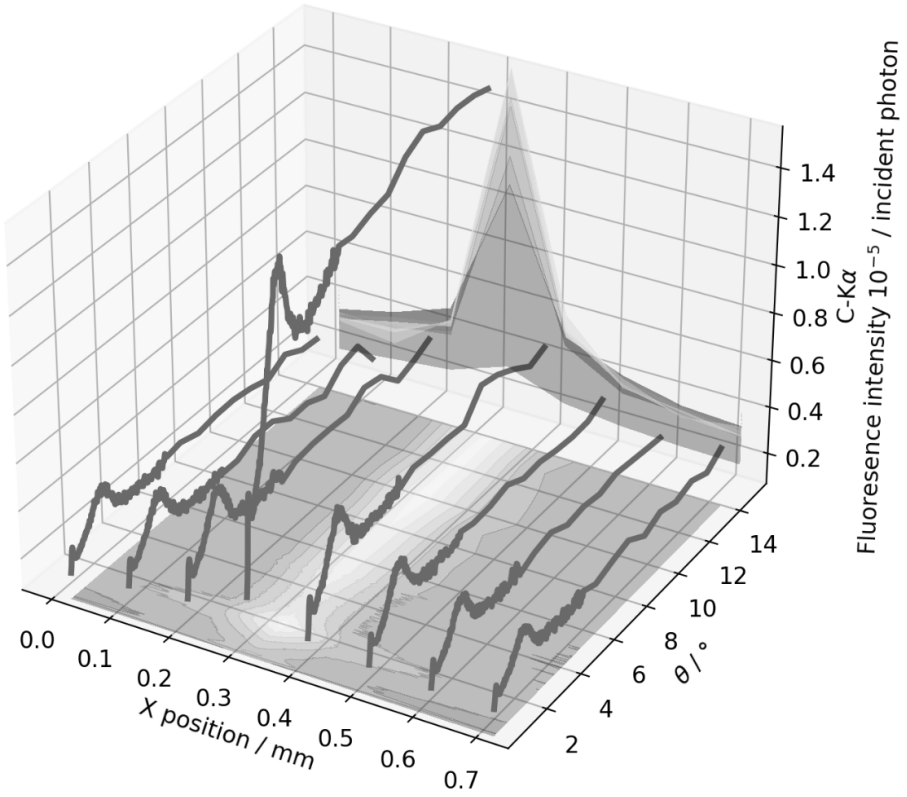


Figure 4.33 Fluorescence measurements perpendicular to the grating lines showing carbon fluorescence as a function of angle of incidence θ and X position on the sample for an incident energy of 680 eV.

The angle of incidence θ was varied between 0° and 15° at all positions on the sample with a constant incident energy of 680 eV. The recorded spectra were deconvoluted as described in Section 4.2. Figure 4.33 shows the carbon fluorescence intensity for the different angles and positions on the sample. A strong increase in carbon fluorescence in the center of the structure is observed for all angles of incidence.

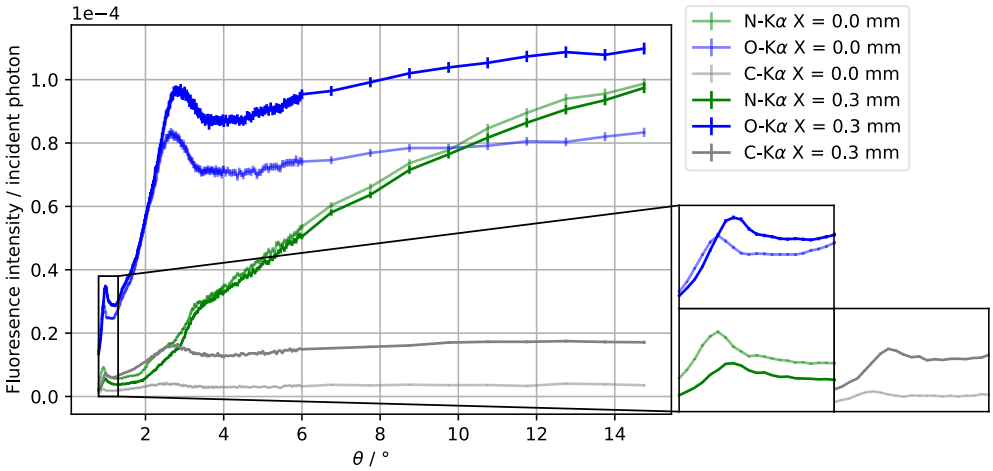


Figure 4.34 Comparison of the angular dependence of oxygen (blue), nitrogen (green) and carbon (gray) fluorescence between the reference measurement (more transparent, X = 0.0 mm) and the measurement of the contaminated line (opaque, X = 0.3 mm) with θ the angle of incidence. The box on the right shows the regime around the first local maxima for the different types of fluorescence.

Figure 4.34 shows the angular dependence for an azimuthal angle $\varphi = 0°$ for the nitrogen, oxygen, and carbon fluorescence for the contaminated part of the sample relative to a reference measurement on the uncontaminated part. The influence of the contamination is significant. It is interesting to note that the contamination seems to mainly consist of carbon and oxygen[7], as I observed an increase in the oxygen signal. The nitrogen signal is more attenuated at small angles of incidence, even below the carbon signal. This indicates that carbon is present closer to the surface of the structure. In addition, the position of the first local maxima of the nitrogen and oxygen fluorescence are shifted, because these peaks are related to the angle where the first mode of the XSW is inside the groves and with the contamination the structure appears thicker. Using the PCE surrogate introduced in Section 4.3.3, an MCMC simulation is run for each position on the sample. The same model (without contamination layer) and model parameters are used throughout.

[7] Probably hydrogen as well, which has not been measured in the spectra.

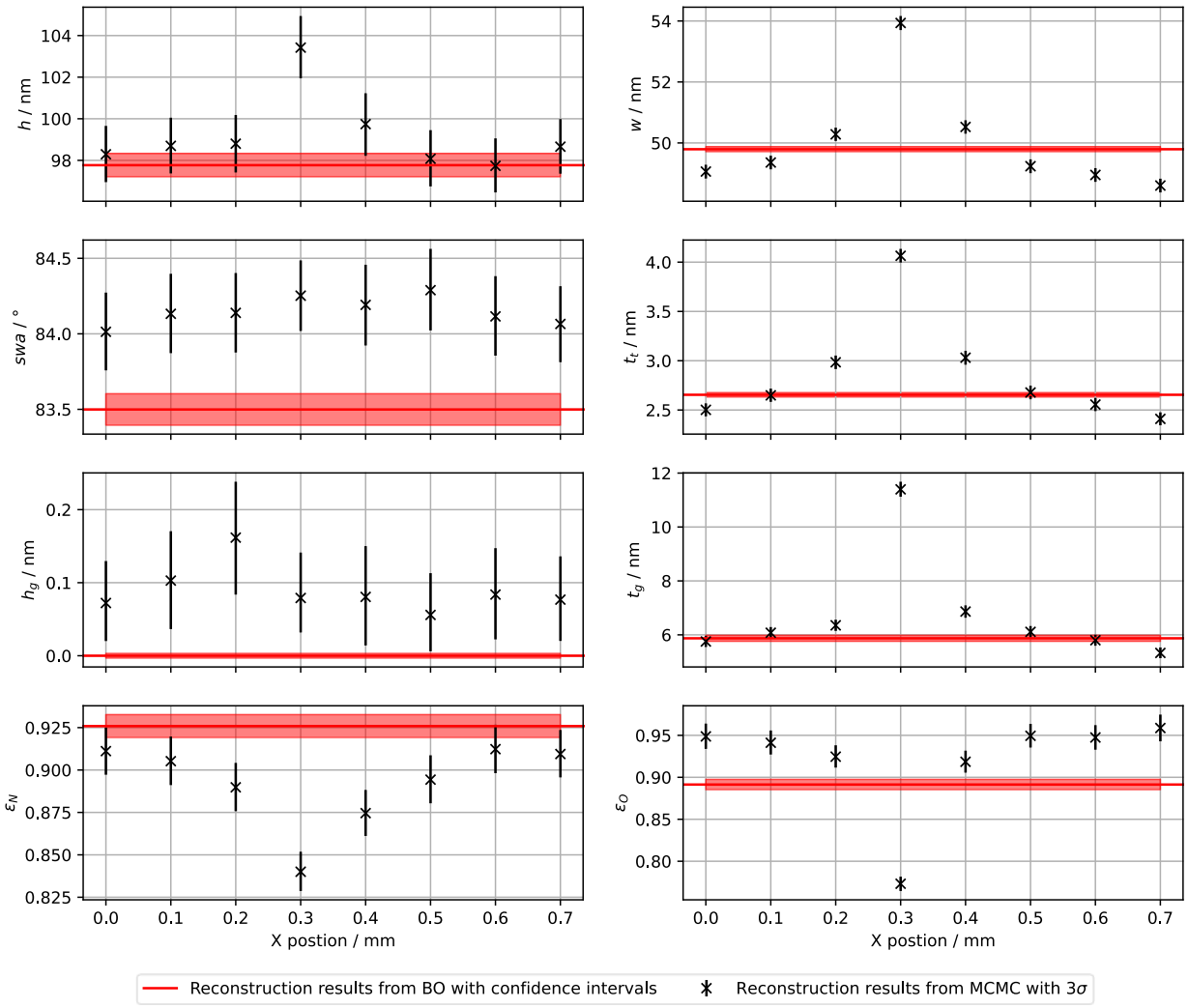


Figure 4.35 Comparison of the reconstruction results from a different measurement with the BO (red) to the results from the MCMC for the different X positions (black) with the error bars indicating the 3σ standard deviation of the posterior distribution. The model parameters are width w, the height h, side-wall angle swa, etch offset h_g, thickness of the oxide t_t, thickness of the oxide in the grove t_g, modeling error ϵ_N and ϵ_O.

Figure 4.35 shows the result of the MCMC simulation (black) with the 3σ standard deviation of the posterior distribution and compares it with the results of the BO reconstruction (red) for all model parameters. At the high carbon fluorescence position (X = 0.3 mm) a significant deviation in the line width w and height h, the oxide layers t_t and t_g and, the modeling errors ϵ_N, ϵ_O is observed. The side-wall angle swa and etch offset h_g are within the standard deviation of the results. It is unlikely that the shape of the nanostructure is as inhomogeneous as suggested by the MCMC simulation results. At X = 0.3 mm (position of contamination) χ^2 is significantly higher compared to the other positions indicating a problem with the model. It is interesting to note that the standard deviation of the model parameters did not increase at the position with the contamination. Here the model used for the reconstruction is not complete because it does not include a contamination layer. It is difficult to include such a layer in the model because the parameters like the optical constants and the exact elemental composition of the contamination layer are unknown. Furthermore, adding another layer would increase the computational time.

Nevertheless, the FEM Maxwell solver can be used to calculate the fluorescence signal for a shape with the model parameters obtained from the BO reconstruction and an estimated contamination layer containing carbon and oxygen. The contamination layer was assumed to have a stoichiometry of C_2O with a density of $\rho_{C_2O} = 1.8 \frac{\mathrm{g}}{\mathrm{cm}^3}$, because this gave a better fit to the experimental data than other stoichiometries or densities. The corresponding values for the optical constants were taken from the CXRO database [45]. The thickness of the contamination layer was chosen to be $t_C = 2.5\ \mathrm{nm}$.

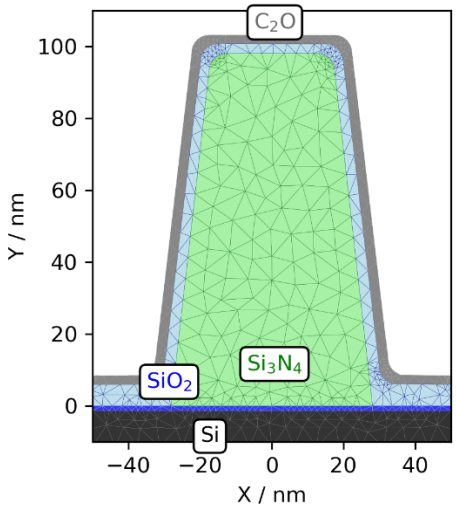


Figure 4.36 Cross section of the reconstructed line profile with a $t_C = 2.5\ \mathrm{nm}$ thick contamination layer (C2O) and periodic boundary condition to the right and left side used for the FEM based Maxwell solver calculation of the fluorescence of the contaminated line.

Figure 4.36 shows the cross section based on the reconstructed line profile with the assumed contamination layer on top of the structure. This shape was used for the calculation of the oxygen, nitrogen, and carbon fluorescence. In addition, a modeling error (or scaling factor) was calculated for all three types of fluorescence. The parameters were chosen by hand to describe the experimental data as well as possible without the need for time-consuming and costly computational optimization.

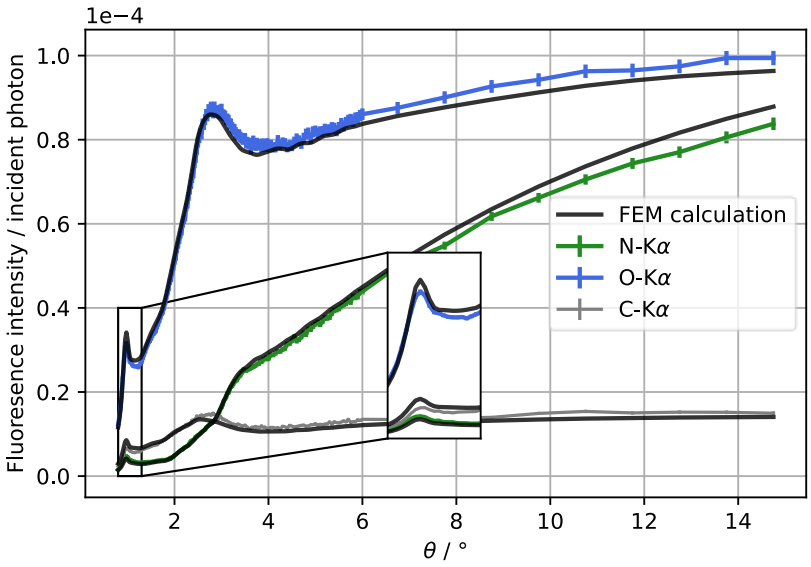


Figure 4.37 Calculation of the fluorescence intensities for the nitrogen, oxygen, and carbon signals compared to the experimental data at the high carbon fluorescence position. The model parameters of the shape are taken from the BO reconstruction with an added contamination layer.

Figure 4.37 shows the results of the calculation with the FEM solver of a nanograting with a contamination layer. The calculation is compared with the measured data of the Si_3N_4 sample at the position of high contamination. For small angles of incidence, the calculation agrees with the experimental data. For higher angles, a difference can be seen. The parameters of the contamination are only chosen by hand and can differ to the real parameters of this layer. Nevertheless, the agreement between measurement and calculation is good at grazing angles of incidence for the assumed parameters. This shows that if contamination is included in the model, the parameters of the reconstructed shape may not be as different as seen in Figure 4.35.

4.5.2 Analyzing the growth of contamination during the measurements

By analyzing the carbon fluorescence of all measurements, I observed an increase of this kind of fluorescence after repeated experiments. In the previous section, it was shown that for a thick contamination layer the model needs to be adapted to give reliable reconstruction results. In this section, the growth of a contamination layer on Sample 1 of the HSQ sample set is investigated. Similar scans with the same angular range were taken during a 48 hours period. In between, the sample was aged by scanning parallel to the lines over the sample with a high grazing angle of incidence.

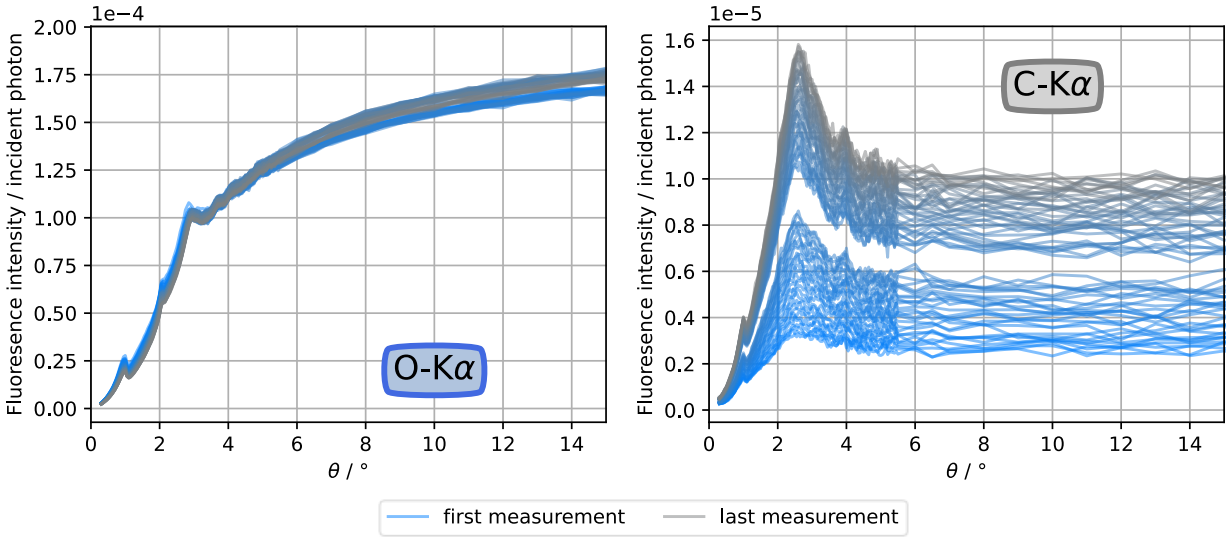


Figure 4.38 Oxygen fluorescence (left) and carbon fluorescence (right) for different angles of incidence θ and the different measurements starting from blue (first measurement) to grey (last measurement).

In Figure 4.38 the measurement results of the oxygen and carbon fluorescence are shown. As the number of measurements increases, the oxygen fluorescence decreases while the carbon fluorescence increases. Both angle dependent fluorescence shows features related to the shape of the structure. For each line, the shape was reconstructed with a PCE surrogate[8] (see Section 4.3.3) and DE[9] as the optimization method. The reconstruction process was performed without including any contamination layer in the model to investigate the influence of an unaccounted contamination on the reconstruction. To analyze the goodness of fit, the reconstructed minimum χ^2 and the deviation from the original shape are examined. The $error_{RMSE\ relative\ shape}$ (adapted from Equation 12)

$$error_{RMSE\ relative\ shape} = \sqrt{\frac{\sum_{i=1}^{n}\left(Y_{original}(x_i) - Y_{reconstructed}(x_i)\right)^2}{\sum_{i=1}^{n}\left(Y_{original}(x_i)\right)^2}} \tag{28}$$

is calculated, which gives information about the difference of the reconstructed to the original shape. Here $Y_{original}(x_i)$ describes the original shape and $Y_{reconstructed}(x_i)$ the reconstructed shape from the interpolation with the B-splines at the position x_i, where n is the total number of x_i. The shape error ($error_{RMSE\ relative\ shape}$) was calculated compared to the results of the BO reconstruction as the reference shape.

[8] For the PCE, the range of the model parameters (B-splines parameterization) for the training data set was chosen to be close to the results of the BO reconstruction.

[9] Due to the reduced computational complexity of the surrogate, it is not necessary to use BO here.

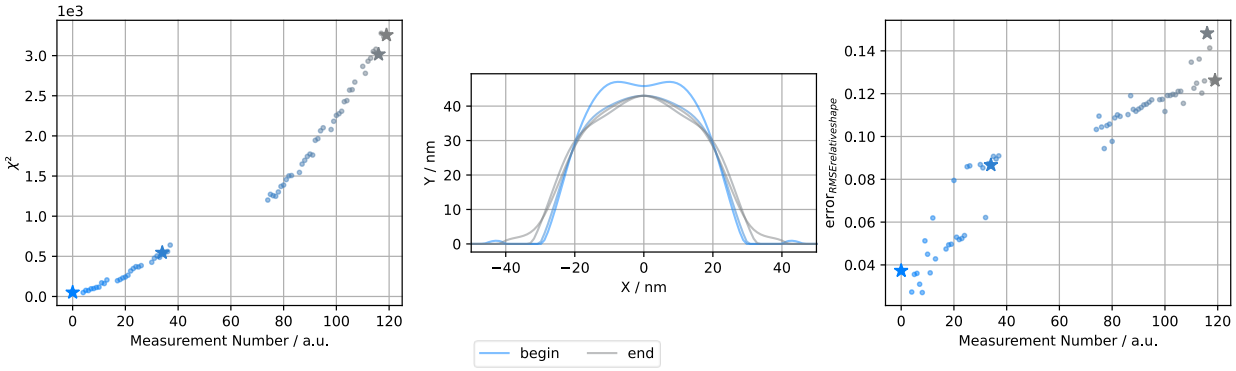


Figure 4.39 On the left side the χ^2 for the measurements (blue first measurement to grey last measurements) is shown. In the middle three reconstructed shapes for the HSQ layer for chosen measurements (visualized with stars in the other plots) are shown. On the right, the relative RMSE for the shape deviation is shown.

Figure 4.39 summarizes the results of all the reconstructions. The left side shows the reconstructed χ^2 for the different measurements. In the middle are three chosen cross-sectional shapes displayed, while on the right the shape error ($error_{RMSE\ relative\ shape}$) is shown. The stars in the right and left pictures are the position of the chosen shapes in the middle. The χ^2 increases steadily with some small outliers. For the first 10 measurements, the shape deviation remains below 5%, but increases up to 15% resulting in a broader shape (grey shapes in the middle of Figure 4.39). The shape reconstructed for the 34th measurement has a hat like feature with a width like the first reconstruction (blue shapes in the middle of Figure 4.39). Even after more than 100 measurements the carbon fluorescence increases as well as the shape deviation and the χ^2. If the shape does not change during the measurements, the contamination layer should be included after 10-20 measurements, depending on the acceptable shape deviations.

To study the influence of contamination in a more controlled environment artificial data are created without any other model errors (e.g., roughness or a change in the shape of the nanostructure). For the artificial data set, the parameters of the contamination layer are well known. The contamination is a carbon layer[10]. The layer thickness was increased from no contamination ($t_C = 0\ nm$) to a maximum thickness of $t_C = 3\ nm$ with a step size of 0.01 nm. The underlying shape was kept constant to results of the BO reconstruction (reference shape previous paragraph).

[10] The optical constants of the carbon layer were taken from the CXRO database [45] with a density of $1.8\,\frac{cm^3}{g}$.

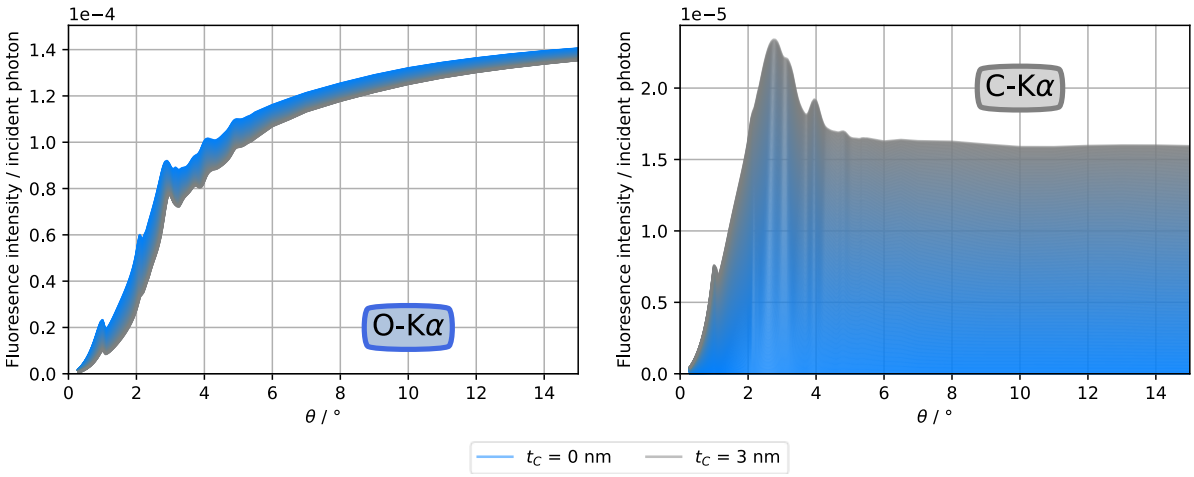


Figure 4.40 Oxygen fluorescence (left) and carbon fluorescence (right) for different angles of incidence θ. The thickness of the contamination layer t_C was varied between 0 nm (blue) and 3 nm (grey).

Figure 4.40 shows the oxygen and carbon fluorescence calculated with the FEM Maxwell solver. As the layer thickness increases, the oxygen fluorescence decreases while the carbon fluorescence increases, which is very similar to the experimental data (Figure 4.38). Both the oxygen and carbon fluorescence show features related to the shape of the structure. To estimate the thickness at which the carbon layer needs to be included in the model, all artificial fluorescence curves are reconstructed with the same model that does not include a contamination layer.

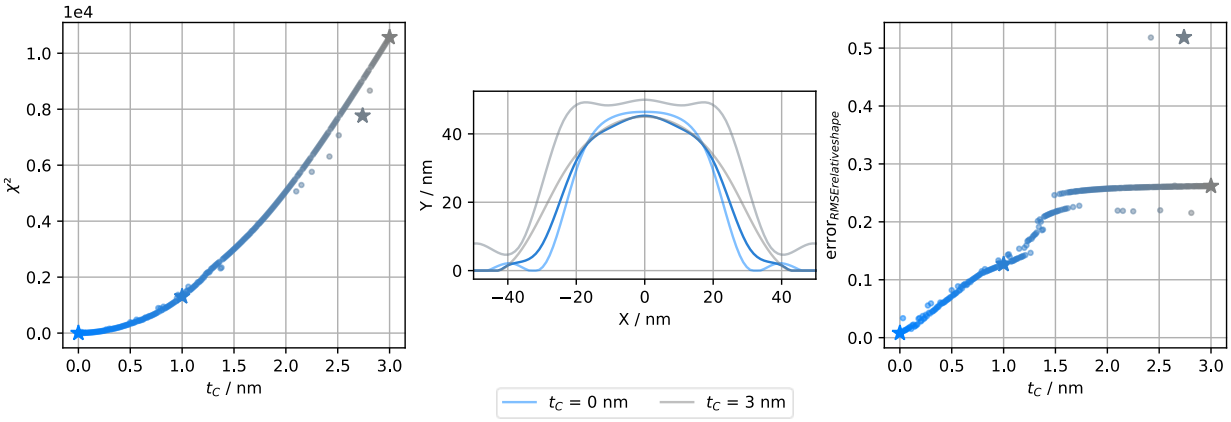


Figure 4.41 On the left side the χ^2 for the different thickness of the contamination layer t_C (blue no contamination to gray 3 nm of contamination) is shown. In the middle chosen reconstructed shapes for the HSQ layer is shown. At the right side the relative RMSE for the shape deviation is shown. The stars in the right and left plot show the corresponding χ^2 and shape deviation for the chosen shapes.

Figure 4.41 summarizes the results of all the reconstructions. The left side shows the reconstructed χ^2 for the different carbon thicknesses. In the middle are chosen cross-section shapes, while on the right is the shape error ($error_{RMSE\ relative\ shape}$). The chosen shapes are marked with stars in the right and left plots.

The χ^2 increases steadily with some outliers. For a small carbon layer, the reconstruction results are mainly unaffected with a shape deviation below 5%. But for a carbon layer of only 1 nm the shape error increases to more than 10%. At this point the line width increased compared to the original shape. At $t_C = 1.5\ nm$ the shape deviation reaches 25% with some outliers. Since the domain of the model parameters is small, the reconstructed model parameters hit the domain boundaries, which reduces the possible shape deviation to the original shape explaining the leveling off the error. Nevertheless, there are two outliners with shape deviations above 50%. Here a shape with increased height and width compared to the original shape is reconstructed. During the optimization, the modeling error was kept as a model parameter. This scaling factor can explain the huge shape deviation without a huge increase in the χ^2. This factor cannot account for the contamination layer, because the χ^2 value and the corresponding shape error increases, nevertheless. The shape at $t_C = 3\ nm$ looks like the original shape with an increased width and a slightly decreased height. The outliners compared to the leveling off the shape error indicate that the unaccounted contamination layer in the optimization process leads to a different minimum or even multi-modalities where more than one shape can describe the data with a similar χ^2 but the high χ^2 values shows that the optimization results were not able to describe the artificial data.

Comparing the results of the artificial data set with the results of the experiment, the results are similar in terms of the change in fluorescence with increasing contamination and in the reconstruction results. This makes it unlikely that the change in the reconstructed shape is due to a physical change in the shape of the nanostructures. This study underlines the importance of investigating the influence of small contamination layers on the reconstruction. Even layers less than 1 nm thickness have a significant impact on the reconstructed shape. Often it is not possible to include the contamination layer in the FEM calculation because the parameters of the contamination are unknown and can change during the measurements. In the future, different options must be explored, such as the inclusion of a model error that can describe the influence of the contamination on the reconstruction, or the reduction of the contamination generated during the measurements (e.g., lubricant-free measuring chambers).

5 Summary

Nanotechnology has found its most prominent application in today's semiconductor devices, which consist of intricate 2D and 3D nanostructures with feature sizes in the single-digit nanometer regime. The performance of these nanostructures is heavily reliant on precise fabrication techniques that control the desired dimensional parameters and 3D element compositions. There is a significant demand for nanometrology techniques that enable the characterization of these parameters with adequate sensitivity. The method developed in this work is a nanometrology tool with the advantages of being element sensitive, non-destructive, and providing average information over large sample areas in comparison to established tools like SEM, TEM, or AFM.

The goal of this method is to characterize the shape of nanostructures with sub nanometer resolution. Therefore, angular resolved GIXRF is measured from the nanostructure under investigation. The shape sensitivity of the measurements can be explained with the XSW field, which originates from the interference between the incident and the reflected X-ray beam. The field inside the structure depends on the angle of incidence and the shape of the structure and influences the XRF intensity of an element depending on its spatial position within the electromagnetic field distribution. The XSW field and the corresponding fluorescence are calculated by utilizing a FEM-based Maxwell solver and subsequently applying the 2D Sherman's equation. Here an inverse problem arises, which is solved with an optimization method. The FEM solver offers high flexibility of possible shapes. The method and its different aspects were tested with a Si_3N_4 grating with an oxide layer and a sample set consisting of four HSQ gratings.

To ensure accurate calculations, it is crucial to know of the optical constants of the materials comprising the nanostructure. These optical constants can be obtained from a database e.g., Henke, provided that the incident energy is not near an absorption edge and the density and elemental composition of the material are known.

In this thesis, soft XRR measurements were used to investigate the anisotropy of a quartz crystal and to study the optical constants of Si_3N_4, amorphous SiO_2, and HSQ near the oxygen K-edge. It was observed that at the absorption edges there is a pronounced sensitivity to the chemical environment of the different materials, leading to deviations between the calculated optical constants and the values from the database. I could verify in my thesis that the incident energy chosen for the GIXRF measurements is not near the absorption edge. Here, the retrieved optical constants are compared to the CXRO database to estimate the density of the material, since nanostructures can have a reduced density compared to bulk materials.

The key aspect of the method is to identify the shape that accurately represents the angle resolved fluorescence measurements. Therefore, the optimization methods BO and PSO were tested and compared in this study. For the samples utilized in this thesis, I could show that BO outperformed PSO in terms of efficiency. Consequently, BO was selected as the standard optimization method.

To perform the reconstruction, the shape needs to be parameterized. For the Si_3N_4 sample, a trapezoid with rounded corners was chosen as the parametrization method. However, for the HSQ sample set, I

compared the use of a Gaussian profile (4 model parameters) and B-splines (8 model parameters) for parameterization. B-splines are flexible allowing to describe a huge variance of shapes. In a preliminary study, both parameterizations were applied to describe the TEM cross section of sister samples. For one out of four samples, the Gaussian profile failed to describe the cross section. Nevertheless, the Gaussian profile exhibited stable convergence during the optimization process but lacked the flexibility required to accurately describe all four samples. With the B-splines parameterization the calculated fluorescence of the reconstructed shapes described the experimental data nearly perfectly. However, the high flexibility of B-splines adds a risk of introducing unrealistic features in the shape. Future improvements should be made to the constraints and the number and positioning of control points.

Depending on the chosen incident energy, nitrogen and oxygen fluorescence can be observed from the Si_3N_4 sample. To make an educated choice of the incident energy, artificial data were generated for an incident energy below and above the oxygen edge and the corresponding shape was reconstructed. For both reconstructions (with and without oxygen fluorescence), the confidence intervals of the model parameters were calculated and compared. Using the fluorescence from different elements in the reconstruction decreases the confidence intervals of the model parameters without increasing the computational effort significantly. For the measurements of the Si_3N_4 sample the incident energy was chosen to be above the oxygen absorption edge to excite oxygen and nitrogen fluorescence. A reconstructed shape closely resembling the experimental data was achieved. Notably, the calculated confidence intervals of the model parameters exhibited varying degrees of sensitivity to different parameters indicating that this method is more sensitive to the line width of nanograting as compared to the line height.

To study the sensitivity in more detail and to reduce the computational complexity of this method, a PCE surrogate was successfully generated to replace the FEM-based forward calculation. From the Sobol coefficients obtained from the PCE, the sensitivity of the reconstruction to the different model parameters was estimated. The sensitivity of the model parameters depends on the type of fluorescence (oxygen or nitrogen), showing that the inclusion of oxygen fluorescence can increase the sensitivity for some model parameters. The highest sensitivity was found for the surface oxide layer for both fluorescence. The method has a high surface sensitivity. The surrogate allowed the use of a MCMC simulation to calculate the uncertainties of the model parameters. The model parameters matched the results from the BO with the 3σ regime and the uncertainties indicating sub nanometer resolution of this method.

To validate the shape of the nanostructure obtained through the method, various microscopic techniques were employed to examine the gratings. The results indicated an agreement between the SEM image and the AFM measurement of the Si_3N_4 sample. However, in the case of the HSQ sample set, a discrepancy was observed between the reconstructed shape and the TEM image of sister samples. This deviation can be attributed to the variability in the samples generated from HSQ, affecting their reproducibility. A TEM image was not taken here, because it would have involved the destruction of the sample. Despite this, the AFM measurement demonstrated a satisfactory agreement with the reconstructed shape. Deviations can be explained by the reproducibility of the sample preparation, the small, examined area of the microscopy images, or the limitations of the microscopy images (e.g., tip size of the AFM).

The model used for the forward calculation is usually a simplified approximation of the real world since it is not possible to include aspects such as contamination in the calculation. In most cases, it is not possible to include such a layer in the model because the parameters (e.g., thickness, optical constants, or exact elemental composition) of the contamination are unknown and this layer can change during measurement. It is important to understand the influence of model imperfections on the reconstruction. The influence of a small contamination layer on the result of this method was studied in detail. To the high sensitivity of this method to surface features, it was found that a contamination layer has an influence on the reconstruction of the shape. This was studied with an artificial dataset and a corresponding experiment, where the sample was illuminated over a long period of time. During this experiment, the growth of a contamination layer on the sample was observed. The contamination layer needs to be included in the reconstruction process to give reliable shapes, if it exceeds a given thickness. In future works, different options to include this model error in the reconstruction process need to be studied.

The method developed in this thesis uses complex simulations and optimization for shape reconstruction. To simulate the measurements accurately, different adjustments can be made to the method, e.g., choice of optimization methods or parameterizations. To facilitate adapting the method developed here for characterizing nanostructures in line during fabrication, the use of a surrogate can significantly reduce computational time. This method is an important and necessary extension of the already exciting nanometrology tools for the characterization of nanostructures.

6 Publications

Parts of this thesis have been published before in peer reviewed journals and conference proceedings. I declare that part of the text and the graphics of the three papers have been reused and adapted in this thesis.

- A. Andrle, P. Hönicke, P. Schneider, Y. Kayser, M. Hammerschmidt, S. Burger, F. Scholze, B. Beckhoff, and V. Soltwisch, "Grazing incidence X-ray fluorescence based characterization of nanostructures for element sensitive profile reconstruction", Proc. SPIE 11057, 110570M (2019); https://doi.org/10.1117/12.2526082

The text of this proceeding was written by me. I participated in the GIXRF measurements. I set up the optimization and performed the reconstruction of the shape of the nanostructures. All co-authors contributed to the final manuscript. All co-authors contributed to the final manuscript.

- A. Andrle, P. Hönicke, J. Vinson, R. Quintanilha, Q. Saadeh, S. Heidenreich, F. Scholze, and V. Soltwisch, “The anisotropy in the optical constants of quartz crystals for soft X-rays”, *J. Appl. Cryst.* (2021). **54**, 402-408, https://doi.org/10.1107/S1600576720016325

I wrote the text of this paper. I prepared and participated in the XRR measurements. From the measurements I calculated the corresponding reflectivity. I set up and performed the optimization of the optical constants. All co-authors contributed to the final manuscript.

- A. Andrle, P. Hönicke, G. Gwalt, P.-I. Schneider, Y. Kayser, F. Siewert, and V. Soltwisch, “Shape- and element-sensitive reconstruction of periodic nanostructures with grazing incidence X-ray fluorescence analysis and machine learning”, *Nanomaterials* (2021), 11, 1647; https://doi.org/10.3390/nano11071647

I wrote the text. I prepared and participated in the GIXRF and AFM measurements. I performed the optimization of the optical constants with the MCMC simulation. I set up and performed the reconstructions of the shape of the nanostructure with the BO including the confidence interval calculations of the result. All co-authors contributed to the final manuscript.

The list of peer-reviewed publications and conference proceedings which I co-authored but not included in this thesis:

- P. Hönicke, A. Andrle, Y. Kayser, K. V. Nikolaev, J. Probst, F. Scholze, V. Soltwisch, T. Weimann and B. Beckhoff, “Grazing incidence-X-ray fluorescence for a dimensional and compositional characterization of well-ordered 2D and 3D nanostructures”, *Nanotechnology* (2020), **31**, 505709; https://doi.org/10.1088/1361-6528/abb557
- P. Hönicke, Y. Kayser, K. V. Nikolaev, V. Soltwisch, J. E. Scheerder, C. Fleischmann, T. Siefke, A. Andrle, G. Gwalt, F. Siewert, J. Davis, M. Huth, A. Veloso, R. Loo, D. Skroblin, M. Steinert, A. Undisz, M. Rettenmayr, B. Beckhoff, “Simultaneous dimensional and analytical characterization of ordered nanostructures”, *Small* (2022), 18, 2105776; https://doi.org/10.1002/smll.202105776
- Q. Saadeh, P. Naujok, V. Philipsen, P. Hönicke, C. Laubis, C. Buchholz, A. Andrle, C. Stadelhoff, H. Mentzel, A. Schönstedt, V. Soltwisch, and F. Scholze, “Time-frequency analysis assisted determination of ruthenium optical constants in the sub-EUV spectral range 8 nm – 23.75 nm”, *Opt. Express* (2021), **29**, 40993-41013; https://doi.org/10.1364/OE.426029
- A. Andrle, N. Farchmin, P. Hagemann, S. Heidenreich, V. Soltwisch, and G. Steidl, Conference paper, “Invertible neural networks versus MCMC for posterior reconstruction in grazing incidence X-ray fluorescence”, In: Elmoataz, A., Fadili, J., Quéau, Y., Rabin, J., Simon, L. (eds) Scale Space and Variational Methods in Computer Vision. SSVM 2021. Lecture Notes in Computer Science, vol 12679. Springer, Cham; https://doi.org/10.1007/978-3-030-75549-2_42
- M. Plock, K. Andrle, S. Burger, P.-I. Schneider, “Bayesian target-vector optimization for efficient parameter reconstruction”, *Adv. Theory Simul.* (2022), **5**, 2200112; https://doi.org/10.1002/adts.202200112
- D. Skroblin, A. Fernández Herrero, T. Siefke, K. Nikolaev, A. Andrle, P. Hönicke, Y. Kayser, M. Krumrey, C. Gollwitzer, and V. Soltwisch, “Challenges of grazing emission X-ray fluorescence (GEXRF) for the characterization of advanced nanostructured surfaces”, *Nanoscale* (2022), **14**, 15475-15483; https://doi.org/10.1039/D2NR03046B
- S. Staeck, A. Andrle, P. Hönicke, J. Baumann, D. Grötzsch, J. Weser, G. Goetzke, A. Jonas, Y. Kayser, F. Förste, I. Mantouvalou, J. Viefhaus, V. Soltwisch, H. Stiel, B Beckhoff and B. Kanngießer, “Scan-free GEXRF in the soft X-ray range for the investigation of structured nanosamples”, *Nanomaterials* (2022), **12**, 3766; https://doi.org/10.3390/nano12213766

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Appendix A. Optimization algorithms

In this section, the different optimizer applied in this work are introduced. I used two different local: optimizers BFGS[11] and Truncated Newton Constrained (TNC) and two heuristic global optimizers: Differential Evolution (DE) and Particle Swarm Optimization (PSO).

This first local method is the quasi-Newton algorithm L-BFGS-B [141]–[144]. A quasi-Newton method only needs the gradient of the objective function without the need of second derivates. The BFGS is the most used method of this kind. The approximation of the Hessian matrix is updated (and not recalculated) during the optimization, which makes this approach more efficient compared to methods recalculating the approximation of the Hessian matrix at every iteration. The “L” in front of BFGS stands for limited memory. For the L-BFGS method the main idea is to only use the most recent iterations (often between 3 and 20) to construct the approximation of the Hessian matrix. The “B” indicates the use of boundaries of the model parameters.

The TNC method is based on a truncated Newton algorithm [145], which is also called Newton Conjugate-Gradient. The conjugated gradient (CG) method is an iterative method to solve linear systems of equations. This method is used to calculate the search direction (find an inaccurate solution to Newton equations).

Differential Evolution (DE) is a heuristic global approach for minimizing nonconvex and non-differentiable continuous functions [146]. The method can be parallelized and can be modified by few control variables. DE starts from a random initial population. New parameters are generated by mutation e.g., adding the weighted difference (differential weight) of two parameter vectors to a third vector. The parameters of the mutation are mixed with the target. Resulting in a trial vector. If the trial vector yields a smaller value of the objective (cost) function, the trial vector replaces the target.

Particle Swarm Optimization (PSO) is a heuristic global optimization approach [147], [148]. PSO is closely connected to the simulation of bird flocking, fish schooling, and swarm theory. The optimum is searched for with a swarm of particles, whose positions are updated each step. The particles are moving in the direction of the global best position (*gbest*) of the swarm and the previously best position (*pbest*). The velocity (the direction and length of the movement) depends on the weighted previous velocity, *gbest* and *pbest*, some random variables within range [0,1], and some positive constant parameters (acceleration coefficients). The velocity is added to the position of the particle to find the new position. The weight of the previous velocity and the acceleration coefficients need to adapt to the optimization problem and can have an influence on the performance of the optimizer.

In some cases, the optimizers described in the previously are not sufficient or too inefficient e.g., for time consuming forward calculations. For time-consuming forward calculations, an "intelligent" way to find the

[11] The algorithm was developed independently by Broyden, Fletcher, Goldfarb, and Shanno.

optimum is needed. Machine Learning (ML) and the resulting surrogates can be used. Here, the Bayesian view on statistics[12] can be applied.

The two following optimization approaches are based on Bayes' Theorem [149]:

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} \tag{29}$$

Here $P(A)$ or $P(B)$ are the probability distribution of the events A or B without any conditions. $P(A|B)$ ($P(B|A)$) is the conditional probability of A (B) occurring given that B (A) happened. In the following $P(A|B)$ is called the posterior distribution, which gives the probability of the proposition A for the observed data B. This quantifies the relative probability of each possible value of A to be the true value [150]. $P(A)$ is the prior distribution of A which must be chosen to be a reasonable probability distribution for A (e.g., a uniform distribution of the model parameters in between physically sensible boundaries). $P(B|A)$ is called the likelihood or the information retrieved from evaluating the data B. $P(B)$ is the evidence likelihood and independent of A. This ensures that the density is normalized. If more information becomes available, the posterior distribution can be updated with the new information and the old posterior becomes the new prior. This is called Bayesian inference (BI).

One way to estimate the posterior distribution in BI is the Markov Chain Monte-Carlo (MCMC) method. The first MCMC algorithm, which is now called the Metropolis algorithm, was published by Metropolis et al. in 1953 [151]. With Monte-Carlo the properties of a distribution are estimated by drawing and examining random samples from the distribution. This approach is useful, when the samples are easy to draw (e.g., short computation time of the objective). The Markov chain in MCMC indicates that random samples are generated by a special sequential process [152]. The new samples depend only on the previous samples ("Markov" property). This can be used in BI to approximate aspects of the posterior distribution, which are normally not possible to calculate directly or analytically. In this thesis I use the Python implementation [153] of an algorithm based on [154]. There, an affine invariant ensemble sampler has been proposed. An ensemble consists of a specific number of walkers. One walker contains a combination of model parameter. To move a walker a proposal is generated with the help of the other walkers in the ensemble. During the optimization the ensemble is updated one walker at a time. The proposal is accepted or rejected by applying the Metropolis Hasting rule [151], [155].

A different approach, that is also based on BI, is the Bayesian Optimization (BO). The BO is very data efficient, therefore it is useful in situations where the evaluation of the objective function is costly (e.g., high computational time of the objective) [36]. The BO uses a stochastic model of an unknown objective function to be minimized in order to determine promising parameter values [156]. The stochastic model gives the prior distribution of the objective and is updated through function evaluations. Most commonly BO is used in conjunction with Gaussian process (GP). GP yield a non-parametric approach to estimate the

[12] For Bernoulli, Bayes, and Laplace probability represented a degree-of-belief instead of a long-run relative frequency at which an event occurs [66, Ch. 1.4].

value of a function (here objective). For BI it is not necessary to describe the objective with a parameterized function[13] [157, Ch. 45], because the prediction of the value of the objective at unknown position depends only on the prior distribution and the assumed noise model. Unsurprisingly the GP is based on a multivariate gaussian distribution. This kind of distribution is defined by a mean vector and a covariance matrix. The GP generalizes the multivariate gaussian distribution to a stochastic process defined by a mean and covariance function (kernel). The covariance function defines the features of the function we want to fit (e.g., if we expect the function to be periodic or the smoothness of the function) and is modified with hyperparameters which need to be learned during the optimization. The prior mean function is often taken as a constant and updated with the already evaluated data. If some knowledge about the objective is available, it can be incorporated into the mean function. With the GP it is possible to predict the value of the unknown objective with its uncertainty at any point. Now we need a mechanism to select the next point. We want to evaluate the function at. In this thesis I use the expected improvement (EI) [158] as an acquisition function. EI balances the global and local search of the optimum of the objective [159]. The next sampling point is chosen to be the point where the EI is maximal. The acquisition functions are often multimodal. For an efficient sampling the acquisition functions need to be cheap to evaluate or approximate [36]. In this thesis I use the BO implemented into the JCMsuite software package [55].

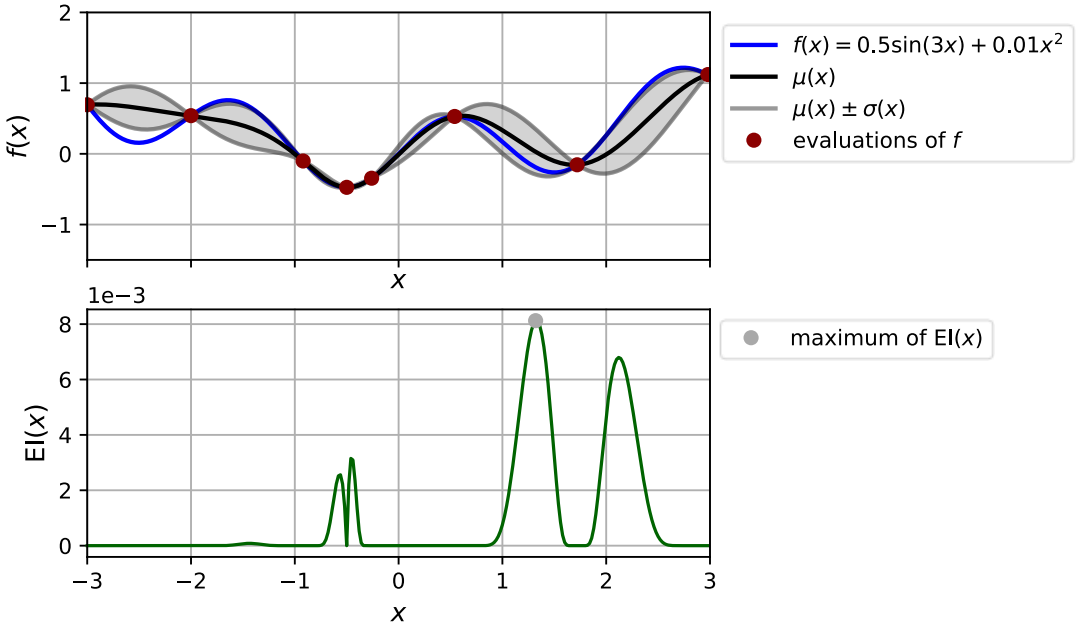


Figure 0.1 Illustration of the BO procedure showing the true values (blue line) of the function $f(x) = 0.5 \sin 3x + 0.01x^2$, the mean $\mu(x)$ (black line) and confidence intervals $\mu(x) \pm \sigma(x)$ (grey area) estimated with the GP for some observation points (red points). In the lower part $\mathrm{EI}(x)$ (green line) and its maximum (grey point) are shown.

Figure 0.1 shows the BO procedure of a random example function. The prediction from the GP is illustrated as well as the EI. The uncertainty of the prediction is smallest at the evaluated points and highest

[13] A parameterized function is defined by a finite number of parameters.

between points with great distance. The EI is highest at regions where the function was not evaluated or close to the minimum of the sampled points. The prediction of the function and the EI are updated with each new evaluation of the function. With the help of the stochastic surrogate (the GP) we can estimate the confidence intervals $\sigma_{\overrightarrow{gp}}$ of the optimized model parameters. This is possible by inverting the Hessian matrix of the error function (Equation 15 or Equation 14) at the optimum found by the optimizer.